

The Margulis–Platonov conjecture over global function fields

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Abstract

We prove the Margulis–Platonov conjecture over every global function field, including characteristic two. For an absolutely almost simple simply connected algebraic group, every noncentral abstract normal subgroup of its rational points is the inverse image of an open normal subgroup of the finite product of its anisotropic local groups.

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1 Introduction

Let k be a global function field: a finite extension of $\mathbb{F}_q(t)$. Let G be an absolutely almost simple simply connected algebraic group over k . The group $G(k)$ is considered as an abstract group. Its

normal subgroups nevertheless reflect the topologies of its local groups. For a place v of k , write k_v for the completion and set

$$A(G) = \{v : \text{rank}_{k_v} G = 0\}, \quad H_A = \prod_{v \in A(G)} G(k_v).$$

The set $A(G)$ is finite. Each of its factors is compact, and H_A carries the product topology. Let $\delta_A : G(k) \rightarrow H_A$ be the diagonal homomorphism. The Margulis–Platonov conjecture asserts that every noncentral normal subgroup of $G(k)$ is obtained by pulling back an open normal subgroup of H_A .

Theorem 1.1. *Let k be a global function field and let G be an absolutely almost simple simply connected k -group. If $N \triangleleft G(k)$ and $N \not\subseteq Z(G(k))$, then*

$$N = \delta_A^{-1}(W)$$

for an open normal subgroup $W \triangleleft H_A$. This holds in every positive characteristic, including when G is anisotropic over k .

If $A(G)$ is empty, the product H_A is trivial, and the theorem says that $G(k)$ has no proper noncentral normal subgroup. In general it identifies all noncentral normal subgroups using finitely many compact local groups. In particular, every finite quotient of $G(k)$ is continuous for the topology induced by these local factors: its finite-index kernel is Zariski dense and therefore noncentral. The substantive issue is this particular local description: finite index alone does not identify which local topology detects a normal subgroup.

1.1 Historical context

The conjecture asks whether the compact groups at the anisotropic places account for every noncentral normal subgroup of the rational-point group. Its origins lie in Kneser’s simplicity question for quaternion norm-one groups and Platonov’s extension of that question to simply connected simple groups. Margulis formulated the description by anisotropic local factors over global fields in [20, Russian original, Section 2.4.8]. The finite-index assertion in his Corollary 2.4.9 is an essential first step. The further problem is to show that each resulting finite quotient comes from the specified local groups. This also explains the conjecture’s role in the congruence subgroup problem, which studies finite-index subgroups of arithmetic groups and needs additional information about their congruence completions [31, pp. 73–76].

For anisotropic inner forms of type A , the problem concerns norm-one groups of division algebras. The quaternion case was completed by Margulis in 1980, as recalled in Tomanov’s historical account [44, p. 895]. Tomanov later proved the conjecture for division algebras of power-of-two index and reduced arbitrary index to odd index [44, Theorem and Corollaries 1–2, p. 896]. Platonov–Rapinchuk and Raghunathan developed the arithmetic control of commutator subgroups [24, 29]; its global-field form is used explicitly in [33, proof of Theorem 6.4]. Rapinchuk–Potapchik reduced the number-field problem to excluding nonabelian finite simple quotients of the multiplicative group of the division algebra [30, Theorem 2.1 and Corollary 2.5]. Segev’s division-algebra argument and Segev–Seitz’s analysis of commuting graphs supplied this exclusion [38, Theorem A][39, Theorems 1–3]. Rapinchuk–Segev–Seitz proved the stronger result that every finite quotient of the multiplicative group of a finite-dimensional division algebra is solvable; their Theorem 6.4 proves the Margulis–Platonov conjecture for inner type A over arbitrary global fields [33]. Rapinchuk subsequently gave a shorter proof using two-generation of finite simple groups [32].

Over a global function field, Harder’s results restrict the anisotropic absolutely almost simple simply connected groups to type A [15][26, p. 914]. The isotropic case follows from the global Kneser–Tits theorem and projective simplicity [13, Theorem 8.1]. Together with the established inner-type- A theorem, these results reduce the present problem to anisotropic outer forms of type A , represented by special unitary groups. The earlier state of the conjecture is discussed in [28, Section 3.5]; general outer-type- A applications were still explicitly conditional on it in [34, Corollary 1.6]. Theorem 1.1 provides the required normal-subgroup description for these groups in every characteristic.

1.2 Relation to the number-field proof

The immediate predecessor is the number-field theorem of [23]. We use its approach through finite quotients of abstract power subgroups, approximation within a prescribed coset, and the distinction between finite characters and nonabelian simple quotients. More specifically, the power-factor construction comes from [23, Proposition 2.6], and the finite lattice calculation for the symmetric norm torus is the argument of [23, Lemmas 2.7–2.9]. The difference-function and exact-transitivity argument for finite characters adapts [23, Section 4]. The commuting approximation and circle-extension reduction follow [23, Section 3.1 and Sections 3.3–3.7]; the invariant probability law follows [23, Section 5.2]. The finite-group obstruction is [23, Lemma 5.2 and Section 6], and the unitary induction follows [23, Section 7]. We reproduce the lattice calculation and the finite-group argument. We also verify an additional subgroup-complement property of the chosen finite-group subsets, which is needed for the present recurrence proof.

Several independent inputs retain their own roles. We use Prasad’s strong approximation theorem over function fields [25]; the local and global cohomological results of Bruhat–Tits and Harder provide the structural reductions [2, 15]. Demarche–Harari’s duality and local-global theorems supply the positive-characteristic arithmetic for tori and homogeneous spaces [6, 7]. The passage from a simple quotient to an action of the full unitary group uses Feit–Seitz’s theorem that class-preserving automorphisms of a finite simple group are inner [9, Theorem C]. The circle-extension argument uses Prasad–Rapinchuk’s metaplectic-kernel computation [27], while the invariant probability law uses Howe–Moore decay in its nonarchimedean form [16][3, Theorem 4.19 and Definition 4.23].

The changes in positive characteristic occur at specific points of this argument. When the exponent is divisible by the characteristic, the power map need not be locally invertible. We adapt the inherited construction by products of powers to this setting, separating the inseparable part of the exponent and using the openness of the relevant local power subgroups. The additive group of the field has finite exponent, so the integer-dilation recurrence used in the number-field proof does not give the required invariance. That proof uses the density theorem of Furstenberg–Katznelson [11, Theorem B] in [23, Section 5.5]. Here we prove a replacement by finite additive Fourier analysis. Odd characteristic uses orthogonality on finite additive subgroups and a quadratic parametrization; characteristic two uses additive polynomials and finite-dimensional coordinates over power subfields. The resulting invariance is then combined with the finite-group obstruction and the unitary induction.

1.3 The argument

The structural reductions leave an anisotropic special unitary group $S = \mathrm{SU}(D, *)$, where D is a central simple algebra of degree $n \geq 3$ over a separable quadratic extension L/k , and $*$ is a unitary involution. Write $U = \mathrm{U}(D, *)$ for the full unitary group. Every noncentral normal subgroup of $S(k)$ has finite index. It therefore contains the subgroup R generated by all e th powers for some

positive integer e . For this e , define

$$P_0 = \overline{\delta_A(R)}, \quad V_0 = \delta_A^{-1}(P_0), \quad V = V_0/R.$$

The closure is taken in the product of the anisotropic local groups. The group V is finite; it measures the possible difference between an abstract power subgroup and the subgroup prescribed by its local closure. We prove $V = 1$ for every e by induction on n , simultaneously over all global function fields.

Two arithmetic constructions permit us to work with this finite group. First, products of conjugated torus powers let us choose a rational point in an *actual prescribed coset* of R , while imposing conditions at finitely many places and controlling the remaining places by a codimension-two exclusion. A line in the parameter space avoids that exclusion on reduction. The anisotropy of the torus makes the parameter map invariant under scalar multiplication; this recovers approximation at the place omitted from strong approximation. Second, a particular norm torus has both the Hasse principle and weak approximation. This turns locally solvable simultaneous norm conditions into exact rational solutions. Sections 3 and 4 establish these constructions.

Suppose first that D is a division algebra and n is odd. Commuting approximation makes the action of $U(k)$ on V preserve conjugacy classes. If $V \neq 1$, a simple quotient and a circle-extension argument then produce one of two obstructions: a finite abelian character of $U(k)$ nontrivial on $S(k)$, or a homomorphism from $U(k)$ onto a finite nonabelian simple group whose restriction to V_0 is surjective and which kills R . Section 5 proves this dichotomy.

The simple quotient gives a finite coloring of the Hermitian elements of D through a Cayley parametrization. Local mixing supplies an invariant probability law for all translated colorings, with uniform color marginals. A finite-group argument selects a nonempty proper conjugacy-invariant subset of the target group and constrains returns between colors inside and outside that subset under fractional transformations. Additive recurrence in characteristic p forces the indicator of membership in this subset to be translation invariant. The fractional transformations then generate left rotations through every target color, forcing the chosen subset to be either empty or the whole group. Sections 6 and 7 develop this contradiction; Appendix A proves the finite-group input. Finite abelian characters require a different argument. Exact transitivity on nonzero pairs in D^2 and the additive span of the full unitary group reduce them to the known inner-type- A theorem in Section 8.

It remains to pass from odd division degree to all degrees. Section 9 does this using smaller unitary groups and quaternion norm-one groups. In even degree, a fixed quadratic Hermitian subfield supplies a centralizer of half the degree. A factorization separates an element into a quaternion factor and a factor in that centralizer. The approximation construction controls the anisotropic places of both factors, and the norm-torus theorem makes the factorization rational. Degree four requires a separate two-norm correction. Uniform control of local power subgroups allows this induction even when p divides e . Finally, Section 10 converts $V = 1$ into Theorem 1.1.

2 Arithmetic reduction and exact cosets

The proof will show that certain finite quotients of a special unitary rational group are trivial. We first explain why these quotients suffice, and why every one of their cosets admits the local approximations used later. Throughout, a global function field has its full finite field of constants, and all quadratic field extensions are separable. We use normalized absolute values at its places. A product indexed by the empty set is the trivial group.

2.1 The remaining unitary groups

We use three established results to isolate the case that requires proof. For a simply connected absolutely almost simple isotropic group over a global field, the global Kneser–Tits theorem and Tits’s simplicity theorem give perfection and simplicity modulo the center of the rational group [13, Theorem 8.1][43]. A noncentral normal subgroup therefore has central, hence abelian, quotient and must be the whole rational group. In this case the algebraic group is isotropic at every completion, so there are no anisotropic places. The Margulis–Platonov theorem is also known for every inner form of type A over every global field [33, Theorem 6.4]; see also [32, Proposition 1]. Finally, Harder’s theorem implies that an anisotropic absolutely almost simple group over a global function field has inner or outer type A [15][26, p. 914]. Consequently it remains to consider an anisotropic outer form of type A .

We write such a group as

$$S = \mathrm{SU}(D, *), \quad U = \mathrm{U}(D, *), \quad \deg_L D = n \geq 3, \quad (2.1)$$

where D is a central simple algebra over a quadratic field extension L/k and $*$ is an involution inducing the nontrivial automorphism of L/k . The algebra D need not be a division algebra. Degree two gives an inner form of type A_1 , already covered by the preceding result. Write $\bar{a}$ for the conjugate of $a \in L$ and

$$L^1 = \{a \in L^\times : a\bar{a} = 1\}.$$

On U , the notation $\det$ will always mean the reduced norm $\mathrm{Nrd}_{D/L}$. It takes values in the norm-one torus $\mathrm{Res}_{L/k}^1 \mathbb{G}_m$, whose group of rational points is L^1 . We retain the notation

$$A = \{v : \mathrm{rank}_{k_v} S = 0\}, \quad H_A = \prod_{v \in A} S(k_v), \quad \delta_A : S(k) \longrightarrow H_A.$$

The set A is finite: outside finitely many places a reductive model has quasi-split fibers and positive local rank. Each factor of H_A is compact.

Proposition 2.1. *For the groups in (2.1), the following statements hold in every positive characteristic.*

- (i) *The determinant maps $U(k) \rightarrow L^1$ and their local analogues are surjective.*
- (ii) *The groups S and U satisfy weak approximation.*
- (iii) *Every place in A splits in L/k . At such a place there is a central division algebra $\mathcal{D}_v$ of degree n over k_v for which*

$$U(k_v) \simeq \mathcal{D}_v^\times, \quad S(k_v) \simeq \mathrm{SL}_1(\mathcal{D}_v).$$

Under this identification the determinant is $\mathrm{Nrd}_{\mathcal{D}_v/k_v}$.

Proof. The exact sequence

$$1 \longrightarrow S \longrightarrow U \xrightarrow{\det} \mathrm{Res}_{L/k}^1 \mathbb{G}_m \longrightarrow 1$$

identifies each determinant fiber with an S -torsor. Harder’s global vanishing theorem and the local vanishing theorem of Bruhat–Tits give $H^1(k, S) = 1$ and $H^1(k_v, S) = 1$. These statements include characteristic two; a convenient uniform reference is [4, Theorem 5.1.1(i)], which recalls [15, Satz A] and [2, Theorem 4.7(ii)]. They prove (i).

Strong approximation for simply connected almost simple groups over global function fields holds away from a place where the group is isotropic [25, Theorem A]. There are infinitely many split places, by Chebotarev applied to a finite splitting field. To approximate at a given finite set of places, omit a split place outside that set. This proves weak approximation for S . For U , fix $\theta \in L \setminus k$. The Cayley map

$$x \mapsto (x + \theta)(x + \bar{\theta})^{-1}, \quad x^* = x, \quad (2.2)$$

is a birational map from the affine space of Hermitian elements of D to U . Its inverse, on the open set where $u - 1$ is invertible, is $(u - 1)^{-1}(\theta - u\bar{\theta})$. The difference $\theta - \bar{\theta}$ is nonzero also in characteristic two. Weak approximation for this affine space, and density of the Cayley open set in each local group, prove (ii).

Suppose that v does not split in L . A unitary involution implies that the Brauer corestriction of D is zero. For a quadratic extension of nonarchimedean local fields, corestriction preserves the local Brauer invariant. Thus $D \otimes_L L_v$ is a matrix algebra. The group S_{k_v} is consequently the special unitary group of an n -dimensional Hermitian form. The local classification of Hermitian forms over a separable quadratic extension gives anisotropic dimension at most two, in all characteristics. Since $n \geq 3$, this group is isotropic. These standard facts about unitary involutions and local Hermitian forms may be found in [17, 35]; for the characteristic-two classification see also [8, Theorem 3.3 and Corollary 3.4].

At a split place, the two simple components of $D \otimes_k k_v$ are interchanged by the involution. Choosing one component B_v identifies $U(k_v)$ with $B_v^\times$ and $S(k_v)$ with $\mathrm{SL}_1(B_v)$. If $B_v \simeq M_r(\mathcal{D}_v)$, this last group has rank $r - 1$. It is anisotropic exactly when $r = 1$, proving (iii). $\square$

2.2 Finite index and the power quotient

We will use the local Kneser–Tits theorem in the following form. If a simply connected absolutely almost simple group over a nonarchimedean local field is isotropic, its group of local points is perfect and is simple modulo its finite center. In particular, a noncentral normal subgroup is the whole group. The perfection clause matters: simplicity modulo the center first gives a central quotient, and perfection then makes that quotient trivial. This local result applies in equal characteristic, including characteristics two and three [43, 2, 13].

Lemma 2.2. *Every noncentral abstract normal subgroup of $S(k)$ has finite index.*

Proof. This is the finite-index reduction in Margulis’s normal subgroup theorem [20, Theorem 2.4.6 and Corollary 2.4.9][21]. We recall the passage from an arithmetic lattice to all rational points. Let $N \triangleleft S(k)$ contain a noncentral element z . Choose a finite set of places T containing A , the places where z is not integral, and two further places at which S is split. Fix compact open subgroups $K_v \subset S(k_v)$ for $v \notin T$, using an integral model outside finitely many places, and arrange that $z \in K_v$ there. Put

$$J_T = \prod_{v \notin T} S(k_v), \quad K^T = \prod_{v \notin T} K_v, \quad \Gamma = S(k) \cap K^T.$$

Here $S(k) \cap K^T$ denotes the inverse image under the diagonal map to J_T , and the prime denotes the restricted product with respect to the K_v .

Reduction theory makes Γ an arithmetic lattice in $\prod_{v \in T} S(k_v)$; compact factors may be projected away. Strong approximation, omitting any one noncompact factor, gives irreducibility. The two added split places have total rank at least four, since $n \geq 3$. The arithmetic normal subgroup

theorem therefore applies. The subgroup Γ is Zariski dense, so a finite normal subgroup of Γ is central. Since $z \in N \cap \Gamma$ is noncentral, the theorem gives

$$[\Gamma : N \cap \Gamma] < \infty. \quad (2.3)$$

The lattice and strong-approximation assertions used here are valid over function fields [14, 25, 21].

Let C be the closure of N in J_T . Strong approximation gives density of $S(k)$ in J_T , hence normality of C . It also gives density of Γ in K^T . By (2.3), the closure of $N \cap \Gamma$ has finite index in K^T and is therefore open. Thus C is open in J_T . For every $v \notin T$, the group $C \cap S(k_v)$ is an open normal subgroup of $S(k_v)$. It is noncentral, and S is isotropic at v because $A \subset T$. Local simplicity and perfection imply $S(k_v) \subset C$. Finite-support products are dense in the restricted product, so $C = J_T$.

Every coset of N in $S(k)$ is now dense in J_T and hence meets the open subgroup K^T . Thus every such coset meets Γ , and

$$[S(k) : N] \leq [\Gamma : N \cap \Gamma] < \infty,$$

as required. $\square$

For an integer $e \geq 1$, let

$$\begin{aligned} R &= S(k)^e := \langle g^e : g \in S(k) \rangle, & P_0 &= \overline{\delta_A(R)} \subset H_A, \\ V_0 &= \delta_A^{-1}(P_0), & V &= V_0/R. \end{aligned} \quad (2.4)$$

The notation $S(k)^e$ means the subgroup generated by powers, not merely the image of the power map. The distinction will be essential when the characteristic divides e .

Proposition 2.3. *The subgroup R has finite index in $S(k)$; P_0 is open and normal in H_A ; and V is finite. Conjugation by $U(k)$ preserves R and V_0 , and hence acts on V . If $V = 1$ for every integer $e \geq 1$, then the Margulis–Platonov assertion holds for $S(k)$.*

Proof. The map $g \mapsto g^e$ is dominant: over an algebraic closure it is surjective on each maximal torus, and conjugates of a maximal torus are dense in S . This dominance does not require separability of the power map. Weak approximation implies that $S(k)$ is Zariski dense, so the set of rational e th powers is Zariski dense as well. The characteristic subgroup R is therefore noncentral, and Theorem 2.2 gives its finite index.

Weak approximation gives density of $\delta_A(S(k))$ in H_A . A finite coset decomposition of $S(k)$ by R then gives a finite coset covering of H_A by the closed subgroup P_0 . A closed subgroup of finite index is open. Normality follows from density and normality of R , and the finiteness of V follows from that of $S(k)/R$. Finally, $U(k)$ normalizes $S(k)$ and preserves its power subgroup. Taking closures at A shows that it also preserves P_0 and V_0 . In fact weak approximation for U shows that P_0 is normalized by $\prod_{v \in A} U(k_v)$.

For the last assertion, let $N \triangleleft S(k)$ be noncentral. By Theorem 2.2, the quotient $S(k)/N$ is finite. Choose e divisible by its exponent; then $R \subset N$. If $V = 1$, we have $R = \delta_A^{-1}(P_0)$, and density induces an isomorphism $S(k)/R \simeq H_A/P_0$. The inverse image in H_A of the subgroup corresponding to N/R is open and normal and has rational inverse image N . This proves the claimed reduction. $\square$

Our task is to prove $V = 1$ for every e . The quotient V measures precisely the failure of the power subgroup R to contain all rational points allowed by its closure at the anisotropic places. The next proposition provides the approximation in each individual coset of R ; passing only to its image in a finite quotient would not suffice. The power-subgroup reduction and this use of exact cosets follow the organization of [23, Section 2.2]; the arithmetic inputs above are their function-field forms.

Proposition 2.4 (Closure of an exact coset). *Let $v_0 \notin A$, and write $S(\mathbb{A}^{v_0})$ for the restricted product of $S(k_v)$ over $v \neq v_0$. Then*

$$\overline{R}^{S(\mathbb{A}^{v_0})} = \{g \in S(\mathbb{A}^{v_0}) : g_A \in P_0\}. \quad (2.5)$$

For any $\gamma \in S(k)$, the closure of γR is obtained by replacing P_0 on the right by $\delta_A(\gamma)P_0$. In particular, all cosets γR with $\gamma \in V_0$ have the same closure.

Consequently one may approximate in an actual prescribed coset γR , meeting prescribed nonempty local open sets at finitely many isotropic places and a prescribed nonempty open subset of $\delta_A(\gamma)P_0$ at A . If $x_ \in U(k)$, the same assertion holds after left translation to the determinant slice x_*S and to the coset $x_*\gamma R$. In each assertion one may also require that the resulting point lie in finitely many nonempty k -Zariski open subsets of S or x_*S , respectively.*

Proof. Strong approximation gives density of $S(k)$ in $S(\mathbb{A}^{v_0})$. Since R is a normal subgroup of finite index, its closure C is a closed normal subgroup of finite index, hence an open subgroup. At every isotropic place $v \neq v_0$, the intersection $C \cap S(k_v)$ is open, normal, and noncentral. The local result recalled above gives $S(k_v) \subset C$. Taking closures of finite-support products, C contains the entire restricted product of these isotropic factors.

The remaining factors form the finite compact product H_A . It follows that $C = P \times \prod'_{v \notin A \cup \{v_0\}} S(k_v)$ for a closed subgroup $P \subset H_A$. Since $R \subset C$, we have $P_0 \subset P$. Conversely, the inverse image of P_0 is closed and contains R , so it contains C and gives $P \subset P_0$. This proves (2.5). Translation gives the assertion for γR , and left translation by x_* gives the determinant-slice version. For finite local conditions that include v_0 , apply the same argument with a different isotropic place outside the specified finite set omitted instead. Finally, a nonempty Zariski open subset of a smooth geometrically irreducible variety over a local field meets every nonempty analytic open subset. Impose the Zariski conditions at one place, adding a place of approximation if necessary, and shrink its local open set. The same density argument then enforces them on the rational point. $\square$

2.3 Finite quotients of groups already understood

The induction will restrict the finite quotient $S(k)/R$ to smaller special unitary groups, often over finite extensions of k . The precise consequence of their Margulis–Platonov theorem is the following.

Proposition 2.5. *Let F be a global function field and H a simply connected absolutely almost simple F -group for which the Margulis–Platonov assertion holds. Put $A_H = \{w : \text{rank}_{F_w} H = 0\}$. Every homomorphism $\psi : H(F) \rightarrow Q$ to a finite group extends uniquely to a continuous homomorphism*

$$\widehat{\psi} : \prod_{w \in A_H} H(F_w) \longrightarrow Q$$

with the same image. In particular, ψ kills every rational point whose coordinates at A_H are sufficiently close to the identity. If $A_H = \emptyset$, every such ψ is trivial. The same conclusions hold factor by factor for finite products of restriction-of-scalars groups.

Proof. Strong approximation for H away from an auxiliary split place gives weak approximation by the same argument as in [Theorem 2.1](#). In particular, $H(F)$ is Zariski dense. The kernel of ψ has finite index, so connectedness of H makes its Zariski closure all of H as well. Thus this kernel is noncentral. By the assumed theorem,

$$\ker \psi = \delta_{A_H}^{-1}(W)$$

for an open normal subgroup W of $\prod_{w \in A_H} H(F_w)$. This product is compact, so its quotient by W is finite. Weak approximation gives density of $H(F)$ in the product, and hence a canonical isomorphism

$$H(F)/\ker \psi \simeq \left(\prod_{w \in A_H} H(F_w) \right) / W.$$

Compose the quotient map with this isomorphism and the induced injection into Q . This constructs $\hat{\psi}$; density proves uniqueness and equality of images. An identity neighborhood contained in W gives the local smallness assertion. For a restriction of scalars, rational points and local points are the corresponding groups over F and products over the places above a given place. For a finite product, the images of the individual factors commute. Their continuous extensions therefore combine to give the stated factorwise conclusion. $\square$

3 Approximation in a prescribed power coset

The closure theorem of Section 2 permits approximation at finitely many places within a prescribed coset of the subgroup generated by powers. We shall also need to control a local property at every other place. The construction in this section achieves that control when its failure at integral points has geometric codimension at least two, provided certain explicitly described nonintegral points satisfy the property as well. The output remains in the prescribed abstract coset throughout.

We adapt the line-avoidance and power-factor constructions of [23, Sections 2.3–2.4]. In positive characteristic, a power map need not have a nonzero differential. We separate its inseparable part before making the geometric codimension argument. We also keep track of the order in which the finite sets of exceptional places are chosen; this will allow the construction to be applied when the local groups arising later depend on the rational point being constructed.

3.1 Avoidance in affine space

For a place v , write $\mathcal{O}_v$ for the valuation ring of k_v , κ_v for its residue field, and $q_v = |\kappa_v|$. An integral model of an affine k -variety means a model over the ring of integers outside a finite set of places. All assertions about reduction below refer to fixed such models. Enlarging the finite set allows us to extend any fixed finite collection of morphisms and equations to these models.

The following elementary form of avoidance leaves one place unrestricted. That freedom will later be removed using a symmetry of the parameter map.

Lemma 3.1 (Avoidance along a line). *Let E be a finite-dimensional k -vector space and let $Z \subset E$ be a closed subset of geometric codimension at least two. Choose $d \in E(k) \setminus \{0\}$ such that its point $[d]$ at infinity does not belong to the closure of Z in $\mathbb{P}(E \oplus k)$. Let*

$$\pi : E \longrightarrow E/kd$$

be the quotient map, and fix a rational linear section s of π . Fix a place v_0 .

There are a finite set of places Σ_0 containing v_0 and an integer $D \geq 1$ with the following property. Let $\Sigma \supseteq \Sigma_0$ be finite, and prescribe nonempty open subsets $\mathcal{W}_v \subset E(k_v)$ for $v \in \Sigma \setminus \{v_0\}$. There exist $y \in E(k)$ and elements $l \in k$ of arbitrarily large v_0 -absolute value such that

- (i) $y + ld \in \mathcal{W}_v$ for every $v \in \Sigma \setminus \{v_0\}$;
- (ii) $y + ld$ is integral outside Σ and its reduction avoids Z at every place outside Σ .

One may take Σ_0 to consist of model exceptions, v_0 , and the places with $q_v \leq D$, where D bounds the cardinalities of the geometric fibers of the reductions of $\pi|_Z$.

Proof. The restriction $\pi|_Z$ is finite. Indeed, projection from $[d]$ extends to a morphism on the projective closure of Z , since its center is disjoint from that closure. A fiber over an affine point can acquire no point at infinity: its projective fiber line meets the hyperplane at infinity only at $[d]$. Thus $\pi|_Z$ is proper. It has finite fibers, because a positive-dimensional closed subset of a fiber line would be the whole line and would have $[d]$ in its projective closure. A proper morphism with finite fibers is finite. Its image $Y \subset E/kd$ is closed, and

$$\dim Y \leq \dim Z \leq \dim E - 2 < \dim(E/kd),$$

so Y is proper.

After excluding finitely many places, the decomposition $E = kd \oplus s(E/kd)$ is integral, d is a basis of its first summand, and $\pi|_Z$ extends to a finite morphism of the models. A finite set of module generators for its coordinate algebra gives a uniform bound D for the cardinality of every geometric fiber. Include these exceptional places and all places with $q_v \leq D$ in Σ_0 .

Linear projection is open over each completion. Additive strong approximation away from v_0 , equivalently the usual consequence of Riemann–Roch for a function field, therefore gives

$$q \in (E/kd)(k) \setminus Y(k)$$

that is integral outside Σ and belongs to $\pi(\mathcal{W}_v)$ at every prescribed place. To ensure $q \notin Y$, shrink one projected opening to avoid Y ; a proper algebraic subset has empty interior over an infinite local field. If there are no prescribed openings, impose such an additional opening inside the integral points at one auxiliary place. This does not change the integrality requirement. Put $y = s(q)$.

The reduction of q belongs to the reduction of Y at only finitely many places outside Σ . For example, choose a polynomial vanishing on Y but not at q , clear its finitely many denominators, and use the fact that its nonzero value at q has only finitely many zeros. At each of these places the forbidden values of the residue of l number at most $D < q_v$. There is therefore a nonempty congruence condition on $l \in \mathcal{O}_v$ for which the reduction of $y + ld$ avoids Z .

At a prescribed place, $q \in \pi(\mathcal{W}_v)$ says precisely that the set of $l \in k_v$ with $y + ld \in \mathcal{W}_v$ is nonempty and open. Additive strong approximation away from v_0 now supplies $l \in k$ satisfying these finitely many conditions and integral at every remaining place outside Σ . Shrink the openings for l to be relatively compact. There are infinitely many such l , since a nonempty open subset of the adeles away from v_0 meets the dense diagonal copy of k in infinitely many points. If their absolute values at v_0 were bounded, these points would lie in a compact subset of the full adèle ring. Its intersection with the discrete subgroup k is finite. Hence $|l|_{v_0}$ is unbounded, as required. $\square$

3.2 Products of torus powers

For the remainder of this section, let $S = \mathrm{SU}(D, *)$ be k -anisotropic of degree $n \geq 3$, with notation as in Section 2. Fix $e \geq 1$ and let $R = S(k)^e$ denote the subgroup generated by all e -th powers. We allow the ambient variety to be S itself or a rational translate $X = x_*S$ in U , where $x_* \in U(k)$.

Choose a maximal k -torus $T \subset S$ and a finite Galois extension M/k splitting T and any fixed auxiliary data needed in an application. Write $p = \mathrm{char} k$. Over M , identify T with the determinant-one diagonal torus in SL_n . Choose a cocharacter

$$\lambda : \mathbb{G}_{m,M} \longrightarrow T_M, \quad \lambda(t) = \mathrm{diag}(t^{w_1}, \dots, t^{w_n}),$$

where the integers w_i are distinct, have sum zero, and are not all congruent modulo p . Such a choice exists for $n \geq 3$: start with a vector in $\mathbb{Z}^n$ of sum zero whose residues are not all equal, and add multiples of p of sum zero to make its entries distinct. In particular $d\lambda(1)$ is noncentral in $\mathfrak{sl}_n$, even if p divides n .

The restriction of scalars of λ , followed by the torus norm, defines a k -morphism

$$\eta : \text{Res}_{M/k} \mathbb{G}_m \longrightarrow T, \quad \eta = N_{M/k} \circ \text{Res}_{M/k}(\lambda). \quad (3.1)$$

Over M the coordinates of its source are indexed by $\sigma \in \text{Gal}(M/k)$, and its formula is $\eta((a_\sigma)_\sigma) = \prod_\sigma \lambda^\sigma(a_\sigma)$. On the scalar copy of $\mathbb{G}_m$, this map has cocharacter $\sum_\sigma \lambda^\sigma$. That cocharacter is defined over k and must vanish, since S is anisotropic. Consequently

$$\eta(ca) = \eta(a) \quad (c \in K^\times, \quad a \in (M \otimes_k K)^\times) \quad (3.2)$$

for every field extension K/k .

For $x \in X(k)$ and $k_1, \dots, k_r \in S(k)$, consider

$$\Phi(a_1, \dots, a_r) = x \prod_{j=1}^r k_j \eta(a_j)^e k_j^{-1}. \quad (3.3)$$

The parameter domain is the open torus $\mathcal{T}_r = (\text{Res}_{M/k} \mathbb{G}_m)^r$ in the affine space $E_r = (\text{Res}_{M/k} \mathbb{A}^1)^r$. For rational parameters,

$$\Phi(\mathcal{T}_r(k)) \subset xR. \quad (3.4)$$

Indeed each factor is the e -th power of $k_j \eta(a_j) k_j^{-1} \in S(k)$.

Lemma 3.2 (Two independent parameter lists). *Write $e = p^a e_0$ with $(e_0, p) = 1$. For a sufficiently long list of rational conjugators k_j , the product map formed with η^{e_0} is smooth at the identity parameter tuple. The conjugator tuples having this property form a nonempty Zariski open subset of the corresponding power of S .*

Choose two disjoint consecutive lists of factors with this property. In the full product map (3.3), allow any further factors before or after these lists. If $Y \subset X$ is closed of geometric codimension at least two, then $\Phi^{-1}(Y) \subset \mathcal{T}_r$ and its closure in E_r have geometric codimension at least two.

Proof. Work first over an algebraic closure. Put $H = d\lambda(1)$, a noncentral diagonal trace-zero matrix. The span of its conjugates contains every root vector E_{ij} . Indeed conjugation by $1 + tE_{ij}$ subtracts from H a nonzero multiple of tE_{ij} whenever its i -th and j -th diagonal entries differ, and permutation conjugations supply every pair of positions. The span is conjugation-invariant. Conjugating E_{ij} by $1 + tE_{ji}$ and subtracting root-vector terms therefore supplies $E_{ii} - E_{jj}$. These root vectors and diagonal differences span $\mathfrak{sl}_n$ in every characteristic. In particular this argument does not require $\mathfrak{sl}_n$ to be a simple Lie algebra.

The differential of η at 1 contains the line spanned by H : vary just the absolute coordinate indexed by the identity of $\text{Gal}(M/k)$. Finitely many conjugates of that line span $\text{Lie}(S)$, and multiplication by e_0 does not change the span. Thus suitable conjugators give a surjective differential for the product map at the identity. Surjectivity is an open condition on the conjugators. Since $S(k)$ is Zariski dense, this nonempty open contains a rational tuple. Both source and target are smooth, so the differential criterion gives the asserted smoothness.

Let Φ_0 denote the product with exponent e_0 in place of e . Each chosen list has a proper closed nonsmooth locus in its own parameter torus. The full map Φ_0 is smooth wherever either list is smooth, since multiplication by the remaining factors is a translation when their parameters are

fixed. The simultaneous nonsmooth locus has codimension at least two, because the two lists use disjoint coordinates. On its complement, smooth pullback preserves the lower bound of two for the codimension of Y . Hence $\Phi_0^{-1}(Y)$ has that bound everywhere.

Finally,

$$\Phi = \Phi_0 \circ [p^a],$$

where $[p^a]$ raises every parameter to its p^a -th power. After splitting the parameter torus, this is coordinatewise Frobenius. It is a finite universal homeomorphism, and therefore preserves the dimensions and codimensions of inverse images of closed subsets. This proves the assertion for $\Phi^{-1}(Y)$. Passing from an open subset of affine space to its closure does not change its dimension. $\square$

We call either of the two lists in Lemma 3.2 a *basic list*. Its actual map with exponent e is dominant, although it need not be smooth. We need one further consequence of dominance, uniform in the translating point.

Lemma 3.3 (Integral parameters on one list). *Fix a basic list and a proper closed subset $Y \subset X$. Outside finitely many places, depending on this list and these fixed models but not on x , the following holds: for every integral $x \in X(k_v)$, there are unit parameters on that list for which the reduction of its product with x avoids Y . All other factors may be given parameter 1.*

Proof. The surjective differential for the exponent- e_0 map remains surjective on reduction outside finitely many places. Its reduction is dominant, as is the exponent- e map obtained by composing with the surjective power map on the parameter torus. We also exclude places where the reduction of Y is not proper.

Choose affine coordinates on the underlying parameter space and equations for Y . Pulling these equations back through the product map, and clearing the fixed torus denominators, gives polynomials of bounded degree. Their degrees are independent of the translating point x . For every residue point $\bar{x} \in X(\kappa_v)$, dominance implies that at least one such polynomial is nonzero. Multiply it by the product of the norm polynomials defining the torus boundary. The result is a nonzero polynomial, of degree bounded by a fixed integer C , whose nonvanishing gives unit parameters and avoidance of Y .

A nonzero polynomial of degree at most C in N variables over a field with q elements has at most Cq^{N-1} zeros; this follows by induction on N . It therefore has a nonvanishing value when $q > C$. Exclude the finitely many places with $q_v \leq C$, and lift the resulting parameters to local units. $\square$

3.3 Boundary conditions and the approximation theorem

The boundary of $\mathcal{T}_r$ in E_r is a union of hyperplanes after extending scalars to M :

$$H_{j,\sigma} = \{a_{j,\sigma} = 0\}, \quad 1 \leq j \leq r, \quad \sigma \in \text{Gal}(M/k).$$

We call a reduction a *single-pole reduction* if it lies on exactly one of these hyperplanes. At a place unramified in M , Frobenius must fix the unique vanishing coordinate. Its action on the embeddings of the Galois extension M/k is regular; hence this place splits completely in M . Near this boundary hyperplane, away from all the others, (3.3) has the form

$$\Phi = g_L \lambda^\sigma(t)^e g_R, \tag{3.5}$$

where $t = a_{j,\sigma}$ and the two side factors are regular. At a single-pole reduction outside model exceptions, t has positive valuation and the side factors are integral and invertible. Thus the necessary local analysis can be done in split matrices, even though S is anisotropic over k .

In the theorem below, a *pole test* means finitely many algebraic nonvanishing conditions on $H_{j,\sigma} \setminus \bigcup_{(i,\tau) \neq (j,\sigma)} H_{i,\tau}$. The conditions may involve x and the conjugators as coefficients. They must be taken with all their Galois conjugates. The theorem states explicitly the nonemptiness requirement that applications must verify.

Proposition 3.4 (Approximation by power factors). *Let k be a global function field, let $S = \mathrm{SU}(D, *)$ be k -anisotropic of degree $n \geq 3$, and put $R = S(k)^e$ for $e \geq 1$. Let $X = x_* S$ with $x_* \in U(k)$, fix $\gamma \in S(k)$, and set*

$$\mathcal{C} = x_* \gamma R \subset X(k).$$

Let $Y \subset X$ be closed of geometric codimension at least two. Fix the torus, Galois splitting field, cocharacter and map η of (3.1). Let B be a finite set containing A , the exceptions for the fixed models, and a place v_0 at which S is isotropic.

At B , prescribe nonempty local open subsets of $X(k_v)$ whose product meets the closure of $\mathcal{C}$ given by Proposition 2.4. Let $(\mathcal{U}_v)_{v \notin B}$ be a family of subsets $\mathcal{U}_v \subset X(k_v)$ satisfying the following conditions:

- (i) *For every $v \notin B$, each integral point whose reduction avoids Y belongs to $\mathcal{U}_v$.*
- (ii) *For every $v \notin B$, the set $\mathcal{U}_v$ contains a nonempty local open subset.*
- (iii) *There are two disjoint consecutive basic lists and, for each number of appended factors, a fixed finite family of algebraic pole tests, independent of v . After the initial point and conjugators have been fixed, these tests imply membership in $\mathcal{U}_v$ at every single-pole reduction outside one finite set of model exceptions. The tests admit choices in the following order. The conjugator tuple of the two basic lists may be chosen in a nonempty Zariski open subset. For each such tuple, the initial point x may be chosen in a nonempty Zariski open subset of X . For every such x and every prescribed finite number of appended factors, their conjugators may be chosen in a nonempty Zariski open subset of the corresponding power of S . For each tuple so chosen, the tests define a nonempty open subset of every absolute boundary hyperplane away from the other hyperplanes. Setting appended parameters to 1 preserves all previously available pole tests.*

Then there is $x' \in \mathcal{C}$ satisfying the prescribed open conditions at B and belonging to $\mathcal{U}_v$ for every $v \notin B$.

Finitely many additional nonempty Zariski open conditions may be imposed at the three stages in (iii), as long as they remain compatible with the stipulated choices.

Proof. We shall first choose the group-valued product and then apply Lemma 3.1 to its affine parameters. There are two reasons to enlarge the initial finite set of places: the initial rational point may have denominators, and the eventual projection may introduce further model exceptions. The former will be handled by extra factors; the latter by the fixed basic lists.

Choosing the basic lists and the initial point. Choose the conjugators of the two basic lists in the open subset supplied by (iii). Nonempty Zariski opens can be intersected here, since the product of copies of S is geometrically irreducible. Put into B the model exceptions for these fixed lists and the finite set from Lemma 3.3. At each newly added place, choose an opening contained in $\mathcal{U}_v$ using (ii). These places lie outside A , so the additional conditions are compatible with the closure of $\mathcal{C}$.

Choose $x \in \mathcal{C}$ meeting these local conditions and the nonempty Zariski open condition on x in (iii). This follows from Proposition 2.4: use an omitted isotropic place different from the finitely many places at which approximation is required. The Zariski condition can be included by shrinking

the opening at v_0 , because a nonempty analytic open subset of a smooth geometrically irreducible variety is Zariski dense.

Let B_1 be the finite set of places outside the enlarged B at which x is not integral. At each $v \in B_1$ choose a nonempty local open subset of $\mathcal{U}_v$.

Moving the denominator places. For $v \in B_1$, the group S is isotropic over k_v . The image

$$\eta((M \otimes_k k_v)^\times)^e \subset S(k_v)$$

is noncentral. Indeed its geometric image contains the noncentral cocharacter image, and local points of its source torus are Zariski dense. Local Kneser–Tits and Tits simplicity, as recalled in Section 2, imply that its normal closure is all of $S(k_v)$. To be explicit, simplicity modulo the center first gives surjectivity onto that quotient; the remaining quotient is abelian and vanishes because $S(k_v)$ is perfect. Every element of this normal closure is a finite product of conjugates of the displayed values; inverses are obtained by inverting the parameters.

Consequently finitely many factors of the form in (3.3) can move x_v into the chosen opening at every $v \in B_1$. Use the largest number needed at these finitely many places, inserting identity factors where necessary. Approximate the local conjugators by rational conjugators, using strong approximation for S away from v_0 [25]; the hypothesis is satisfied because S is simply connected and $S(k_{v_0})$ is noncompact. Choose the rational conjugators integral outside $B \cup B_1$. The nonempty Zariski open condition on the appended conjugators in (iii) can be imposed at the same time: first shrink their local openings into that Zariski open and then apply strong approximation.

With these conjugators fixed, continuity gives nonempty open conditions on the parameter tuple at each place of $B \cup B_1$. At B take all parameters close to 1. At B_1 take the basic parameters close to 1 and the appended parameters close to the values effecting the local moves. Let r be the resulting total number of factors and retain the notation $\Phi : \mathcal{T}_r \rightarrow X$.

The subset to be avoided. In E_r form the union Z of the following closed subsets: the closure of $\Phi^{-1}(Y)$; every intersection of two distinct absolute boundary hyperplanes; and, on each boundary hyperplane, the closure of the locus where its pole tests fail away from the other hyperplanes. Include all Galois conjugates, so this union is defined over k . Lemma 3.2 gives codimension at least two for the first part. The pairwise intersections have codimension two, and the remaining parts do also have codimension at least two, since each test succeeds on a nonempty open subset of its hyperplane.

Outside finitely many model exceptions, integral parameters whose reduction avoids Z have one of two forms. If every absolute coordinate is a unit, Φ is integral and its reduction avoids Y , so (i) applies. Otherwise exactly one coordinate has positive valuation, and its pole tests hold, so (iii) applies. In either case Φ belongs to $\mathcal{U}_v$.

Choosing the line and treating the remaining exceptions. Choose $d = (d_1, \dots, d_r) \in E_r(k)$ whose point at infinity misses the projective closure of Z , and with each d_j sufficiently close to 1 at v_0 that $\Phi(d)$ satisfies the prescribed opening there. These requirements are compatible: the directions excluded at infinity form a proper algebraic subset, whereas the condition at v_0 is a nonempty analytic open. Weak approximation on affine space supplies such a rational direction. Choose a rational linear section of projection along d .

The equations defining Z , the projection, and its fiber bound now give a fixed finite set of additional exceptional places as in Lemma 3.1. Let Σ contain those places, the exceptions in (iii), all exceptions to the preceding reduction argument, and $B \cup B_1$. At a place of $\Sigma \setminus (B \cup B_1)$, the initial point and every conjugator are integral. Lemma 3.3 therefore gives unit parameters on one basic list whose product with x reduces outside Y ; set all other parameters to 1. A sufficiently small local opening around this tuple maps into $\mathcal{U}_v$ by (i). This argument is valid even when the projection or

the pole equations have unsuitable reduction at that place. The small residue fields for which the basic-list argument could fail were already included in B before x was chosen.

Thus there are parameter openings at every place of Σ , with the required local consequence, and no further choices of rational conjugators or initial points are needed. Apply Lemma 3.1, omitting the opening at v_0 . It gives $y \in E_r(k)$ and parameters $a = y + ld$ with $|l|_{v_0}$ arbitrarily large, meeting all prescribed openings away from v_0 and reducing outside Z at every place outside Σ .

Recovering the opening at v_0 and the exact coset. For each component of the parameter tuple,

$$\eta(y_j + ld_j) = \eta(d_j + l^{-1}y_j) \longrightarrow \eta(d_j) \quad \text{as } |l|_{v_0} \longrightarrow \infty,$$

by (3.2). Hence $\Phi(a)$ eventually satisfies the opening at v_0 . For such l all components a_j are nonzero at v_0 , so they belong to $M^\times$ and $a \in \mathcal{T}_r(k)$. At the other places of Σ the chosen openings give the required conclusion, and outside Σ avoidance of Z gives it. Finally (3.4) gives

$$x' = \Phi(a) \in xR = \mathcal{C}.$$

The conditions selected at the places added to B were contained in their original allowed sets $\mathcal{U}_v$, so the conclusion holds for the original finite set as well. $\square$

Remark 3.5. The proposition also applies to local properties that are only formulated on a nonempty Zariski open $X^0 \subset X$. Include $X(k_v) \setminus X^0(k_v)$ in each allowed set, and refine the opening at v_0 to lie in $X^0(k_{v_0})$. Then the global point belongs to $X^0(k)$ and has the asserted properties there. This will be used for regular semisimple elements and for a later open set on which certain auxiliary algebras are defined.

In an application with prescribed determinant but no prescribed abstract coset, first use weak approximation on the determinant slice to meet the finitely many local conditions and then choose the coset containing that point. Conversely, when a coset is prescribed, the proof never replaces it by its closure: every change of the initial rational point is multiplication by explicit e -th powers.

4 Simultaneous norm equations

Prescribing a determinant inside a unitary torus is a norm equation. Later we shall prescribe two norms of the same element: one to a degree- m field and one to a quadratic field. This section proves that the resulting equations satisfy the Hasse principle and weak approximation when the two fields have the indicated symmetric Galois group. The finite-lattice argument is that of [23, Lemmas 2.7–2.9]; we reproduce it and supply the global-function-field duality input.

For a finite separable extension E/K , write $\text{Res}_{E/K}^1 \mathbb{G}_m$ for the kernel of its norm map.

Proposition 4.1 (Simultaneous norms). *Let k be a global function field, let J/k be a separable quadratic extension, and let F/k be separable of degree $m \geq 3$. Suppose that the Galois group of the joint normal closure of F and J is $S_m \times C_2$ under its natural action on the m embeddings of F and its restriction to J . Put $E = FJ$. Then*

$$T = \ker \left[\text{Res}_{F/k}(\text{Res}_{E/F}^1 \mathbb{G}_m) \xrightarrow{N_{E/J}} \text{Res}_{J/k}^1 \mathbb{G}_m \right] \quad (4.1)$$

is a torus. Every torsor under T that has a point over every completion of k has a k -point, and its k -points are dense in the product of its points over any finite set of completions.

In particular, let $h \in F^\times$ and $d \in J^\times$ satisfy $N_{F/k}(h) = N_{J/k}(d)$. If the equations

$$N_{E/F}(a) = h, \quad N_{E/J}(a) = d \quad (4.2)$$

have a solution in $(E \otimes_k k_v)^\times$ for every place v , they have a solution in $E^\times$ approximating any prescribed local solutions at finitely many places.

We first compute the character lattice and then express its arithmetic obstruction in terms of a finite group. A k -lattice is a finitely generated free abelian group with a continuous action of the absolute Galois group of k ; its action factors through a finite quotient. Let P be the joint normal closure in the proposition and let c generate the second factor of $\Gamma = \text{Gal}(P/k) = S_m \times C_2$. The character map dual to the norm in Equation (4.1) is the diagonal inclusion

$$\mathbb{Z}_{\text{sign}} \longrightarrow \mathbb{Z}^m \otimes \text{sign}_{C_2}, \quad 1 \longmapsto \mathbf{1} = (1, \dots, 1).$$

Here S_m permutes the m coordinates and c acts by -1 . The inclusion is primitive, so its cokernel is a lattice. Consequently the norm is surjective with connected kernel T , and

$$X^*(T) = M = (\mathbb{Z}^m / \mathbb{Z}\mathbf{1}) \otimes \text{sign}_{C_2}. \quad (4.3)$$

For any k -lattice M , set

$$\text{Sha}^2(k, M) = \ker \left[H^2(k, M) \longrightarrow \prod_v H^2(k_v, M) \right],$$

and let $\text{Sha}_\omega^2(k, M)$ denote the subgroup of classes whose localizations vanish at all but finitely many places. All Galois cohomology here uses continuous cochains and discrete coefficients.

Lemma 4.2 (Detection on cyclic subgroups). *Suppose that a finite Galois extension P/k splits the k -lattice M , and put $\Gamma = \text{Gal}(P/k)$. Inflation identifies $\text{Sha}_\omega^2(k, M)$ with*

$$\ker \left[H^2(\Gamma, M) \longrightarrow \prod_{C \subseteq \Gamma \text{ cyclic}} H^2(C, M) \right]. \quad (4.4)$$

If this group vanishes and T is the torus with character lattice M , every everywhere locally soluble T -torsor has a rational point and satisfies weak approximation.

Proof. The action on M over P is trivial. A continuous homomorphism from a profinite group to a torsion-free discrete group has finite, hence trivial, image. Thus $H^1(P, M) = 0$; and, since rational coefficients have no positive-degree cohomology, the sequence $0 \rightarrow M \rightarrow M \otimes \mathbb{Q} \rightarrow M \otimes (\mathbb{Q}/\mathbb{Z}) \rightarrow 0$ gives

$$H^2(P, M) = H^1(P, M \otimes (\mathbb{Q}/\mathbb{Z})).$$

An element on the right is a continuous character with finite image. Such a character cannot be locally zero at almost every place unless it is zero, by Chebotarev applied to the finite extension through which it factors. Therefore a class in $\text{Sha}_\omega^2(k, M)$ restricts to zero over P . Inflation–restriction, together with $H^1(P, M) = 0$, then gives a unique preimage in $H^2(\Gamma, M)$.

At a place v unramified in P , the decomposition group is cyclic. The local inflation map from its degree-two cohomology into $H^2(k_v, M)$ is injective, again because degree-one cohomology of the lattice over the splitting field vanishes. Chebotarev supplies infinitely many places with each possible Frobenius conjugacy class. Consequently, a class locally zero almost everywhere restricts to zero on every cyclic subgroup of Γ . Conversely, vanishing on all cyclic subgroups gives local vanishing at every unramified place. This proves Equation (4.4).

For the arithmetic conclusion, global torus duality over function fields gives a perfect pairing of finite groups

$$\text{Sha}^1(k, T) \times \text{Sha}^2(k, M) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

[6, Theorem 5.2]. Here $\text{Sha}^1(k, T)$ is the group of everywhere locally trivial T -torsors. The same theory gives weak approximation for T when $\text{Sha}_\omega^2(k, M) = 0$; explicitly, apply [7, Corollary 4.5] to the reductive group T . In the notation of these references the associated complex is $[0 \rightarrow T]$, and its dual is $[M \rightarrow 0]$, with M in degree -1 , so their Sha_ω^1 of the dual is precisely $\text{Sha}_\omega^2(k, M)$. Since $\text{Sha}^2(k, M) \subseteq \text{Sha}_\omega^2(k, M)$, the vanishing in the statement gives the Hasse principle as well. Translation by a rational point identifies any soluble torsor with T and transfers weak approximation to it. These duality results include the characteristic-primary part; no restriction on the characteristic of k is needed. $\square$

The arithmetic problem has now reduced to showing that Equation (4.4) vanishes for the specific lattice in Equation (4.3). We use the interpretation of $H^2(\Gamma, M)$ as extensions of Γ by the abelian group M inducing the specified action. Restriction to a subgroup is zero exactly when the extension splits over that subgroup. A central element acting by -1 makes these extensions particularly concrete.

Lemma 4.3 (A central sign element). *Let Γ be a finite group acting on a lattice M , and suppose that $c \in Z(\Gamma)$ has order two and acts as -1 on M . Every extension*

$$1 \longrightarrow M \longrightarrow \mathcal{E} \longrightarrow \Gamma \longrightarrow 1$$

defines a class $\beta \in H^1(\Gamma/\langle c \rangle, M/2M)$ that is zero if and only if the extension splits. If the extension splits over an involution $r \in \Gamma$, then the value at the image of r of every cocycle representing β lifts to an element $d_r \in M$ satisfying $(1+r)d_r = 0$.

Proof. Write M additively, with its translations placed to the left of lifts in $\mathcal{E}$. A lift z of c is an involution: z^2 belongs to M and commutes with z , hence belongs to $M^c = 0$. Choose lifts s_g of $g \in \Gamma$, with $s_1 = 1$ and $s_c = z$, and define $f(g, h), d_g \in M$ by

$$s_g s_h = f(g, h) s_{gh}, \quad z s_g z^{-1} = d_g s_g.$$

Conjugating the first equality by z in two ways gives

$$d_{gh} = d_g + g d_h + 2f(g, h). \quad (4.5)$$

Thus $g \mapsto \bar{d}_g$ is a cocycle with values in $M/2M$. Its value at c is zero, and c acts trivially on $M/2M$, so it descends to $\Gamma/\langle c \rangle$. Replacing s_g by $a_g s_g$ changes d_g to $d_g - 2a_g$; replacing z by az changes it to $d_g + (1-g)a$. The cohomology class β is therefore independent of these choices.

If $\beta = 0$, change z so that every d_g is even, and then replace s_g by $(d_g/2)s_g$. These lifts all commute with z . The centralizer of z in $\mathcal{E}$ maps onto Γ and has kernel $M^c = 0$; it is therefore a complement to M and gives a splitting. Conversely, a splitting supplies mutually compatible lifts with $d_g = 0$.

Finally, suppose that r has an involutive lift s_r . Conjugating $s_r^2 = 1$ by z shows that $(1+r)d_r = 0$. Replacing the cocycle by a cohomologous one changes this value modulo $2M$ by $(1-r)a$ for some $a \in M$. The corresponding lift $d_r + (1-r)a$ remains annihilated by $1+r$, since $r^2 = 1$. This proves the last assertion. $\square$

Lemma 4.4 (The symmetric permutation lattice). *Let $m \geq 3$, let $\Gamma = S_m \times \langle c \rangle$ with $c^2 = 1$, and let*

$$M = (\mathbb{Z}^m / \mathbb{Z}\mathbf{1}) \otimes \text{sign}_{C_2}, \quad \mathbf{1} = (1, \dots, 1).$$

Thus S_m permutes coordinates and c acts by -1 . An extension of Γ by M that splits over every cyclic subgroup splits over Γ .

Proof. Fix an extension with the stated cyclic splittings. By Lemma 4.3, it suffices to show that its class

$$\beta \in H^1(S_m, Q_m), \quad Q_m = \mathbb{F}_2^m / \mathbb{F}_2 \mathbf{1} = M/2M,$$

is zero. The exact sequence of S_m -modules

$$0 \longrightarrow \mathbb{F}_2 \mathbf{1} \longrightarrow V_m = \mathbb{F}_2^m \longrightarrow Q_m \longrightarrow 0$$

has connecting homomorphism

$$\partial : H^1(S_m, Q_m) \longrightarrow H^2(S_m, \mathbb{F}_2).$$

Shapiro's lemma identifies $H^i(S_m, V_m)$ with $H^i(S_{m-1}, \mathbb{F}_2)$, where S_{m-1} fixes one coordinate. Under this identification, the map induced by the diagonal inclusion $\mathbb{F}_2 \rightarrow V_m$ is restriction to S_{m-1} . In degree one it is an isomorphism: for both S_m and S_{m-1} , including $S_{m-1} = S_2$, homomorphisms to $\mathbb{F}_2$ are generated by the permutation sign. The cohomology exact sequence therefore shows that ∂ is injective and that its image is the kernel of restriction to $H^2(S_{m-1}, \mathbb{F}_2)$.

Let

$$1 \longrightarrow \langle \zeta \rangle \longrightarrow \hat{S}_m \longrightarrow S_m \longrightarrow 1, \quad \zeta^2 = 1,$$

be the central extension represented by $\partial\beta$. It splits over each point stabilizer, because these are conjugate. Every transposition therefore has involutive lifts; multiplying such a lift by ζ preserves its square. If $m \geq 5$, any two disjoint transpositions fix a common point. Their lifts commute, since their commutator is unaffected by changing a lift by ζ and is trivial in a splitting over that stabilizer. For $m = 3$ there is no disjoint pair of Coxeter generators.

For these degrees choose involutive lifts t_i of the adjacent transpositions $(i, i+1)$, $1 \leq i < m$. Write

$$(t_i t_{i+1})^3 = \zeta^{\epsilon_i}, \quad \epsilon_i \in \mathbb{F}_2.$$

Replace t_i by $\zeta^{a_i} t_i$, where $a_1 = 0$ and $a_{i+1} = a_i + \epsilon_i$. This makes all adjacent-product cubes equal to 1 without changing squares or commutation of distant generators. The Coxeter presentation

$$s_i^2 = 1, \quad (s_i s_{i+1})^3 = 1, \quad s_i s_j = s_j s_i \quad (|i - j| > 1)$$

now gives a section $S_m \rightarrow \hat{S}_m$. Hence $\partial\beta = 0$, and injectivity gives $\beta = 0$.

The degree $m = 4$ requires the integral information retained in Lemma 4.3. The transpositions $\tau = (12)$ and $\nu = (34)$ have no common fixed point, so their lifts might fail to commute. Choose the cocycle $g \mapsto \bar{d}_g$ obtained from the original extension by the lattice, and choose representatives $b(g) \in \mathbb{F}_2^4$ of its values in Q_4 , with $b(1) = 0$. The factor set for $\partial\beta$ is given by

$$\epsilon(g, h) \mathbf{1} = b(g) + gb(h) - b(gh).$$

As $\tau\nu = \nu\tau$, noncommuting lifts of τ and ν would force

$$(1 + \nu)b(\tau) + (1 + \tau)b(\nu) = \mathbf{1}. \tag{4.6}$$

The original lattice extension splits over $\langle \tau \rangle$. Lemma 4.3 consequently gives a lift $d \in M$ of $\bar{d}_\tau$ with $(1 + \tau)d = 0$. Choose a representative $\tilde{d} \in \mathbb{Z}^4$. For some $a \in \mathbb{Z}$,

$$\tilde{d} + \tau\tilde{d} = a\mathbf{1}, \quad 2\tilde{d}_3 = a = 2\tilde{d}_4.$$

It follows that $\tilde{d}_3 = \tilde{d}_4$. Thus every representative of $\bar{d}_\tau$ in $\mathbb{F}_2^4$, in particular $b(\tau)$, has equal third and fourth coordinates. The third and fourth coordinates of $(1 + \nu)b(\tau)$ are zero. Those of $(1 + \tau)b(\nu)$

are also zero because τ fixes these coordinates. This contradicts Equation (4.6). The lifts of τ and ν commute. These form the only disjoint pair among the adjacent transpositions in S_4 , so the same adjustment of adjacent products proves that $\hat{S}_4$ splits.

We have proved $\beta = 0$ in every degree $m \geq 3$. Lemma 4.3 splits the original extension. $\square$

Proof of Proposition 4.1. We computed the character lattice of T in Equation (4.3). Lemma 4.4 shows that its degree-two cohomology is detected on cyclic subgroups. Lemma 4.2 now gives the Hasse principle and weak approximation for T -torsors.

Finally, the solution variety of Equation (4.2) is a T -torsor whenever it is nonempty geometrically: the quotient of any two solutions has both norms equal to 1, which is exactly the defining condition for T . The compatibility of h and d guarantees geometric nonemptiness. Indeed, over a separable closure write the $2m$ coordinates of a as a_i^+, a_i^- , the coordinates of h as h_i , and those of d as d^+, d^- . The equations become

$$a_i^+ a_i^- = h_i \quad (1 \leq i \leq m), \quad \prod_i a_i^+ = d^+, \quad \prod_i a_i^- = d^-.$$

Choose the a_i^+ with product d^+ and set $a_i^- = h_i/a_i^+$. The final equation follows from $\prod_i h_i = d^+ d^-$. The asserted local-to-global and approximation statement thus follows from the result for torsors. $\square$

5 Finite quotients in odd division degree

Assume throughout this section that D is a division algebra of odd reduced degree $n \geq 3$ over the quadratic extension L/k . Put $S = \mathrm{SU}(D, *)$ and $U = \mathrm{U}(D, *)$, as in Section 2. The one-dimensional Hermitian space over D is anisotropic, so S is k -anisotropic. The reduced norm gives a surjective homomorphism

$$\det : U(k) \longrightarrow L^1, \quad L^1 = \{a \in L^\times : a\bar{a} = 1\},$$

with kernel $S(k)$. Recall that, for a fixed integer $e \geq 1$,

$$R = S(k)^e, \quad P_0 = \overline{\delta_A(R)}, \quad V_0 = \delta_A^{-1}(P_0), \quad V = V_0/R.$$

Here $S(k)^e$ denotes the subgroup generated by the e th powers. The group V is finite by Proposition 2.3. Our purpose is to turn a nontrivial quotient of V into a finite quotient defined on the full unitary group.

Proposition 5.1 (Finite quotient dichotomy). *If $V \neq 1$, at least one of the following homomorphisms exists:*

- (i) a homomorphism $\chi : U(k) \rightarrow B$ to a finite abelian group such that $\chi(S(k)) \neq 1$;
- (ii) a homomorphism $\phi : U(k) \rightarrow \mathcal{F}$ to a finite nonabelian simple group such that

$$\phi(V_0) = \mathcal{F}, \quad R \subseteq \ker \phi.$$

The argument follows the quotient construction of [23, Section 3, especially Propositions 3.1 and 3.2]. We give the details because two parts require care in positive characteristic: commuting elements must be constructed in separable field tori, and the treatment of local principal units must not assume that the subgroup of n th powers is open. We first prove the commuting approximation statement that controls the conjugation action on V .

5.1 Commuting elements with prescribed determinants

At a place $v \in A$, the extension L/k splits and

$$U(k_v) = \mathcal{D}_v^\times, \quad S(k_v) = \mathrm{SL}_1(\mathcal{D}_v),$$

where $\mathcal{D}_v$ is a central division algebra of degree n over k_v . A *separable field torus* in $U(k_v)$ means the multiplicative group of a maximal separable subfield of $\mathcal{D}_v$. This qualification matters when the characteristic divides n .

Proposition 5.2 (Commuting approximation). *The following assertions hold.*

- (i) *Let $a, b \in L^1$. For each $v \in A$, let $x_v, t_v \in U(k_v)$ belong to one common separable field torus and satisfy $\det x_v = a_v$ and $\det t_v = b_v$. There exist commuting elements $x, t \in U(k)$ with*

$$\det x = a, \quad \det t = b,$$

whose components at A approximate the prescribed pairs arbitrarily closely.

- (ii) *Given $\gamma \in V_0$ and $h \in U(k)$, there exist commuting elements $x, t \in U(k)$ such that*

$$x \in \gamma R, \quad \det t = \det h,$$

and t_A is arbitrarily close to h_A .

Proof. In the second assertion set $a = 1$ and $b = \det h$. In both assertions we shall construct a regular semisimple first element x whose centralizer has local elements of determinant b at every place. The norm-torus theorem of Section 4 will then give a global second element, together with its approximations at A .

We begin with the local conditions at A . In the first assertion, perturb x_v within the determinant- a_v fiber of its given field torus until it is regular semisimple. The field is separable, so its norm is a smooth map. After extension to an algebraic closure, a norm fiber is a translate of

$$\{(z_1, \dots, z_n) \in \mathbb{G}_m^n : z_1 \cdots z_n = 1\}.$$

No equality $z_i = z_j$, with $i \neq j$, holds identically on this fiber. Its regular elements therefore form a dense open subset, also dense in the local topology. Near a regular semisimple element the conjugation map from the group times its torus is a submersion. Consequently the centralizers of nearby elements are obtained by small conjugations. The same conjugations transport t_v , preserving its determinant and the required approximation.

For the second assertion, first perturb h_v to a regular semisimple element in its determinant fiber. Such a perturbation is possible because the fiber is a smooth translate of S and its regular semisimple open is nonempty. In the resulting separable field torus choose a regular norm-one element close to 1 as the first element. The preceding conjugation argument again gives a local opening for x . These openings near 1 are compatible with the prescribed coset γR : its closure at A is P_0 , and P_0 is open.

Choose a finite set B containing A , an isotropic place to be omitted in strong approximation, all exceptions for integral models, all places ramified in L/k , and the places where a or b is not a unit. At the additional places we need local tori whose determinant images contain both a_v and b_v . At a nonsplit place of L/k , the algebra splits and a diagonal Hermitian torus has determinant image equal to the local norm-one group. At a split place outside A , write a component of the algebra as $M_l(\Delta)$. If Δ is nontrivial, then $l \geq 2$. Use an unramified maximal subfield of Δ in one diagonal

block and a totally ramified separable maximal subfield in a second block. The first norm supplies all units, and the second norm supplies an element of valuation one; their combined image is all of $k_v^\times$. Such separable totally ramified fields exist in every characteristic, for instance by separable Eisenstein polynomials, and embed in Δ by the local invariant criterion. Complete the remaining blocks to a maximal étale subalgebra. If $\Delta = k_v$, use the ordinary diagonal torus. In every case choose the first element regular with determinant a_v and take a sufficiently small opening around it. These constructions also give nonempty local openings at any further exceptional places introduced by the approximation theorem.

We add finitely many auxiliary places, split in L and in D , at which a and b are units. Prescribe unramified regular centralizers having every possible permutation cycle type in S_n , one type at each place. Unramified norms supply both unit determinants. We may use places with sufficiently large residue fields that regular elements with those determinants exist. These conditions will force the Galois group required by Proposition 4.1.

We now verify the hypotheses of Proposition 3.4 on the determinant- a slice X . This slice is a rational translate of S , since the determinant on $U(k)$ is onto. For the first assertion, weak approximation supplies a rational point meeting the finitely many openings; we then apply the proposition to its coset modulo R . For the second assertion, we apply it to γR . The closure statement of Proposition 2.4 permits the openings near 1 at A and places no restriction on the other local openings.

Define $Y \subset X$ over k to be the closure of the geometric locus whose reduced characteristic polynomial has no root of multiplicity one. The root-multiplicity condition is preserved by the descent defining X . We shall require integral reductions outside B to avoid the spread-out model of Y . It has geometric codimension at least two in X . Indeed, on a split matrix component there are at most $\lfloor n/2 \rfloor$ distinct eigenvalues on this locus. The fixed determinant reduces the eigenvalue parameter dimension by one, even if the resulting torus equation is inseparable. A conjugacy class with fixed eigenvalues has dimension at most $n^2 - n$. There are finitely many Jordan types, so

$$\dim Y \leq n^2 - n + \lfloor n/2 \rfloor - 1 \leq (n^2 - 1) - 2.$$

This codimension bound and the description by root multiplicities persist outside finitely many model exceptions.

Suppose that an integral regular semisimple element reduces outside Y , and write $K = k_v$ and $L_K = L \otimes_k K$. A simple absolute residue root belongs to an irreducible residue factor of multiplicity one. Hensel lifting supplies an unramified field factor of the étale centralizer over L_K . Take its full orbit under the involution and denote the resulting algebra by E_j . All its factors are unramified over K , since L_K/K is unramified or split. If the field factor is stable under the involution, its fixed field F_j is unramified over K . If two factors are exchanged, their fixed algebra is the diagonal field under that exchange, and is again unramified. In both cases quadratic descent identifies the algebra with involution as

$$E_j = F_j \otimes_K L_K,$$

with the identity on F_j and quadratic conjugation on L_K .

Its unitary torus contributes local determinant norms as follows. After an unramified extension splitting these algebras, it has one free multiplicative coordinate z_τ for each embedding $\tau : F_j \hookrightarrow \overline{K}$. The determinant is $\prod_\tau z_\tau^{\epsilon_\tau}$ with $\epsilon_\tau \in \{1, -1\}$: each field embedding occurs once in the reduced characteristic representation, and the sign records which member of its involution pair lies in the chosen L_K -component. Passing to the other member inverts the coordinate. The dual map from the rank-one character lattice of L_K^1 therefore has primitive image. Thus the determinant is a surjection

of unramified tori with a torus as kernel, in every characteristic. Lang's theorem makes the map onto on residue-field points, and smooth lifting makes it onto integral points. It supplies the unit b_v .

For a single pole, the approximation theorem places us at a place split in the fixed Galois field and gives

$$x = g_L \operatorname{diag}(t^{ed_1}, \dots, t^{ed_n}) g_R, \quad d_1 < \dots < d_n,$$

with $\operatorname{ord}_v(t) > 0$ and integral invertible side factors. Require the principal minor on the first j indices of $g_R g_L$ to be a unit for each $1 \leq j \leq n$. These are nonempty algebraic open conditions on each pole hyperplane. Indeed, the cyclic product $g_R g_L$ contains one entire undamaged basic parameter list. Keeping all other parameters fixed, that list varies $g_R g_L$ densely in the fixed determinant slice. The principal-minor conditions are nonempty on this slice, as witnessed by a diagonal matrix of the given determinant. Consequently, once the two basic lists have been chosen, every initial x and every appended conjugator tuple admit these pole tests; no further generic restriction on those choices is needed. Identity values of appended parameters preserve the original tests. If c_j is the j th elementary characteristic coefficient, its principal-minor expansion gives

$$\operatorname{ord}_v(c_j) = e(d_1 + \dots + d_j) \operatorname{ord}_v(t). \quad (5.1)$$

The selected minor is the unique term of least valuation. Thus all Newton segments have length one and distinct slopes. The characteristic polynomial splits into distinct linear factors over k_v , so its centralizer supplies every determinant.

For the allowed sets in Proposition 3.4, include all elements outside the regular semisimple open. On that open use the determinant-norm condition just proved. The integral and pole conditions hold, and the local tori constructed above provide the required nonempty openings. Impose regularity at one place of B . The proposition now gives a global regular semisimple x , with the prescribed determinant, local conditions, and, in the second assertion, its exact coset modulo R .

Since D is division, $E = L(x)$ is a maximal separable field in D . It is stable under $*$ because $x^* = x^{-1}$. If $F = E^{*=1}$ is its fixed field, then $E = LF$ and $[F : k] = n$. Let Γ be the Galois group of the joint normal closure of F and L ; its natural action embeds it in $S_n \times C_2$. The prescribed Frobenius classes at places split in L imply that the kernel of $\Gamma \rightarrow C_2$ meets every conjugacy class of S_n . Such a subgroup is all of S_n . For, if it were proper, the transitive action on its cosets would have a derangement: the average number of fixed cosets is one, while the identity fixes more than one. A derangement cannot be conjugate to an element of the subgroup. Since Γ also surjects onto C_2 , we obtain $\Gamma = S_n \times C_2$.

The unitary centralizer of x is

$$T_E = \operatorname{Res}_{F/k}(\operatorname{Res}_{E/F}^1 \mathbb{G}_m), \quad \det|_{T_E} = N_{E/L}.$$

The fiber $N_{E/L}^{-1}(b)$ has points at every completion, by the local openings, integral conditions, and pole conditions. It is a torsor under the norm kernel of Proposition 4.1, whose Galois hypothesis we have just checked. That theorem gives a rational point t with the desired approximations at A . It lies in the centralizer of x and has determinant b . $\square$

5.2 The action of the full unitary group

The group-theoretic passage from a class-preserving action to a quotient of the ambient group is adapted from [30, Corollary 2.3, Theorem 2.4, and the proof of Theorem 2.1]. Here Proposition 5.2 supplies the unitary input. Conjugation by $U(k)$ preserves R , which is characteristic in $S(k)$, and consequently preserves P_0 and V_0 . We show that its action on V preserves every conjugacy class.

Take $\gamma \in V_0$ and $h \in U(k)$. Proposition 5.2(ii) gives $x \in \gamma R$ commuting with t , where $\det t = \det h$. Choose the approximation at A sufficiently close that $th^{-1} \in V_0$. Conjugation by t fixes γR , while conjugation by th^{-1} is inner on V . Hence conjugation by h sends γR to a conjugate of itself.

Suppose now that $V \neq 1$, and choose a simple quotient C of V . Every normal subgroup of V is a union of conjugacy classes, so the kernel of $V \rightarrow C$ is preserved by $U(k)$. The induced automorphisms of C are class-preserving. If C is nonabelian, the theorem of Feit–Seitz [9, Theorem C] makes each of these automorphisms inner. As C has trivial center, the action gives a homomorphism

$$U(k) \longrightarrow \text{Inn}(C) = C.$$

On V_0 it is the original quotient map and it kills R . This proves the second alternative of Proposition 5.1.

It remains to handle the case in which C is cyclic of prime order. Let R' be its kernel upstairs. Then

$$R \subseteq R' \subseteq V_0, \quad C = V_0/R' \subseteq Z(U(k)/R'). \quad (5.2)$$

The rest of the section uses this nonzero central quotient to construct a finite character of $U(k)$ that is nonzero on $S(k)$. We shall use the compact local groups at A to construct and split a circle extension.

5.3 Characters of a local norm-one group

The required local calculation is the division-algebra commutator calculation underlying [36]; we include its filtration proof, using the standard cyclic description of local division algebras [35]. It explains why divisibility of n by the characteristic causes no additional abelian quotient.

Lemma 5.3. *Let K be a nonarchimedean local field of positive characteristic, with residue field $\mathbb{F}_q$, and let $\mathcal{D}$ be a central division algebra of odd degree $n \geq 3$ over K . Set $S_K = \text{SL}_1(\mathcal{D})$ and*

$$T_K = \ker(N_{\mathbb{F}_{q^n}/\mathbb{F}_q} : \mathbb{F}_{q^n}^\times \rightarrow \mathbb{F}_q^\times).$$

Then reduction induces

$$S_K/[\overline{S_K}, \overline{S_K}] \simeq T_K, \quad [\overline{\mathcal{D}^\times}, \overline{\mathcal{D}^\times}] = S_K.$$

There is a generator σ of $\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q)$ such that conjugation by an element of reduced-norm valuation r acts on T_K as σ^r .

Proof. Choose an unramified maximal subfield E_K/K and a prime element Π of $\mathcal{D}$ inducing a generator σ on E_K . We normalize so that $\text{ord}_K(\text{Nrd } \Pi) = 1$; Π^n is a central uniformizer. Let $\mathfrak{P} = (\Pi)$ be the maximal ideal of the maximal order of $\mathcal{D}$, and let $\mathfrak{p}$ be the maximal ideal of K . Every element of S_K is integral. Its reduction has norm one, and the norm-one Teichmüller representatives in E_K give a multiplicative section of the reduction map onto T_K .

Put $S_r = S_K \cap (1 + \mathfrak{P}^r)$ for $r \geq 1$. The norm filtration is

$$\text{Nrd}(1 + \mathfrak{P}^r) = 1 + \mathfrak{p}^{\lceil r/n \rceil}. \quad (5.3)$$

For the inclusion, the roots of the reduced characteristic polynomial of an element of $\mathfrak{P}^r$ have valuation at least r/n ; their elementary symmetric coefficients have the corresponding valuation bounds. For surjectivity, use norms of principal units in the unramified field E_K : its norm maps $1 + \mathfrak{p}_{E_K}^{\lceil r/n \rceil}$ onto the right-hand side. The latter assertion follows successively from surjectivity of the finite-field trace and completeness.

Leading-coefficient coordinates now give

$$S_r/S_{r+1} \simeq \begin{cases} (\mathbb{F}_{q^n}, +), & n \nmid r, \\ \ker \operatorname{Tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}, & n \mid r. \end{cases} \quad (5.4)$$

If $n \nmid r$, the norm images at levels r and $r+1$ agree in (5.3), so any leading coefficient can be corrected at the next level to give norm one. If $r = nm$, the first norm coefficient of $1 + a\Pi^r$ is the residue trace of a at level m ; correction at the next level is possible exactly when that trace is zero. The trace map of a finite-field extension is surjective even when the characteristic divides n .

For $n \nmid r$, choose a Teichmüller element $t \in T_K$ not fixed by σ^r . Such a t exists because $|T_K| = (q^n - 1)/(q - 1)$ is larger than the multiplicative group of any proper subfield of $\mathbb{F}_{q^n}$. The commutator with t multiplies the leading coefficient at level r by the nonzero scalar $t/\sigma^r(t) - 1$. It therefore fills the grade in (5.4).

For $n \mid r$, use commutators between levels 1 and $r-1$. Neither level is divisible by n , since $n \geq 3$. Choose leading coefficient 1 at level 1 and arbitrary coefficient b at level $r-1$. The leading coefficient of their commutator is $\sigma(b) - b$. These fill

$$(\sigma - 1)\mathbb{F}_{q^n} = \ker \operatorname{Tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}.$$

Successive approximation in the complete filtration puts all of S_1 in $\overline{[S_K, S_K]}$. The reverse inclusion holds because the residue quotient is abelian, proving the first assertion.

Finally, commutators of Π with units of E_K have residues $\sigma(z)/z$, which fill T_K by finite-field Hilbert 90. Together with S_1 they are dense in S_K ; all such commutators have reduced norm one. A unit of $\mathcal{D}$ acts trivially on its commutative residue field, whereas Π acts as σ . Writing an arbitrary element as a unit times a power of Π proves the assertion about conjugation. $\square$

5.4 A circle extension associated to the central quotient

Write $I = \mathbb{R}/\mathbb{Z}$, additively, and fix an injective character $\iota : C \hookrightarrow I$. Give L^1 the discrete topology and set

$$Q = \{(y, a) \in U_A \times L^1 : \det y = a_A\}, \quad U_A = \prod_{v \in A} U(k_v). \quad (5.5)$$

The determinant kernel is $S_A = \prod_{v \in A} S(k_v)$. Thus Q is locally compact and second countable, with countably many open determinant fibers, each a translate of the compact group S_A . The map

$$\rho : U(k) \longrightarrow Q, \quad u \longmapsto (u_A, \det u)$$

has dense image on every fiber by weak approximation for S . It follows that P_0 is open and normal in Q and that

$$Q/P_0 \simeq U(k)/V_0. \quad (5.6)$$

For normality, first conjugate by the dense subgroup $\rho(U(k))$ and then pass to its closure; the quotient identification follows from density and openness, with kernel V_0 .

The central quotient in (5.2) gives the discrete central extension

$$1 \longrightarrow C \longrightarrow U(k)/R' \longrightarrow U(k)/V_0 \longrightarrow 1.$$

Pull it back along $Q \rightarrow Q/P_0 \simeq U(k)/V_0$, using (5.6), and push its kernel out by ι . We obtain a locally compact second-countable central extension

$$1 \longrightarrow I \longrightarrow \tilde{Q} \xrightarrow{p} Q \longrightarrow 1. \quad (5.7)$$

The pullback construction gives continuous local sections and an abstract homomorphic lift

$$\ell : U(k) \longrightarrow \tilde{Q}, \quad p\ell = \rho.$$

It also gives a canonical continuous homomorphic section $c : P_0 \rightarrow \tilde{Q}$, obtained by using the identity in the discrete group $U(k)/R'$. For $u \in V_0$,

$$\ell(u) = c(u_A) \iota(uR'). \quad (5.8)$$

We regard values in I as central factors in this multiplicative notation for $\tilde{Q}$. Comparing ℓ with a continuous splitting of p will produce a character of $U(k)$. Identity (5.8) will ensure that its image on $S(k)$ is finite and nonzero.

Lemma 5.4 (Commuting lifts). *Suppose $q_1, q_2 \in Q$ have local components in one common separable field torus at each $v \in A$. Any lifts of q_1 and q_2 to $\tilde{Q}$ commute.*

Proof. Keep their two determinant coordinates fixed. By Proposition 5.2(i), their local components are limits of commuting rational pairs with those exact determinants. The lifts under the homomorphism ℓ of each rational pair commute. For commuting base elements, the commutator of lifts is independent of the choices of lifts. Continuous local sections of (5.7) therefore allow us to take the limit of these commutators, giving the identity. $\square$

The restriction of (5.7) to S_A has a continuous homomorphic splitting. Here is the metaplectic input with its hypotheses made explicit. Choose an isotropic place v_0 of S outside A , and pull the extension back to $S(\mathbb{A}^{v_0})$ by projection to S_A . The lift $\ell|_{S(k)}$ splits this pullback on rational points. The metaplectic kernel away from v_0 is zero, since S is absolutely almost simple and simply connected and v_0 is a nonarchimedean isotropic place [28, Theorem 3]; see also [27]. The pullback extension consequently has zero measurable cohomology class. A measurable homomorphic section is continuous for these locally compact second-countable groups. Restrict it along the subgroup $S_A \subset S(\mathbb{A}^{v_0})$ supported at A to obtain

$$j : S_A \longrightarrow \tilde{Q}. \quad (5.9)$$

This invocation involves only nonarchimedean places and therefore has no omitted-archimedean-place qualification.

We must extend this splitting from S_A to Q . First we will make $j(S_A)$ normal. Its quotient will then be a central extension of the abelian determinant group L^1 ; we will remove that extension's commutators by a further change of j .

5.5 Making the splitting equivariant

For $q \in Q$ and any lift $\tilde{q}$, define $d_q : S_A \rightarrow I$ by

$$\tilde{q}j(s)\tilde{q}^{-1} = j(qsq^{-1})d_q(s). \quad (5.10)$$

This is a continuous character, independent of the lift. It is zero for $q \in S_A$, and composition of conjugations gives

$$d_{q_1q_2}(s) = d_{q_1}(q_2sq_2^{-1}) + d_{q_2}(s).$$

Thus it is a character-valued cocycle factoring through $Q/S_A = L^1$. Lemma 5.3 identifies its coefficient group as

$$\mathrm{Hom}_{\mathrm{cont}}(S_A, I) = \prod_{v \in A} \mathrm{Hom}(T_v, I), \quad T_v = \ker N_{\mathbb{F}_{q_v^n}/\mathbb{F}_{q_v}}.$$

On T_v , a determinant a acts by $\sigma_v^{\text{ord}_v(a_v)}$; on its character group, the action in this cocycle is pullback, $\lambda \mapsto \lambda \circ \sigma_v^{\text{ord}_v(a_v)}$.

The v th component of d_q is zero if $\text{ord}_v(a_v) = 0$. Indeed, since d_q depends only on the determinant, choose a representative whose v th component is a unit in an unramified maximal subfield with reduced norm a_v ; norms from that field supply every unit. It commutes with all Teichmüller lifts of T_v , supported at v in S_A . At the other places choose regular semisimple representatives of the specified determinant, so that these components and the identity belong to separable field tori. Lemma 5.4 then makes (5.10) zero on those Teichmüller lifts, which determine the character. The component and its action therefore factor through the single valuation map $L^1 \rightarrow \mathbb{Z}$ at v . This map is onto by weak approximation in L^1 .

Let d_v be the discrepancy for valuation one. It annihilates $T_v^{\sigma_v}$: the Teichmüller lifts of this subgroup are central scalars, and we may choose regular separable representatives of the determinant and again apply Lemma 5.4. For a finite abelian group T with automorphism σ ,

$$\text{im}(\sigma^* - 1 : \text{Hom}(T, I) \rightarrow \text{Hom}(T, I)) = \{\lambda : \lambda|_{T^\sigma} = 0\}.$$

This is the dual of the kernel-image sequence for $\sigma - 1$; characters of a subgroup extend to I because I is divisible. Choose a correcting character in each local factor, and let $\lambda : S_A \rightarrow I$ be their sum. For $j_\lambda(s) = j(s)\lambda(s)$, direct substitution gives the discrepancy

$$d_q^\lambda(s) = d_q(s) + \lambda(s) - \lambda(qsq^{-1}).$$

The characters may therefore be chosen to make this discrepancy zero at the generator of each valuation group. The cocycle identity then makes every discrepancy zero. Hence we may assume

$$\tilde{q}j(s)\tilde{q}^{-1} = j(qsq^{-1}) \quad (q \in Q, s \in S_A). \quad (5.11)$$

In particular, $j(S_A)$ is normal in $\tilde{Q}$. It is also compact, hence closed, because S_A is compact.

5.6 Removing the determinant commutators

The quotient by $j(S_A)$ is a central extension

$$1 \longrightarrow I \longrightarrow \tilde{Q}/j(S_A) \longrightarrow L^1 \longrightarrow 1. \quad (5.12)$$

Commutators define an alternating biadditive pairing $\omega : L^1 \times L^1 \rightarrow I$, with multiplicative arguments. Lemma 5.4 shows that

$$\omega(a, b) = 0 \quad (5.13)$$

whenever, at every $v \in A$, both a_v and b_v are norms from one common maximal separable subfield of $\mathcal{D}_v$.

We reduce this pairing to characters of local residue units. Put

$$U_0 = \{a \in L^1 : \text{ord}_v(a_v) = 0 \text{ for every } v \in A\}.$$

Using unramified maximal subfields in (5.13) gives $\omega(U_0, U_0) = 0$. The valuation map $L^1 \rightarrow \mathbb{Z}^A$ is onto by weak approximation. For each $v \in A$, choose a maximal totally ramified separable field E_v/k_v inside $\mathcal{D}_v$. Its norm group is open: the separable norm map has surjective trace differential, and hence has open image on local points. That norm group contains an element of valuation one. Weak approximation now gives elements $\pi_i \in L^1$, $i \in A$, whose valuation vectors are the standard basis vectors and all of whose components at v belong to $N_{E_v/k_v}(E_v^\times)$. The common-norm test gives

$\omega(\pi_i, \pi_j) = 0$ for all i, j . Every element of L^1 is a product of an element of U_0 and powers of the π_i . Only the pairings $\omega(\pi_v, u)$, $u \in U_0$, remain to be considered.

For fixed v , this pairing vanishes whenever $u_v \in N_{E_v/k_v}(E_v^\times)$: use E_v at v and unramified maximal subfields at the other places, where both determinants are units. Put

$$H_v = \mathcal{O}_v^\times \cap N_{E_v/k_v}(E_v^\times).$$

The group H_v is open in the compact unit group, so its index is finite. Weak approximation makes the projection $U_0 \rightarrow \mathcal{O}_v^\times/H_v$ onto. The vanishing just proved says that its kernel is killed by $u \mapsto \omega(\pi_v, u)$. Thus this pairing factors through that finite quotient. Composing the quotient character with the projection from $\mathcal{O}_v^\times$ defines a continuous character η_v extending the pairing. Since the norm group contains all n th powers, $n\eta_v = 0$. This argument uses openness of a separable norm group, not openness of the n th-power subgroup.

We claim that η_v kills the principal units. Let $c \in k_v$ have valuation one. Perturb the Eisenstein polynomial $X^n - c$, with its constant coefficient fixed, to separable Eisenstein polynomials f_j tending to it. If the characteristic divides n , one may take $f_j = X^n + \varepsilon_j X - c$ with nonzero $\varepsilon_j \rightarrow 0$. If it does not divide n , no perturbation is necessary. Let α_j be a root and $E'_j = k_v(\alpha_j)$. These degree- n separable fields embed in $\mathcal{D}_v$, and oddness of n gives

$$N_{E'_j/k_v}(\alpha_j) = c, \quad N_{E'_j/k_v}(1 + \alpha_j) \rightarrow 1 + c.$$

For each j , choose $\pi'_v \in L^1$ with the same valuation vector as π_v , with v th component in $N_{E'_j/k_v}(E'_j{}^\times)$. At the other places choose unit components. Weak approximation and openness of the norm groups permit this choice. Since $\pi'_v/\pi_v \in U_0$, their pairings with U_0 agree. Choose elements of U_0 whose v th components belong to the same norm group and tend to $N_{E'_j/k_v}(1 + \alpha_j)$. At the other places their components, and those of π'_v , are norms from unramified maximal subfields. The common-norm test and continuity give

$$\eta_v(N_{E'_j/k_v}(1 + \alpha_j)) = 0.$$

Letting j tend to infinity gives $\eta_v(1 + c) = 0$. These units generate $1 + \mathfrak{p}_v$: if $\text{ord}_v(d) \geq 2$, then

$$1 + d = \frac{(1 + c)(1 + d)}{1 + c},$$

and both numerator and denominator have the form $1 + c'$ with $\text{ord}_v(c') = 1$. Thus η_v is a character of $\mathbb{F}_{q_v}^\times$, killed by n .

We can remove these residual characters without losing (5.11). The permissible changes of j are characters of S_A invariant under Q , whose v th factors are the characters of $T_v/(\sigma_v - 1)T_v$. If a section is multiplied by such a character λ , the commutator pairing changes by $-\lambda$ evaluated on the ordinary local commutator of determinant representatives. To see this, write the commutator of any two lifts as j of their base commutator times the old value of ω ; substitution of $j\lambda$ gives the asserted formula.

To compute the effect on η_v , represent the determinant $(\pi_v)_v$ by a division prime multiplied by a unit of the fixed unramified maximal subfield. Such a unit adjusts the reduced norm to the specified value because the ratio of the two norms is a unit. It commutes with every other unit of that subfield, so this adjustment does not affect their commutator. For an unramified unit with residue r , the residue of that commutator is therefore $\sigma_v(r)/r$. Changing the norm preimage r alters it by $(\sigma_v - 1)T_v$, so there is a well-defined homomorphism

$$\mathbb{F}_{q_v}^\times \rightarrow T_v/(\sigma_v - 1)T_v, \quad N(r) \mapsto \sigma_v(r)/r. \quad (5.14)$$

It is onto by finite-field Hilbert 90. Its target has order $\gcd(n, q_v - 1)$: on the cyclic group T_v the cokernel of $\sigma_v - 1$ has the same order as its kernel, and

$$T_v^{\sigma_v} = \{z \in \mathbb{F}_{q_v}^\times : z^n = 1\}.$$

The domain in (5.14) is cyclic. Its dual therefore identifies all characters killed by n with characters of that target. Choose the invariant local character that cancels η_v , and make these changes for every $v \in A$. At other places the determinant representatives for a mixed pair can be taken in unramified subfields, so no other local component is introduced. Unit-unit pairings and pairings of the chosen π_i still vanish by (5.13), which holds for every equivariant choice of j . All the generating pairings now vanish, so $\omega = 0$.

The extension (5.12) is consequently abelian. The divisible group I is injective as an abelian group, so the extension splits, continuously since its base L^1 is discrete. Take in $\tilde{Q}$ the inverse image of the image of such a section. This subgroup meets I trivially and maps onto Q ; hence it maps isomorphically to Q . The inverse map is continuous on each open determinant fiber, where it is a translate of j . We have constructed a continuous homomorphic splitting

$$s : Q \longrightarrow \tilde{Q} \tag{5.15}$$

of the original extension (5.7).

5.7 Extracting a finite character

We finish the proof of Proposition 5.1 by comparing (5.15) with the rational lift. Define $\chi_0 : U(k) \rightarrow I$ by

$$\ell(u) = s(\rho(u)) \chi_0(u). \tag{5.16}$$

Both maps are homomorphisms lifting ρ , and their difference is central, so χ_0 is a character.

Its image on $S(k)$ is finite. For $u \in R'$, Equation (5.8) reads $\ell(u) = c(u_A)$. Thus $\chi_0|_{R'}$ is the restriction of the continuous character $p_0 \mapsto c(p_0)s(p_0)^{-1} \in I$ on the profinite group P_0 . Such a character has finite image: its compact image is a profinite quotient, while a closed subgroup of the circle is either finite or the circle itself. Since R' has finite index in $S(k)$, the image $\chi_0(S(k))$ is also finite.

This image is nonzero. Choose a nonidentity element $\beta \in C$ and a representative $u \in V_0$. The subgroup R is dense in P_0 , so there are $r_j \in R$ with $(ur_j)_A \rightarrow 1$. All these elements still represent β modulo R' . If χ_0 vanished on $S(k)$, Equations (5.8) and (5.16) would imply

$$c((ur_j)_A) \iota(\beta) = s(\rho(ur_j)).$$

Both continuous sections tend to the identity. This would give $\iota(\beta) = 0$, contradicting injectivity of ι .

Finally, we make the character finite on all of $U(k)$ without changing its restriction to $S(k)$. Let $\mathbb{F}_L$ be the full constant field of L . The divisor map on $L^\times$ has kernel $\mathbb{F}_L^\times$ and image in the free abelian divisor group. Every subgroup of a free abelian group is free, and an extension of a free abelian group splits. Therefore

$$L^1 = T \oplus \bigoplus_{j \in J} \mathbb{Z}a_j$$

for a finite torsion group T and a possibly infinite basis indexed by J . Choose $u_j \in U(k)$ with $\det u_j = a_j$ and a character $\psi : L^1 \rightarrow I$ satisfying $\psi(a_j) = \chi_0(u_j)$ and $\psi|_T = 0$. Then

$$\chi = \chi_0 - \psi \circ \det$$

vanishes on the chosen u_j and equals χ_0 on $S(k)$. If m annihilates $\chi_0(S(k))$ and t annihilates T , then mt annihilates $\chi(U(k))$: remove the free determinant part by products of the u_j , and the t th power of what remains belongs to $S(k)$. The subgroup of I annihilated by mt is finite. Thus χ has finite image and is nonzero on $S(k)$, proving the first alternative.

All empty-product cases are included. In particular, if A is empty, then $Q = L^1$ and $S_A = 1$; Lemma 5.4 makes the circle extension abelian immediately, and the character extraction above is unchanged. This completes the proof of Proposition 5.1.

6 Palettes and simultaneous color returns

Throughout this section and Section 7, D is a division algebra of odd reduced degree $n \geq 3$. Suppose that the second alternative of Proposition 5.1 occurs: there is a homomorphism

$$\phi : U(k) \longrightarrow \mathcal{F}, \quad \phi(V_0) = \mathcal{F}, \quad R \subseteq \ker \phi,$$

where $\mathcal{F}$ is finite, nonabelian, and simple. We call the values of ϕ *colors*. Every central unitary scalar has color 1, since its image centralizes the surjective image $\mathcal{F}$.

We shall rule out this homomorphism by comparing multiplicative colors with additive translations on the Hermitian elements of D . Two facts make the comparison possible. First, a probability law on color configurations will retain every color while being invariant under all additive translations. Second, the following finite-group obstruction will prohibit certain simultaneous returns of those colors.

The following finite-group theorem provides the distinction between colors that the recurrence argument will preserve. Its double-coset assertion is [23, Lemma 5.2 and Section 6]. We include the proof, together with the additional assertion about abelian subgroups, in Appendix A.

Theorem 6.1 (Finite simple-group obstruction). *Let F be a finite nonabelian simple group. There is a nonempty proper conjugacy-invariant subset $\mathcal{S} \subset F$, with $1 \notin \mathcal{S}$, having the following properties.*

- (i) *If $W \in \mathcal{S}$ and $Z \notin \mathcal{S}$ commute, then there exists $B \in F$ such that*

$$ZW \notin C_F(B)(ZW^{-1})C_F(BZ^2).$$

- (ii) *For every abelian subgroup $A \subset F$ with $A \cap \mathcal{S} \neq \emptyset$, the complement $A \setminus \mathcal{S}$ is a subgroup of A .*

In particular, $\mathcal{S}$ is invariant under inversion.

Fix a set $\mathcal{S} \subset \mathcal{F}$ supplied by Theorem 6.1. We shall use both its double-coset obstruction and its assertion that, on an abelian subgroup meeting $\mathcal{S}$, the complement of $\mathcal{S}$ is a subgroup. In particular, $\mathcal{S}$ is inverse-stable. The probability construction and the fractional transformations below adapt [23, Sections 5.2–5.4]. Section 7 supplies the additive argument in positive characteristic.

6.1 A compact space of color configurations

Let $\tilde{G}$ be the special unitary group of the hyperbolic Hermitian form on D^2 with coefficients 1, -1 , and let X be its projective variety of isotropic right D -lines. This is a flag variety for a minimal k -parabolic of $\tilde{G}$: the division hypothesis makes its relative k -rank one. Every rational isotropic line is represented uniquely by a column $(u, 1)$ with $u \in U(k)$. Indeed, its second coordinate cannot be zero, and every nonzero element of D is invertible. We henceforth identify $X(k)$ with $U(k)$ in this way.

Choose $\theta \in L \setminus k$, with $\bar{\theta} = -\theta$ when $\text{char } k \neq 2$, and put

$$J = \{x \in D : x^* = x\}, \quad q(x) = (x + \theta)(x + \bar{\theta})^{-1}.$$

The space J is an additive k -vector group; its rational points will also be denoted by J . The column $(x + \theta, x + \bar{\theta})$ is isotropic, and the Cayley parametrization gives

$$X(k) = J \sqcup \{\infty\}, \quad q(\infty) = 1. \quad (6.1)$$

For example, $x + \bar{\theta}$ cannot vanish for Hermitian x , and the inverse parametrization is defined at every $u \neq 1$ because $u - 1$ is invertible. Over a completion, the unitary and additive parametrizations are open charts in $X(k_v)$; their complements are proper algebraic subvarieties and have measure zero in its smooth measure class.

Let $\mathcal{H}$ be the countable group of rational transformations of X induced by form similitudes. Besides $\tilde{G}(k)$, it contains

$$L_a(u) = au, \quad R_a(u) = ua \quad (a \in U(k)), \quad T_b(x) = x + b \quad (b \in J).$$

It also contains the central affine maps $x \mapsto lx + s$, with $l \in k^\times$ and $s \in k$. We write $\tau_s = T_s$ for scalar translations. For $p \in X(k)$ define

$$c(p) = (c_h(p))_{h \in \mathcal{H}}, \quad c_h(p) = \phi(u(hp)).$$

The closure of these configurations in $\mathcal{F}^{\mathcal{H}}$ is denoted by $\mathcal{C}$; its members are called *palettes*. It is a compact metrizable space. The action and its relation to rational points are

$$(gc)_h = c_{hg}, \quad c(gp) = gc(p).$$

Every finite coordinate pattern occurring in $\mathcal{C}$ occurs at a rational point. Consequently identities involving finitely many colors pass to all palettes. For instance,

$$c_{L_a h} = \phi(a)c_h, \quad c_{R_a h} = c_h \phi(a). \quad (6.2)$$

6.2 An invariant law retaining all colors

A translation-invariant probability is easy to obtain by averaging. What requires proof is that averaging can be arranged to retain every color at every coordinate. We first construct a joint probability on local flags and palettes, whose conditional color distributions are uniform.

Proposition 6.2. *There is a probability measure μ on $\mathcal{C}$ such that*

$$(T_b)_* \mu = \mu \quad (b \in J), \quad \mu\{c : c_h = a\} = |\mathcal{F}|^{-1} \quad (h \in \mathcal{H}, a \in \mathcal{F}).$$

We divide the proof into the construction of a joint law, identification of its base measure, and uniformity of its conditional colors.

6.2.1 Rational weights and a joint law

For each place v , choose an Iwasawa compact subgroup $\mathcal{K}_v \subset \tilde{G}(k_v)$ and its invariant probability m_v on $X(k_v)$. The compact group is transitive on this flag variety. At almost all places use the integral models. Set

$$Y = \prod_v X(k_v), \quad m = \prod_v m_v.$$

Each local probability belongs to the smooth measure class. A rational similitude preserves m_v at all but finitely many places: away from its denominators and the places where its multiplier is a nonunit, it normalizes the chosen integral compact. Thus

$$\rho(h, y) = \frac{d(h_*m)}{dm}(y)$$

is a continuous positive function given by a finite product of local densities. With this pushforward convention its cocycle identity is

$$\rho(ag, y) = \rho(a, y)\rho(g, a^{-1}y). \quad (6.3)$$

Assign a positive weight $w(p)$ to each rational point by transporting weight 1 from a fixed base point under $\tilde{G}(k)$, multiplying by the forward local volume Jacobians. Rational conjugacy of parabolics gives transitivity. The weight is independent of the transporter: at a rational fixed point the product of tangent determinants is 1 by the product formula. The same argument gives the weight rule for every $h \in \mathcal{H}$, including similitudes.

We need convergence of these weights before they can define a probability. The following proof uses the discreteness of the function-field modulus and makes the exponent explicit.

Lemma 6.3. *For every real $t > 1$,*

$$\sum_{p \in X(k)} w(p)^t < \infty.$$

Proof. Let P be the stabilizer of the base point, with unipotent radical N_P . Write δ_P for its adelic modulus, the absolute value of the determinant of its action on $\text{Lie}(N_P)$. Extend it to $\tilde{G}(\mathbb{A}_k)$ by Iwasawa decomposition with a right compact factor:

$$g = p_0 k_0, \quad \delta_P(g) = \delta_P(p_0), \quad k_0 \in \mathcal{K} := \prod_v \mathcal{K}_v.$$

For a stabilizing element the forward tangent Jacobian is the inverse modulus. Taking the rational transporter to be γ^{-1} therefore identifies the required sum with

$$\sum_{\gamma \in P(k) \backslash \tilde{G}(k)} \delta_P(\gamma)^t. \quad (6.4)$$

The values $\delta_P(\gamma)$ in this sum are bounded above. To see this, take a rational vector v_P spanning the top exterior power of $\text{Lie}(N_P)$, considered inside the corresponding exterior-power representation of $\tilde{G}$. Choose local norms integral almost everywhere. The Iwasawa decomposition gives

$$\prod_v \|\gamma^{-1} v_P\|_v \leq C_1 \delta_P(\gamma)^{-1}.$$

The product norm of a nonzero rational vector is bounded below by a fixed positive constant: one nonzero rational coordinate and the product formula suffice. Hence $\delta_P(\gamma) \leq C$ uniformly.

Choose a compact open subgroup $K' \subset \mathcal{K}$ so small that $K' \cap \tilde{G}(k) = \{1\}$. The images of the sets $\gamma K'$ in $P(k) \backslash \tilde{G}(\mathbb{A}_k)$ are disjoint for distinct cosets in (6.4), and all have the same positive volume. Indeed an overlap would put $\gamma_2^{-1} p \gamma_1 \in \tilde{G}(k)$ in K' for some $p \in P(k)$. The same argument excludes stabilizers on each such neighborhood. Since the modulus is right- K' -invariant, it suffices to show that

$$\int_{P(k) \backslash \tilde{G}(\mathbb{A}_k)} \mathbf{1}_{\{\delta_P \leq C\}} \delta_P(g)^t dg < \infty.$$

Iwasawa integration reduces this to left Haar integration over $P(k) \backslash P(\mathbb{A}_k)$. One can normalize Haar measures by the compact groups $\mathcal{K}$ and $P(\mathbb{A}_k) \cap \mathcal{K}$; the resulting double cosets and their stabilizer weights agree. Let M_P be a Levi subgroup, and use the decomposition $P = N_P M_P$, with the unipotent coordinate first. The left Haar measure contains the factor $\delta_P(m_0)^{-1}$. The quotient of the unipotent radical has fixed finite volume, so integration along it leaves

$$\mathbf{1}_{\{\delta_P(m_0) \leq C\}} \delta_P(m_0)^{t-1}$$

on $M_P(k) \backslash M_P(\mathbb{A}_k)$. The product formula makes the fiber volumes independent of rational changes of coordinates.

The group $\tilde{G}$ has relative rank one and P is minimal. Thus the kernel of the modulus on the adelic Levi is its norm-one adelic group, whose rational quotient has finite volume by reduction theory; see [14]. The nonempty modulus levels all have this same finite volume, since Levi Haar measure is also right invariant. The modulus values form a discrete subgroup of the powers of the cardinality of the full constant field. Summing δ_P^{t-1} over the levels bounded above is a convergent geometric series precisely when $t > 1$. This proves the lemma. $\square$

For $t > 1$, give $(p, c(p)) \in Y \times \mathcal{C}$ normalized mass $w(p)^t$, and denote the resulting probability by μ_t . The weight rule gives

$$h_* \mu_t = \rho(h, \cdot)^t \mu_t.$$

The reciprocal implicit in the pushforward density is accounted for here: the mass at hp after pushforward is the old mass at p . By compactness take a weak limit μ^0 as $t \downarrow 1$. Uniform convergence of $\rho(h, \cdot)^t$ for each fixed h gives

$$h_* \mu^0 = \rho(h, \cdot) \mu^0 \quad (h \in \mathcal{H}). \quad (6.5)$$

6.2.2 The base projection is the product probability

Let ν be the projection of μ^0 to Y . We show that $\nu = m$; in particular, no singular base probability enters the later conditional argument. Fix a sufficiently large nonempty finite set T of places. Subscripts T denote products over T , and superscripts T denote the complementary products. An arithmetic lattice $\Gamma \subset \tilde{G}_T$ integral outside T preserves m^T . Consequently (6.5) gives

$$\gamma_* \nu_T = \rho_T(\gamma, \cdot) \nu_T \quad (\gamma \in \Gamma).$$

Define

$$f(g) = \int_{Y_T} \rho_T(g, y) d\nu_T(y).$$

The cocycle rule and this transformation law imply $f(\gamma g) = f(g)$. The function is positive and continuous on $\tilde{G}_T$.

Choose a probability η of full support on $\tilde{G}_T$ that is invariant under left multiplication by $\mathcal{K}_T$. Averaging first over that compact group shows that

$$\int \rho_T(h, z) d\eta(h) = 1 \quad (z \in Y_T) :$$

compact transitivity turns the inner average into the integral of a probability density over m_T . By (6.3), it follows that

$$f(g) = \int f(gh) d\eta(h).$$

Apply Jensen's inequality to the bounded strictly concave function $a \mapsto a/(1+a)$ on the positive ray. Its nonnegative Jensen discrepancy has integral zero on the finite-volume quotient $\Gamma \backslash \tilde{G}_T$, by right invariance of quotient measure. Equality in strict Jensen implies that $f(gh)$ is constant for η -almost every h , for almost every g . Full support and continuity then make f constant everywhere. Its value at 1 is 1. This argument requires no integrability assumption on f itself.

These density integrals determine the base probability. Choose $a_j \in \tilde{G}_T$ contracting a big cell to a point $x_0 \in Y_T$, using a parabolic cocharacter in each local factor. The complement of the cell has measure zero, so $(a_j)_* m_T \rightarrow \delta_{x_0}$. For $b \in C(Y_T)$ set

$$B_j(y) = \int_{\mathcal{K}_T} b(k_0 x_0) \rho_T(k_0 a_j, y) dk_0.$$

For fixed y , the density in this integral makes $k_0^{-1}y$ have law $(a_j)_* m_T$. This follows by pushing compact Haar measure to Y_T and using $\rho_T(k_0 a_j, y) = \rho_T(a_j, k_0^{-1}y)$. Compact equicontinuity therefore gives $B_j(y) \rightarrow b(y)$ uniformly. Since $f(k_0 a_j) = 1$,

$$\int B_j d\nu_T = \int_{\mathcal{K}_T} b(k_0 x_0) dk_0 = \int b dm_T.$$

It follows that $\nu_T = m_T$. Exhausting the places proves $\nu = m$.

Disintegrate the joint law as

$$d\mu^0(y, c) = dQ_y(c) dm(y). \quad (6.6)$$

The compact metrizable spaces admit regular conditional probabilities. Equation (6.5) and uniqueness of disintegration give $Q_{hy} = h_* Q_y$ almost everywhere. Since $\mathcal{H}$ is countable, we may use all these identities simultaneously.

6.2.3 Uniform conditional colors

Let $\mathbf{p}(y)$ be the c_1 -marginal of Q_y , regarded as a probability vector on $\mathcal{F}$, and put $N = \ker \phi \cap S(k)$. The palette rules (6.2) imply that $\mathbf{p}$ is invariant under the separate left and right rotations by every element of N . We first prove invariance under a whole local special unitary group.

Enlarge T to contain a place v_0 at which S is split, as well as the required integral-model exceptions. On the unitary chart over T , the base measure is in the Haar class of U_T . Disintegrate that class along the smooth determinant map, and choose measurable representatives d_λ of its fibers. Write

$$u_T = g d_\lambda, \quad g \in S_T.$$

The conditional measure class in g is Haar measure. For almost every determinant λ , each component of $\mathbf{p}(g d_\lambda, y^T)$ descends, by its left invariance, to

$$Z_\lambda = \Gamma' \backslash (S_T \times Y^T), \quad \Gamma' = N \cap \{a \in S(k) : a \text{ is integral outside } T\}. \quad (6.7)$$

The group Γ' has finite index in an arithmetic lattice. The quotient has finite measure, obtained from Haar measure on a lattice fundamental domain times m^T ; the integral left tail actions preserve the latter. The bounded components of $\mathbf{p}$ thus belong to $L^2(Z_\lambda)$.

Right invariance by $a \in \Gamma'$ acts on this quotient through

$$(g, y^T) \mapsto (g d_\lambda a_T d_\lambda^{-1}, R_a y^T).$$

The tail rotations commute with the left action and belong to a compact group of measure-preserving transformations. There are elements $a_j \in \Gamma'$ escaping every compact set at v_0 while remaining bounded at the other factors of S_T . For example, use Proposition 2.4 to find infinitely many elements of $R \subseteq N$ in a fixed compact-open cylinder away from v_0 . Adelic discreteness forces an escaping subsequence at v_0 . After another subsequence the bounded components and compact tail rotations converge. Conjugating by the fixed d_λ does not change these properties.

Decompose $L^2(Z_\lambda)$ into the invariant subspace for the right $S(k_{v_0})$ action and its orthogonal complement. The local group is the split group $\mathrm{SL}_n(k_{v_0})$, with its finite center, and local Kneser–Tits identifies it with its root-generated group. Hence Howe–Moore coefficient decay applies on the complement in every characteristic; see [16] and [3, Theorem 4.19]. If v is the projection of a component of $\mathbf{p}$ to that complement, the operators fixing it have the form $\pi(g_j)A_j$, where g_j escapes in $S(k_{v_0})$ and the commuting unitaries A_j converge strongly to a unitary A . Therefore

$$\|v\|^2 = \langle \pi(g_j)A_j v, v \rangle = \langle \pi(g_j)A v, v \rangle + o(1) \longrightarrow 0.$$

The local and compact actions used here are strongly continuous, as one first checks on continuous functions with compact support and then by density in L^2 . We conclude that $\mathbf{p}$ is invariant under the full local right group $S(k_{v_0})$.

Lifting back and using Haar measure and Fubini, $\mathbf{p}$ consequently depends on u_{v_0} only through its determinant, up to null sets. More explicitly, restrict at v_0 to the conull unitary chart and consider the measurable projection

$$\pi : Y \longrightarrow \bar{Y} := \det(U(k_{v_0})) \times \prod_{v \neq v_0} X(k_v), \quad y \longmapsto (\det u_{v_0}, (y_v)_{v \neq v_0}).$$

Give $\bar{Y}$ the pushforward probability $\bar{m}$. The local invariance just proved says that $\mathbf{p} = \bar{\mathbf{p}} \circ \pi$ almost everywhere for a bounded measurable function $\bar{\mathbf{p}}$ on this space. Every rational left rotation from $S(k)$ acts trivially on the first coordinate of $\bar{Y}$.

Take $g \in V_0$. Proposition 2.4 supplies $r_j \in R$ tending to g in the restricted adelic product away from v_0 . Their left actions on $\bar{Y}$ converge to that of g . This convergence passes to bounded measurable functions: outside one fixed finite set the actions preserve the factor probabilities, and on the remaining local factors their densities are uniformly bounded. The determinant factor of $\bar{m}$ is unchanged. For a bounded continuous cylinder function, convergence follows from local continuity and bounded convergence; approximation in $L^1(\bar{m})$ then proves it for $\bar{\mathbf{p}}$. Since every r_j belongs to $R \subseteq N$, invariance passes to the limit and gives $\mathbf{p}(L_g y) = \mathbf{p}(y)$ for every $g \in V_0$, almost everywhere.

Conditional equivariance also gives $\mathbf{p}(L_g y) = \phi(g)\mathbf{p}(y)$. Since $\phi(V_0) = \mathcal{F}$, the probability vector is uniform. Equivariance identifies the c_h -marginal at y with the c_1 -marginal at hy , so all conditional coordinate marginals are uniform.

Completion of the proof of Proposition 6.2. Let η_0 be the palette marginal of μ^0 . The countable additive group J is an increasing union of finite additive subgroups J_j . Average

$$\eta_j = |J_j|^{-1} \sum_{b \in J_j} (T_b)_* \eta_0.$$

Every summand has uniform coordinate marginals, because a translation merely changes the coordinate index. Any weak limit μ has the same marginals. For each fixed b , all sufficiently late averages are T_b -invariant, so the limit is invariant under every translation. $\square$

6.3 A translation obstructed by a change of membership

We now connect this law to Theorem 6.1. Given $w \in U(k)$ and a scalar affine map $x \mapsto lx + s$, its image in unitary coordinates agrees, up to a central norm-one scalar, with

$$z = (w - \bar{t})(1 - tw)^{-1}. \quad (6.8)$$

Here, putting

$$a = l\bar{\theta} - s - \theta, \quad b = s - (l - 1)\theta,$$

one takes $\bar{t} = -b/a$ when $a \neq 0$. Direct substitution in (6.1) proves this formula. Moreover

$$a\bar{a} - b\bar{b} = -l(\theta - \bar{\theta})^2, \quad d := 1 - t\bar{t} \neq 0.$$

If $a = 0$, then $(l, s) = (-1, -\theta - \bar{\theta})$, and the affine transformation inverts colors. It therefore preserves membership in $\mathcal{S}$ and will never cause an exceptional membership change.

For $a \neq 0$ set

$$C = tw, \quad R_C = (1 - C^*)(1 - C)^{-1}, \quad z = wR_C.$$

These definitions agree with (6.8). The elements C, C^* commute and $CC^* = t\bar{t}$ is central. Division and $d \neq 0$ make the denominators invertible; R_C is unitary and commutes with w . In particular z commutes with w .

Consider the similitude

$$D_C(u) = (u + C)(C^*u + 1)^{-1}.$$

Its matrix has multiplier d . Its denominator cannot vanish at a unitary element, since that would force $t\bar{t} = 1$. Writing $u = w\alpha$ gives the useful identity

$$w^{-1}D_C(u) = (\alpha + t)(\bar{t}\alpha + 1)^{-1}, \quad [w^{-1}D_C(u), w^{-1}u] = 1. \quad (6.9)$$

No commutation between u and w is being assumed.

For a variable $u \in U(k)$ put

$$v = D_{C^*}(uR_C), \quad u' = R_CD_{-C}(v).$$

Apply (6.9) first with parameters $w^{-1}, \bar{t}$, and then with parameters $w, -t$. Conjugating the first resulting commutation gives

$$[vw, uz] = 1, \quad [w^{-1}v, z^{-1}u'] = 1. \quad (6.10)$$

On the other hand, matrix multiplication gives the fractional matrix of $u \mapsto u'$ as

$$\begin{pmatrix} d + C - C^* & -(C - C^*) \\ C - C^* & d - (C - C^*) \end{pmatrix}.$$

Applying it to $(x + \theta, x + \bar{\theta})$ and dividing by d shows that it is the additive translation by

$$b_w(l, s) = \frac{(\theta - \bar{\theta})(C - C^*)}{d} \in J. \quad (6.11)$$

The matrix calculation also applies at infinity.

To detect simultaneous returns, take one independent copy of $(\mathcal{C}, \mu)$ for each color, and in the product space let E_0 be the event that the first coordinate of the copy indexed by $a \in \mathcal{F}$ equals a . Then

$$\delta := \mathbb{P}(E_0) = |\mathcal{F}|^{-|\mathcal{F}|} > 0.$$

For the diagonal translation action define its return correlation

$$K(b) = \mathbb{P}(E_0 \cap T_b^{-1}E_0) \quad (b \in J). \quad (6.12)$$

This is nonnegative and positive definite. The spectral theorem for the discrete abelian group J gives a positive measure σ on its compact character group $\hat{J}$ such that

$$K(b) = \int_{\hat{J}} \chi(b) d\sigma(\chi), \quad \sigma(\hat{J}) = \delta, \quad \sigma(\{1\}) \geq \delta^2. \quad (6.13)$$

The last inequality follows by projecting $\mathbf{1}_{E_0}$ onto invariant vectors; its projection onto constants alone has squared norm δ^2 .

Proposition 6.4. *If the colors of w and its scalar affine image satisfy $\phi(w) \in \mathcal{S}$ and $\phi(z) \notin \mathcal{S}$, then*

$$K(b_w(l, s)) = 0.$$

Proof. If the correlation were positive, every color would occur as $\phi(u) = \phi(u')$ for the fixed translation in (6.11). Indeed a point of the positive intersection supplies the two-coordinate return pattern in each palette copy, and every such finite pattern occurs rationally. The rational points for different colors may differ, which is all that is needed.

Write $W = \phi(w)$, $Z = \phi(z)$, and, for a return point, put $X' = \phi(u) = \phi(u')$ and $Y' = \phi(v)$. With $B = Z^{-1}X'$, Equation (6.10) yields

$$W^{-1}Y' \in C_{\mathcal{F}}(B), \quad Z^{-1}Y'WZ \in C_{\mathcal{F}}(BZ^2).$$

As $WZ = ZW$, direct multiplication gives

$$(W^{-1}Y')^{-1}(ZW^{-1})(Z^{-1}Y'WZ) = ZW.$$

The return colors make B range over all of $\mathcal{F}$, contradicting Theorem 6.1. The exceptional affine map preserves membership and hence does not arise under the stated hypotheses. $\square$

It remains to record the precise dependence of this translation on (l, s) . This will allow us to apply additive Fourier analysis to scalar affine positions without treating multiplicative colors as continuous functions. Put

$$T = \theta + \bar{\theta}, \quad P_{\theta} = \theta\bar{\theta}.$$

For fixed w define the k -linear map

$$\mathcal{B}_w(\alpha, \beta) = \frac{\alpha(w - w^*) + \beta(\bar{\theta}w - \theta w^*)}{\theta - \bar{\theta}} : k^2 \longrightarrow J.$$

Expanding $-a\bar{b}$ and using the preceding denominator identity gives

$$\frac{t}{d} = \frac{\alpha + \bar{\theta}\beta}{(\theta - \bar{\theta})^2}, \quad \alpha = P_{\theta}l - \frac{s^2 + Ts + P_{\theta}}{l}, \quad \beta = 2s + T(1 - l).$$

Consequently

$$b_w(l, s) = \mathcal{B}_w \left(P_{\theta}l - \frac{s^2 + Ts + P_{\theta}}{l}, 2s + T(1 - l) \right). \quad (6.14)$$

Both scalar arguments vanish at the exceptional affine parameter, so this formula extends there by $b_w = 0$. Equations (6.13) and (6.14), together with Proposition 6.4, are the inputs to the recurrence argument.

7 Additive recurrence and the exclusion of simple colors

We retain the odd division degree hypothesis and the hypothetical quotient $\phi : U(k) \rightarrow \mathcal{F}$ from Section 6. Our objective is to show that membership in $\mathcal{S}$ is unchanged by scalar translations on the additive chart. Once this is known, fractional transformations generate left rotations whose colors generate $\mathcal{F}$, contradicting the fact that $\mathcal{S}$ is nonempty and proper. This follows the organization of [23, Sections 5.5–5.6]. The recurrence proofs below are specific to positive characteristic: odd characteristic permits a quadratic Fourier argument, whereas characteristic two requires additive polynomials and the subgroup-complement property of Theorem 6.1.

Write τ_s for the scalar translation $x \mapsto x + s$, with $s \in k$. For a palette $c \in \mathcal{C}$, put

$$I_c(h) = \mathbf{1}_{\mathcal{S}}(c_h), \quad h \in \mathcal{H}.$$

We shall produce a palette satisfying

$$I_c(\tau_s h) = I_c(h) \quad (s \in k, h \in \mathcal{H}). \quad (7.1)$$

Recall that the nonnegative function K constructed in Section 6 has a spectral representation

$$K(b) = \int_{\widehat{\mathcal{J}}} \chi(b) d\sigma(\chi), \quad \sigma(\widehat{\mathcal{J}}) = \delta, \quad \sigma(\{1\}) \geq \delta^2, \quad \delta = |\mathcal{F}|^{-|\mathcal{F}|}.$$

Here additive groups have the discrete topology, and their duals are compact groups of characters with values in the complex unit circle.

7.1 Two averaging facts

We first isolate the Fourier argument shared by the two characteristics. An average on a countable set Ω will mean integration against a finitely supported probability measure on Ω , denoted by $\mathbb{E}_j$. A subset has density zero for these averages if its indicator has average tending to zero.

Lemma 7.1. *Let G be a countable abelian group, let $v : \Omega \rightarrow G$, and suppose that*

$$\mathbb{E}_j \chi(v(y)) \longrightarrow \begin{cases} 1, & \chi = 1, \\ 0, & \chi \neq 1 \end{cases} \quad (\chi \in \widehat{G}). \quad (7.2)$$

Let $E \subseteq \Omega$. For each $x \notin E$, suppose that there are $a_x \in G$ and a positive measure ν_x on $\widehat{G}$, of mass δ and with $\nu_x(\{1\}) \geq \delta^2$, such that

$$F_x(y) = \int \chi(v(y) - a_x) d\nu_x(\chi) = 0 \quad (y \in E).$$

If $\Omega \setminus E$ is nonempty, its lower density is at least δ . If, along a subsequence, E has density tending to $d > 0$, then along a further subsequence there is a finite set $\Lambda \subset \widehat{G} \setminus \{1\}$ such that

$$\nu_x(\Lambda) \geq d\delta^2/2 \quad (x \notin E). \quad (7.3)$$

Proof. By dominated convergence and (7.2), $\mathbb{E}_j F_x \rightarrow \nu_x(\{1\})$. Since $|F_x| \leq \delta$ and F_x vanishes on E , this gives the lower-density assertion.

The union of the atoms of the measures ν_x is countable. Pass to a subsequence on which

$$c(\chi) = \lim_j \mathbb{E}_j(\mathbf{1}_E(y) \chi(v(y)))$$

exists for every character in that union. In particular, $c(1) = d$. For any finite list of distinct characters, their Gram matrices for $\mathbb{E}_j$ converge to the identity by (7.2). Bessel's inequality therefore gives $\sum_{\chi} |c(\chi)|^2 \leq 1$. Consequently the set of nontrivial atoms for which $|c(\chi)| \geq d\delta/2$ is finite; denote it by Λ .

The nonatomic part of ν_x contributes zero to the limiting average against $\mathbf{1}_E$. Indeed, the mean square of its Fourier transform, evaluated at $v(y) - a_x$, tends to zero: expand the square, apply (7.2) to $\chi\bar{\eta}$, and integrate over two copies of the nonatomic measure. The diagonal has product measure zero. Cauchy-Schwarz now proves the assertion about the weighted average. The atomic part may be averaged term by term because its weights are summable. Thus $F_x|_E = 0$ implies

$$0 = d\nu_x(\{1\}) + \sum_{\chi \neq 1} c(\chi)\chi(-a_x)\nu_x(\{\chi\}).$$

The absolute contribution from characters outside $\Lambda \cup \{1\}$ is at most $d\delta^2/2$. The trivial character contributes at least $d\delta^2$, and $|c(\chi)| \leq 1$. The remaining atoms must therefore have total mass at least $d\delta^2/2$, as required. $\square$

We shall also use the following elementary form of polynomial avoidance. It applies to finite additive subgroups without any compatibility with the coordinates of the ambient vector space.

Lemma 7.2. *Let k be an infinite field of positive characteristic. A proper algebraic subset of k^r has density zero on every increasing sequence of finite additive subgroups exhausting k^r . It also has density zero on products of finite subsets whose smallest cardinality tends to infinity.*

Proof. It suffices to consider the zero set of a nonzero polynomial f of total degree D . The elementary grid bound, obtained by induction from the one-variable root bound, says that its zero proportion on $A_1 \times \cdots \times A_r$ is at most $D/\min_i |A_i|$. This proves the second assertion.

For the first assertion, choose any finite additive subgroup $H \subset k$. An exhausting sequence $U_j \subset k^r$ eventually contains H^r , and each such U_j is a disjoint union of its H^r -cosets. On a coset $u + H^r$, apply the grid bound to the nonzero polynomial $f(u + X)$. The zero proportion is at most $D/|H|$, uniformly in the coset. Finite additive subgroups of k have arbitrarily large cardinality, so this bound proves the claim. $\square$

7.2 Odd characteristic

Assume first that $\text{char } k \neq 2$. Our choice of Cayley parameter gives $\bar{\theta} = -\theta$, so that $T = 0$. Let $\mathcal{A} = k^\times \ltimes k$ be the scalar affine group, with (l, s) acting by $z \mapsto lz + s$. Associate to its elements the vectors

$$v(l, s) = \left(l, s, \frac{s^2 + P_\theta}{l} \right) \in k^3. \quad (7.4)$$

For a fixed initial affine position, the scalar parameters in Equation 6.14 become linear functions of these three coordinates. We average over $l \in I_j \setminus \{0\}$ and $s \in H_j$ independently and uniformly, where I_j, H_j are increasing finite additive subgroups exhausting k , and

$$\{ab : a, b \in I_j\} \subseteq H_j.$$

Such sequences exist by taking finite-dimensional vector spaces over the prime field and enlarging H_j as necessary.

Lemma 7.3. *For every nontrivial additive character χ of k^3 , $\mathbb{E}_j \chi(v(l, s)) \rightarrow 0$. Every proper polynomial locus in (l, s) also has density zero for these averages.*

Proof. Write $\chi(l, s, r) = \chi_1(l)\chi_2(s)\chi_3(r)$. If $\chi_3 = 1$, orthogonality on H_j proves the assertion when $\chi_2 \neq 1$; when $\chi_2 = 1$, the average of a nontrivial χ_1 on $I_j \setminus \{0\}$ is eventually $-1/(|I_j| - 1)$.

Suppose $\chi_3 \neq 1$. Fix distinct $h_1, \dots, h_m \in k$, which belong to H_j for all sufficiently large j . For fixed $l \in I_j \setminus \{0\}$, set

$$f_l(s) = \chi_2(s)\chi_3((s^2 + P_\theta)/l).$$

The average of f_l equals the average of $m^{-1} \sum_a f_l(s + h_a)$. In the squared modulus of the latter expression, the correlation for $a \neq b$ is a constant of modulus one times the average on H_j of

$$s \mapsto \chi_3(2(h_a - h_b)s/l).$$

Choose $z_{ab} \in k$ with $\chi_3(2(h_a - h_b)z_{ab}) \neq 1$. Once all z_{ab} belong to I_j , the point $lz_{ab} \in H_j$ detects this character, uniformly for every allowed l . All these correlations are then zero. Cauchy–Schwarz gives $|\mathbb{E}_{s \in H_j} f_l(s)|^2 \leq 1/m$, uniformly in l . Letting m grow proves the character assertion. Polynomial avoidance follows from the product-grid assertion of Lemma 7.2. $\square$

Proposition 7.4. *In odd characteristic, μ -almost every palette satisfies (7.1). In fact, for each $h \in \mathcal{H}$, membership in $\mathcal{S}$ is constant on all positions xh , $x \in \mathcal{A}$.*

Proof. Fix a palette c , a position h , and the set

$$E = \{y \in \mathcal{A} : c_{yh} \notin \mathcal{S}\}.$$

For $x = (l_x, s_x) \notin E$, define the surjective linear map $L_x : k^3 \rightarrow k^2$ by

$$L_x(l, s, r) = \left(\frac{P_\theta - s_x^2}{l_x} l + 2s_x s - l_x r, 2s - \frac{2s_x}{l_x} l \right). \quad (7.5)$$

Its kernel is $kv(x)$. Substituting the parameters of yx^{-1} into Equation 6.14 gives exactly $L_x v(y)$. First suppose $c = c(p)$ is a rational palette, and let $w_x = u(xhp)$ be the unitary representative at position xh . The position yh is obtained from it by the scalar affine map yx^{-1} . Using the map $\mathcal{B}_{w_x} : k^2 \rightarrow J$ of Section 6, define

$$\nu_x = (\chi \mapsto \chi \circ \mathcal{B}_{w_x} \circ L_x)_* \sigma.$$

This positive measure on $\widehat{k^3}$ has mass δ and trivial atom at least δ^2 . Its Fourier transform satisfies $\widehat{\nu}_x(v(y)) = K(\mathcal{B}_{w_x}(L_x v(y)))$. Proposition 6.4 consequently gives

$$\widehat{\nu}_x(v(y)) = 0 \quad (y \in E), \quad \text{supp } \nu_x \subseteq (kv(x))^\perp. \quad (7.6)$$

The support assertion holds because every character in the image annihilates $\ker L_x$.

These measures exist for arbitrary palettes as well. Approximate c by rational palettes on successively larger finite sets of positions. For each fixed $x \notin E$, take a weak limit of the corresponding measures on the compact dual. The equations in (7.6) pass to the limit, as does support in the closed annihilator $(kv(x))^\perp$. The lower bound on the trivial atom persists because $\{1\}$ is closed.

We claim that either $E = \mathcal{A}$, or E has density zero. By Lemmas 7.1 and 7.3, a nonempty complement has lower density at least δ . If E had positive density along a subsequence, those lemmas would give a finite set of nontrivial characters such that, for every $x \notin E$, at least one of them annihilates $kv(x)$.

For any fixed nontrivial χ , this annihilation condition is a proper polynomial condition on (l_x, s_x) . To treat arbitrary additive characters, fix a primitive p -th root of unity and regard characters as $\mathbb{F}_p$ -linear functionals. The space $\text{Hom}_{\mathbb{F}_p}(k, \mathbb{F}_p)$ is a k -vector space under

$$(a\lambda)(t) = \lambda(at).$$

If χ has component functionals $\lambda_1, \lambda_2, \lambda_3$, annihilation of $kv(l, s)$ says

$$l\lambda_1 + s\lambda_2 + \frac{s^2 + P_\theta}{l}\lambda_3 = 0.$$

This is a rational vector equation in the finite-dimensional span of the three functionals. It is not an identity: the vectors $v(l, s)$ span k^3 , as follows on multiplying any putative linear relation among their coordinates by l . Clearing denominators therefore places its solutions in a proper polynomial locus. A finite union of these loci has density zero, contradicting the positive lower density of $\mathcal{A} \setminus E$. The claim follows.

It remains to turn this density dichotomy into actual invariance. Let $q_0 = |\mathcal{F} \setminus \mathcal{S}|/|\mathcal{F}|$. For every fixed h , uniform marginals give $\int \mathbb{E}_j \mathbf{1}_E d\mu = q_0$. The claim and dominated convergence show that $\mu(E = \mathcal{A}) = q_0$. For each $x \in \mathcal{A}$, the event $E = \mathcal{A}$ is contained in $\{c_{xh} \notin \mathcal{S}\}$, whose probability is also q_0 . Their difference is null. There are only countably many x and h , so outside a single null set membership is constant on every one of the asserted families of positions. $\square$

7.3 Characteristic two

Assume now that $\text{char } k = 2$, so $T \neq 0$. We shall prove scalar-translation invariance directly on rational points. A unitary element w belongs to the field $E = L[w] \subset D$, which is stable under $*$ because $w^* = w^{-1}$. With $F = E^{(*)}$, we have $E = LF$. The degree $[E : L]$ divides the odd degree of D , so E/L , and hence F/k , is separable. The Cayley chart identifies $(E/F)^1$ with $F \cup \{\infty\}$.

We first prove a density statement on each of these fields. Throughout this subsection, densities on k are taken over an arbitrary fixed increasing sequence of finite additive subgroups exhausting k .

Lemma 7.5. *Let $F \subset D$ be as above, let $Y \in F$, and let $H_0 \in k[X]$ be a nonconstant additive polynomial. For*

$$w_x = q(Y + H_0(x)), \quad x \in k,$$

either $\phi(w_x) \notin \mathcal{S}$ for every x , or the set of such x has density zero.

Proof. Put $E_0 = \{x : \phi(w_x) \notin \mathcal{S}\}$ and $Q_0 = H_0^2 + TH_0$. The polynomial Q_0 is nonconstant and additive. Equation 6.14, applied to scalar translations, gives

$$K(B(x)Q_0(y - x)) = 0 \quad (x \notin E_0, y \in E_0), \tag{7.7}$$

where

$$B(x) = \frac{w_x + w_x^*}{T} = \frac{T}{(Y + H_0(x))^2 + T(Y + H_0(x)) + P_\theta} \in J. \tag{7.8}$$

In any fixed k -basis of J , the coordinates of $B(x)$ are rational functions of x tending to zero at infinity. This follows also directly by expanding the inverse denominator in $F((x^{-1}))$. The denominator never vanishes for $x \in k$, since its two factors are $Y + H_0(x) + \theta$ and $Y + H_0(x) + \bar{\theta}$, and $\theta \notin F$.

Push σ forward by the character map

$$\chi \longmapsto [y \longmapsto \chi(B(x)Q_0(y))]$$

to obtain a measure ν_x on $\widehat{k}$. It has mass δ and trivial atom at least δ^2 . Equation (7.7) is precisely the vanishing required by Lemma 7.1, with $v(y) = y$ and $a_x = x$. Ordinary orthogonality on exhausting additive subgroups supplies (7.2). Thus, if $k \setminus E_0 \neq \emptyset$, it has positive lower density. If E_0 also had density tending to some $d > 0$ along a subsequence, there would be a fixed finite set $\Lambda \subset \widehat{k} \setminus \{1\}$ such that

$$\sigma\{\chi : \chi(B(x)Q_0(\cdot)) \in \Lambda\} \geq d\delta^2/2 \quad (x \notin E_0). \quad (7.9)$$

We prove that, for every fixed $\chi \in \widehat{J}$ and every fixed nontrivial $\lambda \in \widehat{k}$, the equality

$$\chi(B(x)Q_0(\cdot)) = \lambda \quad (7.10)$$

holds on a set of density zero. This is the point where the additive polynomial and the use of arbitrary additive characters both matter. Regard the characters as $\mathbb{F}_2$ -linear functionals, with the k -vector space structure used in the preceding subsection. Write $Q_0(y) = \sum_i a_i y^{2^i}$, and choose $M = 2^r \geq \deg Q_0$. The extension k/k^M is finite. For a basis $e_1, \dots, e_N$ over k^M , the representation

$$x = \sum_{j=1}^N e_j x_j^M$$

is unique and defines an additive group isomorphism $k^N \rightarrow k$. Let $b_\ell(x)$ be the coordinates of $B(x)$, and χ_ℓ the component functionals of χ . For each i , choose a basis $(f_{im})_m$ of k/k^{2^i} . Expansion in that basis gives rational functions $R_{\ell im} \in k(X_1, \dots, X_N)$ such that

$$a_i b_\ell \left(\sum_j e_j X_j^M \right) = \sum_m f_{im} R_{\ell im}(X)^{2^i}. \quad (7.11)$$

Indeed the left side belongs to $k(X_1^{2^i}, \dots, X_N^{2^i})$, and this field has basis $(f_{im})_m$ over $k^{2^i}(X_1^{2^i}, \dots, X_N^{2^i})$.

Define fixed functionals $\psi_{\ell im}(y) = \chi_\ell(f_{im} y^{2^i})$. The left side of (7.10), in functional notation, is now

$$\sum_{\ell, i, m} R_{\ell im}(X) \psi_{\ell im}. \quad (7.12)$$

It is a rational map into a finite-dimensional k -vector space. Moreover, it tends to zero when all X_j are multiplied by a common indeterminate tending to infinity. For completeness, the left side of (7.11) tends to zero under this scaling because each b_ℓ vanishes at infinity. If some $R_{\ell im}$ did not tend to zero, the terms of largest Laurent degree on the right could not cancel: the f_{im} are linearly independent over $k^{2^i}(X_1^{2^i}, \dots, X_N^{2^i})$. Thus every one of the rational functions in (7.12) tends to zero.

It follows that this rational map cannot be identically equal to the fixed nonzero functional λ . If λ is outside its finite-dimensional span the equality is impossible; otherwise, taking coordinates and clearing denominators puts its solutions in a proper polynomial locus. Any exceptional denominator locus is proper as well. Lemma 7.2, transported through the additive isomorphism $k^N \rightarrow k$, proves the asserted density zero.

Average (7.9) over x . Its right side, restricted to $k \setminus E_0$, has positive lower average. Its left side has average tending to zero: for each fixed χ , the finite union of the sets (7.10), with $\lambda \in \Lambda$, has density zero, and dominated convergence applies to the finite measure σ . This contradiction proves the density dichotomy. $\square$

The density statement alone allows exceptional points. The abelian structure of the colors on a field removes them.

Proposition 7.6. *In characteristic two, membership in $\mathcal{S}$ is unchanged by every scalar translation on $X(k)$. Consequently every palette satisfies (7.1).*

Proof. Fix $E = LF$ as above and let $A_E = \phi((E/F)^1)$, a finite abelian subgroup of $\mathcal{F}$. If $A_E \cap \mathcal{S}$ is empty there is nothing to prove. Otherwise Theorem 6.1 says that $A_E \setminus \mathcal{S}$ is a subgroup, say A_E^0 . Define the homomorphism, with additive target notation,

$$\psi : E^\times \longrightarrow A_E/A_E^0, \quad \psi(a) = \phi(a/\bar{a}) \bmod A_E^0,$$

where bar denotes the involution on E . Membership of $\phi(q(Z))$ in $\mathcal{S}$ is equivalent to $\psi(Z + \theta) \neq 0$.

For $Y \in F$, consider the finite-valued function $g(s) = \psi(Y + s + \theta)$ on k . We show that its additive period group

$$\Pi = \{t \in k : g(s + t) = g(s) \text{ for every } s \in k\}$$

has finite index. For $t \neq 0$, agreement of these two values is equivalent to vanishing of ψ on

$$\begin{aligned} (Y + s + t + \theta)(Y + s + \bar{\theta}) \\ = Y^2 + (T + t)Y + P_\theta + s^2 + (T + t)s + t\bar{\theta}. \end{aligned}$$

Here $\psi(\bar{a}) = -\psi(a)$, so the product measures the difference of the two values of g . Division by $t \in k^\times$ does not change ψ . With

$$Z_t = \frac{Y^2 + (T + t)Y + P_\theta}{t}, \quad H_t(s) = \frac{s^2 + (T + t)s}{t},$$

the agreement condition becomes $\psi(Z_t + H_t(s) + \bar{\theta}) = 0$. Replacing $\bar{\theta}$ by θ negates this value and preserves its zero set. The polynomial H_t is nonconstant and additive, so Lemma 7.5 says that either agreement holds for every s , or its set has density zero. Thus every $t \notin \Pi$ gives density-zero agreement.

Choose translations of g indexed by distinct cosets of Π . Any two agree on a set of density zero, because the averages are eventually invariant under each fixed additive shift. If there were more such translates than values of g , then at every point two would agree. A finite union of density-zero sets cannot cover k . This proves that $[k : \Pi]$ is finite.

Finally apply Lemma 7.5 with $H_0(s) = s$. The zero set of g is either all of k or has density zero. Since g is constant on cosets of Π , any nonempty zero set contains a coset of density $[k : \Pi]^{-1}$. Hence g is either identically zero or nowhere zero. Membership is therefore constant on $\{q(Y + s) : s \in k\}$. Every finite rational point belongs to such a field chart, and ∞ is fixed by translations. This proves the rational-point assertion. For fixed $s \in k$ and $h \in \mathcal{H}$, the equality of membership indicators $\mathbf{1}_{\mathcal{S}(c_{\tau_s h})} = \mathbf{1}_{\mathcal{S}(c_h)}$ is a closed condition. It holds for every rational palette, hence for every palette in their closure. $\square$

7.4 Generating rotations

In either characteristic, choose a palette c satisfying (7.1). Define

$$\mathcal{G}_c = \{g \in \mathcal{H} : I_c(gh) = I_c(h) \text{ for every } h \in \mathcal{H}\}.$$

This is a subgroup. In addition to all τ_s , it contains left rotations L_a with $\phi(a) = 1$, and all unitary conjugations: these assertions follow from the palette rotation identities and the conjugacy invariance of $\mathcal{S}$. We now explain how these transformations generate squares of arbitrary separable field rotations. The construction adapts [23, Section 5.6], with the field-spanning step made explicit in positive characteristic.

Lemma 7.7. *Let $E = LF \subset D$ be a field stable under $*$, with $F = E^{(*)}$ and E/L separable. Then $L_{\xi^2} \in \mathcal{G}_c$ for every $\xi \in (E/F)^1$.*

Proof. We first obtain all F -translations, then use a central rotation to generate $\mathrm{SL}_2(F)$, and finally express a square rotation in Cayley coordinates. Choose $n_0 = q(x_0) \in (E/F)^1 \cap \ker \phi$, with x_0 generating F/k , such that the polynomials

$$(s + x_i + \theta)(s + x_i + \bar{\theta}) \quad (7.13)$$

have pairwise disjoint root sets as x_i runs over the conjugates of x_0 under embeddings fixing L . These are nonempty Zariski open conditions: over a splitting field the conjugate coordinates of $\mathrm{Res}_{F/k} \mathbb{A}^1$ are independent, and all prohibited equalities are proper linear conditions. They can be met inside $\ker \phi$ because powers annihilating $\mathcal{F}$ are Zariski dense in the norm-one torus. Indeed its rational points are Zariski dense by the Cayley parametrization, and every positive power map is dominant, even when it is inseparable.

Since the color set is finite, there are $t_0 \in k$ and an infinite set $\Sigma \subset k$ for which $\phi(\tau_s(n_0)) = \phi(\tau_{t_0}(n_0))$ for every $s \in \Sigma$. Here translations on unitary elements are understood through the Cayley chart. Put

$$g_s = L_{n_0}^{-1} \tau_s^{-1} L_{\tau_s(n_0) \tau_{t_0}(n_0)^{-1} \tau_{t_0}} L_{n_0} \in \mathcal{G}_c.$$

Each g_s fixes the unitary representative 1, which is the point ∞ in additive coordinates. Its fractional matrix has entries in E and is upper triangular. To compute its diagonal entries, the exact matrix of $L_q(z)$, for $z \in F$, is

$$M_z = (z + \bar{\theta})^{-1} \begin{pmatrix} z & -P_\theta \\ 1 & z + T \end{pmatrix}.$$

The factor $(z + \bar{\theta})^{-1}$ is retained here because it need not be central in D . Multiplying the matrices in the expression for g_s gives diagonal entries

$$a_s = \frac{x_0 + t_0 + \bar{\theta}}{x_0 + s + \bar{\theta}}, \quad d_s = \frac{x_0 + s + \theta}{x_0 + t_0 + \theta}.$$

Thus its affine multiplier on the field line $F \cup \{\infty\}$ is

$$m_s = \frac{(x_0 + t_0 + \theta)(x_0 + t_0 + \bar{\theta})}{(x_0 + s + \theta)(x_0 + s + \bar{\theta})} \in F^\times. \quad (7.14)$$

This is also the ratio of the logarithmic derivatives of q at $x_0 + s$ and $x_0 + t_0$.

The same multiplier controls conjugation of scalar translations on the whole chart J . Indeed an upper triangular fractional matrix $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ acts by $x \mapsto (ax + b)d^{-1}$. Conjugating $x \mapsto x + r$, $r \in k$, gives $x \mapsto x + rad^{-1}$. Here $ad^{-1} = m_s$, so $\mathcal{G}_c$ contains translations by km_s .

The elements m_s , $s \in \Sigma$, span F over k . Otherwise a nonzero k -linear functional on F would annihilate them. After passing to a normal closure, such a functional is a linear combination of the distinct embeddings of F . It would give a linear relation among the rational functions

$$s \mapsto \frac{(x_i + t_0 + \theta)(x_i + t_0 + \bar{\theta})}{(x_i + s + \theta)(x_i + s + \bar{\theta})}$$

at infinitely many s , and therefore an identity. Each summand has a pole that occurs in no other summand by (7.13); its numerator is nonzero. Every coefficient must consequently vanish, a contradiction. Composing the translations already obtained now gives every F -translation on J .

For a central rotation, take the particular scalar $a = q(0) = \theta/\bar{\theta} \neq 1$. Its color is trivial, so $L_a \in \mathcal{G}_c$. In additive coordinates it acts by $x \mapsto -P_\theta(x + T)^{-1}$. Consequently $W = L_a\tau_{-T}$ is represented by $\begin{pmatrix} 0 & -P_\theta \\ 1 & 0 \end{pmatrix}$, and direct multiplication gives

$$W \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} W^{-1} = \begin{pmatrix} 1 & 0 \\ -b/P_\theta & 1 \end{pmatrix} \quad (b \in F).$$

Thus all lower as well as upper unipotent matrices over F act in $\mathcal{G}_c$. Hence $\mathcal{G}_c$ contains the fractional action on J of $\mathrm{SL}_2(F)$, since upper and lower elementary matrices generate that group.

Finally set

$$C = \begin{pmatrix} 1 & \theta \\ 1 & \bar{\theta} \end{pmatrix}.$$

For $\xi \in (E/F)^1$, the matrix

$$C^{-1} \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix} C = \frac{1}{\bar{\theta} - \theta} \begin{pmatrix} \bar{\theta}\xi - \theta\xi^{-1} & P_\theta(\xi - \xi^{-1}) \\ \xi^{-1} - \xi & \bar{\theta}\xi^{-1} - \theta\xi \end{pmatrix} \quad (7.15)$$

has all entries in F and determinant one. It belongs to the $\mathrm{SL}_2(F)$ just generated. In unitary coordinates its action is $u \mapsto \xi u \xi$, as is seen by applying the diagonal matrix to the column $(x + \theta, x + \bar{\theta})^\dagger$. Composing this action with the unitary conjugation $u \mapsto \xi u \xi^{-1}$ gives $u \mapsto \xi^2 u$. This is the required left rotation. $\square$

Proposition 7.8. *There is no homomorphism $\phi : U(k) \rightarrow \mathcal{F}$ to a finite nonabelian simple group that kills R and satisfies $\phi(V_0) = \mathcal{F}$.*

Proof. Every color is represented by an element $\xi \in V_0$ lying in a separable field of the kind used in Lemma 7.7. Indeed, start with any representative and perturb it within its R -coset to a regular semisimple element, using Proposition 2.4 and the nonempty local regular semisimple open subset. In a division algebra its generated algebra over L is a field, stable under the involution, and regular semisimplicity makes that field separable. This also covers characteristics dividing the odd degree n .

Lemma 7.7 and the left-rotation palette rule therefore make the membership indicator invariant under left multiplication by the square of every element of $\mathcal{F}$. These squares generate $\mathcal{F}$: their generated subgroup is normal, and if it were trivial then $\mathcal{F}$ would have exponent two and hence be abelian. Since ϕ is surjective, the positions obtained from any one position by left rotations realize every color. The indicator must be constant on $\mathcal{F}$, contrary to the choice of the nonempty proper set $\mathcal{S}$. $\square$

8 Excluding finite abelian characters

Throughout this section, D is a central division algebra of odd degree $n \geq 3$ over the quadratic extension L/k , with unitary involution $*$. We write $U = \mathrm{U}(D, *)$ and $S = \mathrm{SU}(D, *)$. The remaining alternative in Proposition 5.1 is excluded by the following result.

Theorem 8.1. *Every homomorphism from $U(k)$ to a finite abelian group is trivial on $S(k)$.*

The difference-function argument below adapts the corresponding argument of [23, Section 4]. Its two arithmetic ingredients have particularly useful function-field forms: every nonsingular Hermitian element admits a scalar normalization, and the additive span of $U(k)$ is all of D . The latter is an equality of abstract additive groups. Local density alone would not suffice for the

exact transformations used here. Fix a homomorphism $\chi : U(k) \rightarrow B$, with B finite and written additively. We construct a function on D whose differences are controlled by χ , and prove that exact transformations of pairs make those differences constant. Scalar normalization then gives a function on all of $D^\times$, invariant under a noncentral normal subgroup of the inner group $\mathrm{SL}_1(D)(L)$. The known inner-type theorem turns this invariance into continuity of $\chi|_{S(k)}$ at A ; local unitary commutators then force its vanishing.

8.1 Hermitian normalization and a cyclic maximal field

We first arrange a commutative field inside D on which both conjugation and the involution can be calculated explicitly.

Lemma 8.2. *If $h = h^* \neq 0$ and $s = \mathrm{Nrd}(h) \in k^\times$, there exists $a \in D^\times$ such that*

$$a^*a = sh, \quad \mathrm{Nrd}(a) = s^{(n+1)/2}. \quad (8.1)$$

There is also a $$ -stable maximal subfield $E \subset D$ of the form $E = LF$, where F/k is cyclic of degree n and $*$ restricts to the nonidentity automorphism of E/F .*

Proof. The variety defined by Equation 8.1 is a torsor under S : over a separable closure the Hermitian equation has a solution, and the prescribed determinant is compatible because

$$\mathrm{Nrd}(sh) = s^{n+1} = N_{L/k}(s^{(n+1)/2}).$$

The simply connected H^1 -vanishing theorem over global function fields therefore gives a rational point [15, 4]. This applies without any restriction on the characteristic or on its divisibility into n .

The cyclic approximation theorem for global fields supplies a cyclic extension F/k of degree n with full local degree at each place below the Brauer support of D [18, Theorems 1–2 and Section 4]. Prescribing a degree- n local field at one of these places ensures that the global cyclic Galois algebra is a field. The characteristic-primary part is covered by Artin–Schreier–Witt approximation, and the Wang condition is automatic in positive characteristic. Since n is odd, F and L are disjoint. The local invariant criterion shows that LF splits D . Its degree over L is n , so the embedding criterion gives an embedding $\alpha : LF \hookrightarrow D$ [35].

Let bar on LF fix F and induce the nonidentity automorphism of L/k . Skolem–Noether provides $h \in D^\times$ with

$$\alpha(\bar{z})^* = h\alpha(z)h^{-1} \quad (z \in LF).$$

Taking adjoints shows that h^* is another intertwiner, whence $c = h^{-1}h^* \in \alpha(LF)^\times$. On this field $x^* = h\bar{x}h^{-1}$, and consequently $\bar{c} = c^{-1}$. Hilbert 90 gives $t \in \alpha(LF)^\times$ with $t/\bar{t} = c$. Replacing h by ht makes it Hermitian, since $(ht)^* = h\bar{t}c = ht$, and preserves the intertwining identity. Choose a as in Equation 8.1 for this h . Then the embedding $z \mapsto a\alpha(z)a^{-1}$ is $*$ -stable:

$$(a\alpha(\bar{z})a^{-1})^* = (a^*)^{-1}h\alpha(z)h^{-1}a^* = a\alpha(z)a^{-1}.$$

Its image is the required field E . □

Write $\bar{z} = z^*$ on E , and put $E^1 = \{z \in E^\times : z\bar{z} = 1\}$. This group has infinitely many points, by the Hilbert 90 parametrization. We shall use parameters $d \in E^\times$ with the following properties:

$$\begin{aligned} t_1, \dots, t_n, \text{ the } L\text{-conjugates of } d, \text{ are distinct,} \\ \{t_1, \dots, t_n\} \cap \{\bar{t}_1, \dots, \bar{t}_n\} = \emptyset, \quad d\bar{d} \notin k. \end{aligned} \quad (8.2)$$

The last condition implies $t_j\bar{t}_j \neq 1$ for every j . It ensures separability of a degree-two parameter map even in characteristic two.

8.2 A difference function and its exact identities

Let $\mathcal{D}$ be the set of nonzero $b \in D$ such that $b + b^* = a^*a$ for some $a \in D$. Define

$$m_b = 1 - ab^{-1}a^*, \quad f(b) = \chi(m_b). \quad (8.3)$$

Multiplication gives $m_b m_b^* = 1$. If $a'^*a' = a^*a \neq 0$, then $a' = va$ for $v \in U(k)$, so the resulting elements m_b are conjugate. When $a = 0$, $m_b = 1$. Thus f is well-defined, including on skew elements. Using ax in the defining equation also gives

$$f(x^*bx) = f(b) \quad (x \in D^\times). \quad (8.4)$$

Fix d satisfying Equation 8.2. For $(a, c) \in D^2 \setminus \{(0, 0)\}$ there is a unique $e \in D$ with

$$e + e^* = c^*c, \quad ed + d^*e^* = a^*a. \quad (8.5)$$

Indeed, solve the linear equation

$$ed - d^*e = a^*a - d^*c^*c.$$

After splitting D , its linear part is a Sylvester operator whose eigenvalues are the differences of the disjoint reduced spectra of d and d^* . It is therefore invertible. Taking adjoints shows that $c^*c - e^*$ is a second solution of the same linear equation, proving both identities in Equation 8.5. The solution is nonzero, since $e = 0$ would force $a = c = 0$. We may consequently define

$$F_d(a, c) = f(ed) - f(e).$$

Lemma 8.3. *The function F_d is invariant under independent left multiplications of its two arguments by elements of $U(k)$ and under the transformation*

$$(a, c) \mapsto (a - cd, a - c). \quad (8.6)$$

Proof. The unitary multiplications leave Equation 8.5 unchanged. For the pair transformation put $b = ed$ and $z = b^* + e - a^*c$. Expanding in the indicated order gives

$$z + z^* = (a - c)^*(a - c), \quad zd + d^*z^* = (a - cd)^*(a - cd).$$

The pair transformation is invertible, since $d \neq 1$, so $z \neq 0$. Set $P = ce^{-1} - ab^{-1}$ and $Q = b^{-*}ze^{-1}$, where $b^{-*} = (b^*)^{-1}$. Direct multiplication gives

$$P^*P = Q + Q^*, \quad Q = (b^{-1})^*(zd)b^{-1}.$$

The multiplicative identity that relates their characters is

$$1 - PQ^{-1}P^* = m_b m_z m_e^*, \quad (8.7)$$

where the defining vectors for m_b, m_z, m_e are $a, c - a, c$, respectively. One can check the identity without commuting any factors: use $Q^{-1} = ez^{-1}b^*$ and

$$\begin{aligned} Pe &= -ab^{-1}z + m_b(c - a), \\ b^*P^* &= ze^{-*}c^* + (c - a)^*m_e^*. \end{aligned}$$

In multiplying these expressions, the terms not containing $-m_b(c - a)z^{-1}(c - a)^*m_e^*$ sum to $m_b m_e^*$; here one uses $m_b a = -ab^{-1}b^*$ and $c^*c = e + e^*$. By congruence invariance the left side of Equation 8.7 has character $f(zd)$. Hence $f(zd) = f(b) + f(z) - f(e)$, which is precisely the assertion. $\square$

To make this invariance useful, we must show that the transformations act transitively on nonzero pairs. We first prove the additive assertion that will turn one nonzero elementary shear into arbitrary addition.

Lemma 8.4. *The additive subgroup generated by $U(k)$ equals D :*

$$\mathbb{Z}U(k) = D.$$

Proof. Put $M = \mathbb{Z}U(k)$; it is a subring because $U(k)$ is a group. Choose a place of k split in L . Hilbert 90 and weak approximation give a central phase $z \in L^1$ with a simple pole at one of the two places above it and a simple zero at the other. Then $t_0 = z + z^{-1} \in k \cap M$ has a simple pole at the chosen place. In particular k/k_0 is finite separable for $k_0 = \mathbb{F}_p(t_0)$, where $p = \text{char}(k)$. The group M is a module over $R = \mathbb{F}_p[t_0]$.

Fix a finite prime π of R , let K be the corresponding completion of k_0 , and let C be the additive closure of M in $D_K = D \otimes_{k_0} K$. It is a closed module over the valuation ring $\mathcal{O}_K$. Weak approximation for U puts the product of all its local groups above π inside C . The largest K -vector subspace contained in C is

$$W = \bigcap_{r \geq 0} \pi^r C.$$

Indeed the right side is closed, is an $\mathcal{O}_K$ -module, and is stable under division by π . Left and right multiplication by each local unitary group preserve W .

Their K -linear span is the full algebra D_K . To check this, view $\text{Res}_{k/k_0} U$ as a group over k_0 and extend scalars from K to an algebraic closure. Separability of k/k_0 and L/k gives independent pairs of matrix factors, on which a unitary element has the form (g, g^{-t}) with $g \in \text{GL}_n$. These pairs span both matrix factors: varying a scalar multiple of g separates the weights λ and λ^{-1} , and invertible matrices span M_n . Independent factors give the whole product. The local points of the smooth unitary groups are Zariski dense, so the same linear-span assertion holds over K . It follows that W is a two-sided ideal of D_K .

Every simple factor occurs in this ideal. At a nonsplit place of k the algebra over L splits and the unitary group is isotropic because $n \geq 3$. Take an unbounded sequence in that local group and the identity in every other component. For suitable integers $r_\nu \rightarrow \infty$, multiplication by π^{r_ν} gives a subsequence converging to a nonzero vector in that factor and to zero elsewhere. For every fixed r , this sequence eventually belongs to $\pi^r C$; closedness therefore puts its limit in W . At a split place, the unitary group contains reciprocal central scalars. The same argument with (π^{-r}, π^r) , and then with the factors reversed, isolates each of the two simple factors. Thus $W = D_K$, and $C = D_K$.

It remains to pass from all these local closures to an exact equality. The k_0 -linear span of M is D , since a proper finite-dimensional subspace would remain proper after completion. Choose a full R -lattice $\Lambda \subset M$. For $x = \pi^{-j}\lambda$, with $j \geq 0$ and $\lambda \in \Lambda$, local density gives $a \in M$ with $a - x \in \Lambda \otimes_R \mathcal{O}_K$. Choose a polynomial $q \in R$ which clears all denominators of a away from π and satisfies $q \equiv 1 \pmod{\pi^j}$. The Chinese remainder theorem permits both conditions. Then $qa - qx$ belongs to the local lattice at every finite prime of R , hence to Λ , and $(q - 1)x \in \Lambda$. It follows that $x \in M$. Partial fractions now give every element of $k_0\Lambda = D$ inside M , proving the lemma. $\square$

8.3 Separating the blocks by actual shears

Skolem–Noether gives elements $w_j \in D^\times$ and a decomposition

$$D = \bigoplus_{j=1}^n Ew_j, \quad w_j d = t_j w_j.$$

Thus $D \oplus D$ is the direct sum of n two-dimensional E -blocks. Let H be the subgroup of the invariance group of F_d generated by $\text{diag}(\lambda, \mu)$, with $\lambda, \mu \in E^1$, acting simultaneously on every block, and by the simultaneous matrices

$$B_h(t_j) = \begin{pmatrix} 1 & t_j \bar{h} \\ h & 1 \end{pmatrix} \quad (h \in E^1). \quad (8.8)$$

The phases act independently on the two D coordinates; no independent choice between blocks is assumed. These transformations are available: changing the sign of the second output in Equation 8.6 gives B_{-1} , and diagonal phase conjugation gives every B_h . Their determinants $1 - t_j$ are nonzero.

Lemma 8.5. *For every block i , H contains a nonidentity upper elementary unipotent supported only on block i , and a nonidentity lower elementary unipotent supported only on block i .*

Proof. We first construct a transformation triangular at a single block. The first-column slope of $B_h(t)B_{hz}(t)$ is $h(1+z)/(1+tz)$. At $t = t_j$, its norm from E to F is

$$\ell_j(z) = \frac{(1+z)^2}{(1+t_j z)(z + \bar{t}_j)}. \quad (8.9)$$

This is a separable rational map of degree two. In characteristic two, its derivative is nonzero because $t_j \bar{t}_j \neq 1$; in other characteristics separability is automatic. For a generic $z \in E^1$, the other point of the fiber is

$$z' = \frac{(t_j - \bar{t}_j)z + 1 + t_j \bar{t}_j - 2\bar{t}_j}{(1 + t_j \bar{t}_j - 2t_j)z + \bar{t}_j - t_j}.$$

It is distinct from z and lies in E^1 : conjugate reciprocation preserves the two roots and fixes z , so also fixes z' . Writing $s_t(z) = (1+z)/(1+tz)$, choose $h' = s_{t_j}(z)/s_{t_j}(z') \in E^1$ and set

$$g_j(t) = (B_{h'}(t)B_{h'z'}(t))^{-1}B_1(t)B_z(t).$$

Its lower-left numerator is

$$(1 + tz')(1 + z) - h'(1 + z')(1 + tz).$$

This nonzero linear polynomial has its unique root at t_j ; it is nonzero as a polynomial because the two fractional slope functions have distinct poles. Thus this entry is nonzero at every other t_ℓ and every $\bar{t}_\ell$.

Matrices of the form

$$\begin{pmatrix} \alpha(t) & t\bar{\beta}(t) \\ \beta(t) & \bar{\alpha}(t) \end{pmatrix},$$

with bar acting on coefficients and fixing t , are closed under multiplication and inversion. Therefore the upper-right entry of $g_j(t)$ is nonzero at all t_ℓ , including t_j . All diagonal entries can also be kept nonzero. Indeed at $z = -1$ the fiber interchange in Equation 8.9 fixes -1 and has derivative -1 (equal to 1 in characteristic two). Hence $h' \rightarrow -1$, both boost products tend to $(1-t)I$, and $g_j(t) \rightarrow I$. The excluded conditions are consequently proper algebraic conditions on the rational norm-one curve, whose rational points are infinite. We obtain an upper triangular nondiagonal matrix at j and a matrix with all four entries nonzero at each other block. Using $\bar{t}_j$ in the same construction gives the lower triangular analogue.

Fix i . For $j \neq i$, the subgroup generated by g_j and diagonal phases is triangular at j and projectively Zariski dense at i . For the second assertion, its closure contains the diagonal torus and a matrix with four nonzero entries; a proper algebraic subgroup of PGL_2 containing that torus lies

in a Borel subgroup or in the normalizer of the torus, and neither contains such a matrix. The second derived subgroup is trivial at j but remains projectively dense at i , by algebraic perfectness of PGL_2 .

Use the full normal kernels

$$K_j = \ker(H \longrightarrow \mathrm{GL}_2(E) \text{ on block } j).$$

Each K_j has projectively dense image at i . Successive commutators of these normal kernels are supported on the intersection of their supports and still have projectively dense image at i , since the commutator map is regular and its image generates the derived algebraic group PGL_2 . More explicitly, successive groups $N_{r+1} = [N_r, K_j]$, beginning with one K_j , remain in every kernel already used because those kernels are normal. Consequently the image P_i of $\bigcap_{j \neq i} K_j$ is projectively dense and is normal in the full image H_i of H at i .

We next produce actual unipotents in P_i . Commutators of the triangular matrices at i with diagonal phases give upper and lower elementary unipotents in H_i with nonzero coefficients. Conjugating by phases and adding coefficients shows that both coefficient groups contain scalar multiples of $\mathbb{Z}E^1$. We claim that they contain a common nonzero ideal in a suitable ring of S -integers of E . Choose a place of F split in E and a phase z with a simple pole above it, as in the proof of Lemma 8.4. Then $t_0 = z + z^{-1} \in F$ has a simple pole, so $E/\mathbb{F}_p(t_0)$ is finite separable and $\mathbb{Z}E^1$ is a module over $R = \mathbb{F}_p[t_0]$. The k_0 -linear span of E^1 is E , where now $k_0 = \mathbb{F}_p(t_0)$. Indeed the group $\mathrm{Res}_{F/k_0}(\ker(N_{E/F} : \mathrm{Res}_{E/F} \mathbb{G}_m \rightarrow \mathbb{G}_m))$ has rational points E^1 . After extending scalars to an algebraic closure of k_0 , its coordinates are independent pairs (x_ℓ, x_ℓ^{-1}) . A linear relation $\sum_\ell (a_\ell x_\ell + b_\ell x_\ell^{-1}) = 0$ forces all coefficients to vanish, so these pairs span their ambient product. Hilbert 90 gives Zariski density of its rational points. Thus $\mathbb{Z}E^1$ contains a full R -lattice. If $\mathcal{O}$ is the integral closure of R in E , any two scalar multiples of such a lattice contain a common nonzero ideal $I \subset \mathcal{O}$: clear the finitely many denominators of generators of the finite R -module $\mathcal{O}$ relative to both lattices. There are distinct places P, Q of E above the pole of t_0 . The nonzero divisor $\deg(Q)P - \deg(P)Q$ has degree zero. Finiteness of the degree-zero divisor class group over the finite constant field gives a positive multiple equal to the divisor of a nonconstant function. That function is an infinite-order unit of $\mathcal{O}$ [37, Chapter 14].

Choose $g \in P_i$ carrying ∞ to a finite point r of $\mathbb{P}^1(E)$. Fix $0 \neq s \in I$ and choose an infinite-order unit $u \in \mathcal{O}^\times$ sufficiently congruent to 1 that

$$u = 1 + st \quad \text{for some } t \in I, \quad (1 - u^2)r \in I.$$

A power of an infinite-order unit satisfies the finitely many required congruences. With $U(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ and $L(x) = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}$, the identity

$$U(s)L(t)U(-s/u)L(-tu) = \mathrm{diag}(u, u^{-1})$$

shows that H_i contains the affine transformation $h : x \mapsto u^2x + (1 - u^2)r$. It fixes r and ∞ . Normality puts $q = h^{-1}g^{-1}hg$ in P_i . This element fixes ∞ and has affine slope u^{-4} : the derivatives of h at ∞ and r are u^{-2} and u^2 , respectively. Hence, for $0 \neq x \in I$,

$$qU(x)q^{-1}U(-x) = U((u^{-4} - 1)x)$$

is a nonidentity upper unipotent in P_i . By definition of P_i it lifts to a transformation supported only on block i . Interchanging the two axes proves the lower assertion. $\square$

Proposition 8.6. *For $d \in E^\times$ satisfying Equation 8.2, the invariance group of F_d is transitive on $D^2 \setminus \{(0,0)\}$. Consequently, whenever $b, bd \in \mathcal{D}$,*

$$f(bd) - f(b) = \chi(\bar{d}/d). \quad (8.10)$$

Proof. If $c \neq 0$, one of its E -block coordinates is nonzero. The pure upper shear from Lemma 8.5 changes a by some fixed nonzero $y \in D$, leaving c unchanged. Conjugating by left multiplication of the first coordinate by $v \in U(k)$ replaces this increment by vy . Products and inverses therefore add every element of $(\mathbb{Z}U(k))y = D$, by Lemma 8.4. The lower shears similarly permit arbitrary changes of c when $a \neq 0$. These operations connect every nonzero pair to one with both entries nonzero, and connect any two such pairs. Thus F_d is constant.

Take $e = (d - \bar{d})^{-1}$, $c = 0$ and $a = 1$ in Equation 8.5. The elements defining $f(e)$ and $f(ed)$ are 1 and $\bar{d}/d$, respectively. This evaluates the constant. Any $b, bd \in \mathcal{D}$ yield a pair (a, c) through their defining equations; it is nonzero since the Sylvester operator has zero kernel. This proves Equation 8.10. $\square$

8.4 From the difference identity to continuity

Lemma 8.2 extends f to every nonzero element of D . For $b + b^* \neq 0$, choose $s \in k^\times$ with $sb \in \mathcal{D}$ and put $f^\#(b) = f(sb)$; for skew b put $f^\#(b) = 0$. This does not depend on the choice of s . If both sb and tb are permitted, reduced norms of their defining Hermitian equations show that $(t/s)^n$ is a norm from L . The quotient $k^\times/N_{L/k}(L^\times)$ has exponent two, so oddness of n implies that t/s itself is a norm. Central congruence in Equation 8.4 then gives the same value. Consequently

$$f^\#(x^*bx) = f^\#(b), \quad f^\#(sb) = f^\#(b) \quad (x \in D^\times, s \in k^\times). \quad (8.11)$$

Let $m \geq 1$ annihilate B , and let $T = \ker(N_{E/L} : \text{Res}_{E/L} \mathbb{G}_m \rightarrow \mathbb{G}_m)$. Choose $d \in T(L)^m$ satisfying Equation 8.2. Such choices are Zariski dense in a nonempty open subset of $\text{Res}_{L/k} T$. Over a separable closure of k , the two embeddings of L give two independent lists $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$, each with product one. These determinant-one eigenvalue lists can avoid all stated equalities, and $n \geq 3$ also allows their coordinatewise products to be unequal. Hilbert 90 for the cyclic extension E/L gives Zariski density of $T(L)$, and the power map is dominant even when $p \mid m$.

For every such d and every $b \in D^\times$,

$$f^\#(bd) = f^\#(b). \quad (8.12)$$

Indeed normalize b and bd separately into $\mathcal{D}$. Their ratio replaces d by a nonzero k -multiple, preserving Equation 8.2 and the ratio $\bar{d}/d$. If $d = y^m$, then $\bar{d}/d = (\bar{y}/y)^m$ has zero character, so Proposition 8.6 applies. Congruence invariance extends Equation 8.12 to every $D^\times$ -conjugate of d .

Let J be the subgroup of $\text{SL}_1(D)(L)$ generated by all these conjugates. It is noncentral and normal. The established inner-type Margulis–Platonov theorem over L implies that J contains every rational point sufficiently close to 1 at the places

$$A_D = \{w : D \otimes_L L_w \text{ is division}\}$$

[28, Sections 3.4–3.5][32]. These are precisely the two places of L over each place of A . Since $f^\#$ is defined on all of $D^\times$, products of the invariant steps give

$$f^\#(br) = f^\#(b) \quad (b \in D^\times, r \in J). \quad (8.13)$$

We now use this identity to obtain the continuity of χ on $S(k)$.

Proof of Theorem 8.1. Choose $u \in U(k)$ with $\text{Nrd}(u) \neq 1$; a central phase suffices, since the norm-one torus of L/k has infinitely many rational points. Division makes $1 - u$ invertible. For $b = (1 - u)^{-1}$,

$$b + b^* = 1, \quad b^* = -ub, \quad f^\#(b) = \chi(u). \quad (8.14)$$

If $u' \in U(k)$ has the same reduced norm as u , put $b' = (1 - u')^{-1}$. Oddness gives $\overline{\text{Nrd}(b)} = -\text{Nrd}(u)\text{Nrd}(b)$, and the same identity for b' , so

$$q = \text{Nrd}(b')/\text{Nrd}(b) \in k^\times.$$

For u' sufficiently close to u at A , choose $c \in D^\times$ with $\text{Nrd}(c) = q^{(n-1)/2}$ and c close to 1 at A_D . Existence without approximation follows from $H^1(L, \text{SL}_1(D)) = 1$. Locally the reduced norm is smooth, including when $p \mid n$: its differential is the nonzero reduced trace. It therefore has solutions near 1 for norms near 1. Weak approximation for $\text{SL}_1(D)$ adjusts the global solution to these local solutions. Set $b'' = q^{-1}c^*b'c$. Then

$$\text{Nrd}(b'') = q^{-n}q^{n-1}\text{Nrd}(b') = \text{Nrd}(b), \quad f^\#(b'') = f^\#(b').$$

Thus $r = b^{-1}b'' \in \text{SL}_1(D)(L)$ is close to 1 at A_D and belongs to J . Equation 8.13 gives $\chi(u') = \chi(u)$. Taking $u' = us$ proves continuity of $\chi|_{S(k)}$ for the topology induced by S_A .

For clarity, this continuous finite-valued homomorphism extends to S_A . Weak approximation makes $S(k)$ dense. If K is its kernel, then $\overline{K} \cap S(k) = K$ by continuity. The finite cosets of K cover $S(k)$; their closures therefore cover S_A . Thus $\overline{K}$ is an open normal subgroup, and $S(k)/K \simeq S_A/\overline{K}$, giving the extension.

At $v \in A$ write $U(k_v) = D_v^\times$ and $S(k_v) = \text{SL}_1(D_v)$. Approximate any two elements of $D_v^\times$ by rational unitary elements, prescribing the identity at the other places of A . Their rational commutators have character zero, so continuity kills every local commutator. By Lemma 5.3, these commutators generate a dense subgroup of $\text{SL}_1(D_v)$. The extension is therefore zero on every factor of S_A , proving $\chi(S(k)) = 0$. If A is empty, the preceding continuity assertion already gives the result. $\square$

9 Induction on the degree

We now prove that the power defect vanishes in every degree. The induction is simultaneous over all global function fields: passing to a finite separable extension changes the ground field, but every group to which we apply the induction hypothesis has smaller absolute degree. The factorization used in the even step is adapted from [23, Section 7]. We give its algebra and approximation conditions in detail, since the local groups occurring in the factorization vary with the element being factored.

Theorem 9.1. *Let k be a global function field, let L/k be a separable quadratic extension, and let $(D, *)$ be a central simple L -algebra of degree $n \geq 3$ with unitary involution. Suppose that $S = \text{SU}(D, *)$ is k -anisotropic. For every integer $e \geq 1$, put*

$$R = S(k)^e, \quad P_0 = \overline{\delta_A(R)}, \quad V_0 = \delta_A^{-1}(P_0).$$

Here $S(k)^e$ denotes the subgroup generated by the e -th powers, and A is the set of places where S is anisotropic. Then $V_0 = R$.

Throughout the proof, e is fixed and

$$\pi : S(k) \longrightarrow S(k)/R$$

denotes the finite quotient map of Proposition 2.3. Its target has exponent dividing e . We say that an element is *killed* if its image under π is trivial. The inner type A case, including degree two, is already known. Isotropic groups also satisfy the assertion by the reductions in Section 2. These cases are available whenever they occur in the induction.

9.1 Identity neighborhoods for smaller groups

Assume the induction hypothesis for a fixed smaller-degree special unitary group embedded in S . Proposition 2.5 then extends the restriction of π to its product of anisotropic local groups. An element sufficiently close to 1 at those places is consequently killed. We need a version of this observation for binary groups which have not yet been chosen.

Lemma 9.2. *Let K be a local field of positive characteristic, and fix positive integers e and r . There is a constant $c = c(K, e, r)$ with the following property. Let K'/K be a finite extension of degree at most r , let B be a quaternion division algebra over K' , and let $g \in \mathrm{SL}_1(B)$. If both reduced eigenvalues λ of g satisfy*

$$\mathrm{ord}_K(\lambda - 1) > c,$$

then g belongs to the subgroup of $\mathrm{SL}_1(B)$ generated by its e -th powers. The valuation in this inequality is extended from K to an algebraic closure.

In particular, suppose that such binary groups are embedded in a fixed unitary group so that their reduced eigenvalues occur among the ambient reduced eigenvalues. A sufficiently small ambient neighborhood of 1 works for every one of these groups and every continuous finite quotient of exponent dividing e .

Proof. First let H be any local form of SL_2 . Choose $g_0 \in H(K)$ such that g_0^e is regular semisimple. Such elements exist: rational points are Zariski dense, and the power map is dominant, including when the characteristic divides e . For a regular semisimple element a , the differential at 1 of $x \mapsto axa^{-1}x^{-1}$ has image $(\mathrm{Ad}(a) - 1)\mathrm{Lie}(H)$. Over an algebraic closure this image contains both root lines. Their conjugates span $\mathfrak{sl}_2$. This remains true in characteristic two: conjugating the upper root vector by a lower unipotent gives

$$E_{12} + tI + t^2E_{21},$$

so the scalar direction is in the span as well. Zariski density permits a finite set of rational conjugates of g_0^e with these spanning properties. The product of the corresponding commutator maps therefore has surjective differential at the identity. Its image contains an identity neighborhood by the local submersion theorem, and every value belongs to the subgroup generated by e -th powers. That subgroup is thus open.

For the division form, let ν_B be the division valuation on B . For $g \neq 1$,

$$\nu_B(g - 1) = \frac{1}{2} \mathrm{ord}_{K'} \mathrm{Nrd}_B(g - 1) = \frac{1}{2} \sum_{\lambda} \mathrm{ord}_{K'}(\lambda - 1).$$

Consequently eigenvalues sufficiently close to 1 put g in any specified identity neighborhood of $\mathrm{SL}_1(B)$, in particular in its open power subgroup.

There are only finitely many possibilities for K' up to topological field isomorphism when $[K' : K] \leq r$: these fields are Laurent series fields over finite fields, and their residue degrees

are bounded by r . Over each there is exactly one quaternion division algebra. Topological field isomorphisms preserve the normalized valuation and the subgroup generated by powers. Taking a maximum of the finitely many thresholds, and using the bound on the ramification index to convert $\text{ord}_{K'}$ to ord_K , proves the first assertion. Finally, sufficiently small ambient matrices have all their reduced eigenvalues close to 1. A homomorphism of exponent dividing e kills the generated power subgroup, which proves the last assertion. $\square$

In our applications a binary group over a finite extension F/k acts, after passage to an algebraic closure, by ordinary two-dimensional blocks. The degrees $[F : k]$ are bounded by n . We may therefore use Lemma 9.2 at a place of k before constructing F or its binary group. At split binary factors there is no condition: the inductive Margulis–Platonov assertion makes every finite quotient of the rational group depend only on its anisotropic factors.

9.2 Odd degree

Suppose first that D is division and n is odd. If V_0/R were nontrivial, Proposition 5.1 would give either a finite abelian character of $U(k)$ nontrivial on $S(k)$, or a nonabelian finite simple quotient as in that proposition. The first possibility is excluded by Theorem 8.1, and the second by Proposition 7.8. Thus $V_0 = R$ in odd division degree, without any induction hypothesis.

Now write $D = M_l(\Delta)$, where Δ is a unitary division algebra of degree d , and suppose that $n = ld$ is odd and $l > 1$. Realize S as the special unitary group of an l -dimensional Hermitian space over Δ . Both d and l are odd, and $l \geq 3$. If $d = 1$, Hermitian forms of fixed dimension and determinant over L/k are isometric: after choosing a compatible determinant for an isometry, its existence is an instance of Harder’s vanishing for a special-unitary torsor. There is a form of the prescribed determinant containing a hyperbolic plane. This would make S isotropic, so the anisotropic case has $d > 1$.

Every subspace of this anisotropic Hermitian space is nondegenerate. Given a special unitary transformation and a nonzero vector, choose a two-dimensional subspace containing that vector and its image. Witt extension gives an isometry of the plane carrying one to the other. Its determinant can be corrected on the perpendicular line inside the plane: the determinant map on that line’s unitary group is surjective, since each fiber is a torsor under a simply connected special unitary group and has a rational point. Extending this special plane isometry by the identity, and dividing the original transformation by it, leaves a transformation fixing the chosen vector. Repeat in its orthogonal complement, ending in dimension two. Hence $S(k)$ is generated by special plane groups.

Each plane group has degree $2d < n$. It has no anisotropic place. At a place nonsplit in L , it is the group of an ordinary Hermitian form of dimension $2d \geq 6$, and is isotropic. At a split place it is an inner form represented by a matrix algebra with matrix size at least two. The induction hypothesis therefore makes every plane group trivial in the finite quotient $S(k)/R$. This proves the assertion in all odd degrees.

9.3 A fixed quadratic subfield in even degree

Let $n = 2m$ with $m \geq 2$. We will factor a suitably chosen element of each coset of V_0/R as a product of an element in a binary group and an element in a fixed unitary centralizer of degree m . The first step is to construct that centralizer.

Lemma 9.3. *There is a separable quadratic field $S_0 \subset D$, disjoint from L , which is fixed pointwise by $*$. Its two geometric eigenspaces in the standard representation of D both have dimension m .*

Proof. In odd characteristic choose $S_0 = k(s)$ with $s^2 = r$ and $r \in N_{L/k}(L^\times)$. In characteristic two choose $s^2 + s = r$, with $r \in k$. We specify finitely many local conditions on this quadratic algebra. At the exceptional nonsplit places of L/k , require it to split. The exceptional set is chosen so that at every remaining nonsplit place, L/k is unramified and the involution is represented, up to scalar, by a unimodular Hermitian form. At a split place of L/k where the index of D does not divide m , require S_0 to be a field. Require the same at one further place split in L . Weak approximation supplies these conditions. In odd characteristic one chooses an element of $L^\times$ first and takes its norm; at split places there is no norm restriction, and at the specified nonsplit places a square value is available. The extra split place ensures that S_0 is a field and is distinct from L .

We check local embeddability with the stated multiplicities. Where L splits, this is the usual embedding criterion for central simple algebras. If S_0 splits, the local index divides m , so there are two blocks of degree m . If S_0 is a field, the index after quadratic extension divides m : multiplying a local invariant by two removes precisely the possible extra factor of two in the index of an algebra of degree $2m$. The involution pairs the two L -components and supplies the embedding on the second component. Where L is a field and S_0 splits, use an orthogonal decomposition into two m -dimensional spaces.

It remains to consider a nonexceptional place where both are fields. The local Hermitian form has norm determinant. If S_0 and L agree locally, the form is hyperbolic, since L is unramified; let S_0 act through its two embeddings on dual totally isotropic spaces. If the two quadratic fields differ, use m copies of the Hermitian plane obtained by tracing a line over their composite. In odd characteristic use half the trace; in the basis $1, s$ its determinant is r , a norm from L . In characteristic two use the trace; its matrix is

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

with determinant 1. Local Hermitian classification identifies these forms with the required one. This classification applies also in characteristic two; see Section 2.

To pass from local embeddings to a global embedding, let X be their variety, with the multiplicities fixed as above. It is a homogeneous space under S , with smooth connected reductive geometric stabilizer

$$S(\mathrm{GL}_m \times \mathrm{GL}_m).$$

Over a separable closure, an embedding is just a decomposition into complementary subspaces of dimension m , which gives this description. The toric quotient of the stabilizer has a rank-one character lattice with its induced Galois action, either trivial or quadratic sign. Its Sha^1 is zero. In the trivial case this follows from the absence of nonzero continuous homomorphisms from a profinite group to $\mathbb{Z}$. In the sign case a class factors through the quadratic quotient, and a nonzero class is detected at an inert Frobenius place by Chebotarev. Global duality therefore gives $\mathrm{Sha}^2(k, T) = 0$ for this toric quotient T [6, Theorem 5.2].

The local embeddings give adelic points on X ; integral points at almost all places follow by spreading out and Lang's theorem for the connected stabilizer. For a simply connected semisimple acting group, the obstruction in the proof of [7, Theorem 2.5] lies in $\mathrm{Sha}^2(k, T)$, and its vanishing gives a rational point. Applying that theorem proves the lemma. $\square$

Fix s as in Lemma 9.3, and put

$$C_s = C_D(s), \quad J_0 = \text{the third quadratic field in } LS_0/k.$$

Then C_s is a central simple algebra of degree m over LS_0 , and $*$ restricts to a unitary involution with fixed ground field S_0 . The unitary groups of C_s are defined over S_0 . Whenever they are embedded in a group over k , restriction of scalars from S_0 is understood; their rational elements are thus the ordinary unitary elements of C_s .

9.4 The algebraic factorization

The fixed field gives two blocks. Comparing them with their images under $u \in S$ produces a quaternion algebra over a field that depends on u . The factorization below reduces the global problem to one or two norm equations in that field.

Over a splitting field, let P be either spectral idempotent of s . It is fixed by the extended involution. Define

$$\begin{aligned} P' &= uPu^{-1}, \\ p &= PuP + (1 - P)u(1 - P), \\ z &= up^* = P'P + (1 - P')(1 - P), \\ h &= pp^*. \end{aligned} \tag{9.1}$$

The expressions p, z, h are unchanged when P is replaced by $1 - P$, so they descend to k . Direct multiplication gives

$$h = PP'P + (1 - P)(1 - P')(1 - P) = 1 - P - P' + PP' + P'P = zz^*. \tag{9.2}$$

In particular h commutes with both P and P' .

Write a split matrix component of u in blocks as

$$\begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix}.$$

The two blocks of h have the same characteristic polynomial, denoted by Ψ . For example, where the relevant blocks are invertible,

$$h_+ = A'(u^{-1})_{++} = (1 - B'D'^{-1}C'A'^{-1})^{-1}; \tag{9.3}$$

the corresponding other product is conjugate to this one. The identity extends to all u by density. Exchanging the labels preserves Ψ , and $h^* = h$; hence $\Psi \in k[T]$. Similarly, complementary-minor identities and $\det u = 1$ show that

$$d_0 = \text{Nrd}_{C_s}(p^*) \in J_0. \tag{9.4}$$

Indeed label exchange sends this determinant to $\text{Nrd}_{C_s}(p)$, as does the involution on the center over S_0 . Their composite fixes d_0 , and its fixed field is J_0 .

Let $\Omega \subset S$ be the open set on which Ψ is separable and $h(1 - h)$ is invertible. For $u \in \Omega$, put $F = k(h)$. This is an étale algebra of degree m , and its regular characteristic polynomial is Ψ . Moreover, F and LS_0 generate their full tensor product inside D : in either s -block the element h has m distinct eigenvalues, so its minimal polynomial over LS_0 still has degree m . The same notation applies over a completion, component by component. We will later choose u so that F and LS_0F are fields globally. Taking determinants of $h = pp^*$ gives the compatibility

$$N_{F/k}(h) = N_{J_0/k}(d_0). \tag{9.5}$$

Lemma 9.4. *For $u \in \Omega$, there is a quaternion algebra Q over F with an embedding $Q \subset C_D(F)$ such that*

$$J_0F \subset Q \cap C_s, \quad z \in Q, \quad \text{Nrd}_Q(z) = h.$$

The involution $$ on Q is its canonical quaternion conjugation. The algebra Q is split at any field factor at which $-h(1-h)$ is a norm from J_0F .*

Suppose that $a \in (J_0F)^\times$ satisfies

$$N_{J_0F/F}(a) = h, \quad N_{J_0F/J_0}(a) = d_0. \quad (9.6)$$

Then

$$u = wq, \quad w = za^{-1} \in \text{SL}_1(Q), \quad q = a(p^*)^{-1} \in \text{SU}(C_s). \quad (9.7)$$

If only the first equation in (9.6) holds, the same factorization has $q \in \text{U}(C_s) \cap S$.

Proof. The centralizer $C_D(F)$ is a quaternion algebra over LF , understood componentwise. Its canonical quaternion conjugation commutes with $*$. Their composite is a semilinear algebra automorphism of order two over LF/F , and descent gives a quaternion algebra Q/F . On LS_0F , canonical conjugation exchanges the two s -labels. Composing with $*$ therefore fixes J_0F , proving the stated inclusion.

On the two-dimensional block associated with an eigenvalue of h , both P and P' have rank one. The rank assertion for P' follows from the two terms defining h and the invertibility of h and $1-h$. With P the first-coordinate projection, write

$$P' = \begin{pmatrix} h & x \\ y & 1-h \end{pmatrix}, \quad z = \begin{pmatrix} h & -x \\ y & h \end{pmatrix}.$$

The equation $P'^2 = P'$ gives $xy = h(1-h)$. Thus z has quaternion trace $2h$ and norm h , and $z + z^* = 2h$ shows that its canonical conjugate equals z^* . Hence $z \in Q$. The element $z - h$ is invertible, exchanges the J_0F -lines, and satisfies

$$(z - h)^2 = -h(1-h).$$

It realizes Q as the cyclic quaternion algebra for the separable quadratic extension J_0F/F and this scalar. The usual norm criterion for a cyclic algebra proves the splitting assertion. This argument also applies in characteristic two: the quadratic extension is separable and $z - h$ implements its nontrivial automorphism.

The first norm equation says $aa^* = h$, so w has quaternion norm one and belongs to S . Also

$$qq^* = a(p^*)^{-1}p^{-1}a^* = ah^{-1}a^* = 1, \quad wq = z(p^*)^{-1} = u.$$

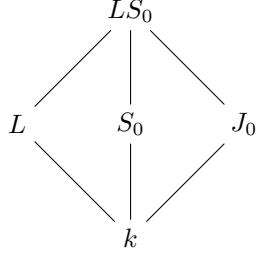
Therefore $q \in \text{U}(C_s) \cap S$. Finally,

$$\text{Nrd}_{C_s}(q) = N_{J_0F/J_0}(a)d_0^{-1},$$

so the second norm equation puts q in $\text{SU}(C_s)$. □

For $m \geq 3$ we will solve both equations in (9.6) using Proposition 4.1. When $m = 2$ we solve only the first and then correct the determinant of q . Figure 1 records the fixed fields and the varying norm equations. The following construction makes the degree-four correction possible with neighborhoods fixed before the element q is chosen.

Fixed biquadratic extension



Every edge has degree 2.

Norms for the varying algebra $F = k(h)$

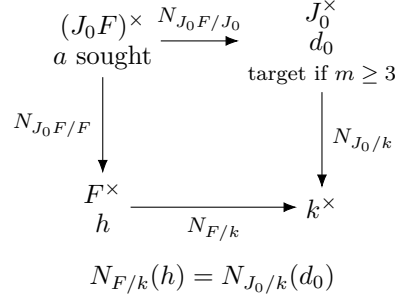


Figure 1: The fields and norms in the even-degree factorization. The left diagram is fixed before u is chosen; its edges denote field inclusions. On the right, for $u \in \Omega$, the étale algebra $F = k(h)$ has degree m and depends on u ; here $J_0 F = J_0 \otimes_k F$. The norm square commutes. For $m \geq 3$, the sought element a must map to both h and d_0 ; their norms to k already agree. For $m = 2$, only the norm condition to F is imposed at this stage.

9.5 Auxiliary fields for the degree-four correction

Suppose in this subsection that D has degree four over L . Retain the Hermitian quadratic field $S_0 = k(s)$ already constructed, put $E = LS_0$ and $C_s = C_D(s)$, and let J_0 be the third quadratic subfield of the biquadratic extension E/k . Thus C_s is a quaternion algebra over E , with a unitary involution over S_0 . When its unitary groups are viewed inside the ambient groups over k , restriction of scalars from S_0 is understood. We will correct the determinant of an element of $U(C_s, *) \cap S$ by elements in binary special unitary groups. The auxiliary fields in the following lemma let us prescribe where those binary groups are split. This construction is the degree-four determinant correction of [23, Section 7], with separable quadratic descent used also in characteristic two.

Lemma 9.5. *There is a finite set B_s of places of k , depending only on $(D, *, S_0)$, with the following property. For every finite set T of places disjoint from B_s , there is a separable quadratic field $F' = k(f) \subset C_s$ such that $f^* = f$, F' is linearly disjoint from E , and, for every $v \in T$,*

$$F' \otimes_k k_v \simeq k_v \times k_v, \quad \mathrm{SU}(C_D(F'), *) \otimes_{F'} F'_w \simeq \mathrm{SL}_2 \quad \text{for both } w \mid v.$$

Here $\mathrm{SU}(C_D(F'), *)$ is a group over F' : its central simple algebra has center LF' and degree two. The embeddings can be chosen so that f has reduced trace zero in C_s in odd characteristic, and reduced trace one in characteristic two.

Proof. Write κ for canonical quaternion conjugation on C_s . It commutes with $*$, since reduced trace commutes with $*$. Consequently $* \circ \kappa$ is a semilinear algebra automorphism of order two relative to E/S_0 . Its fixed algebra B_0 is a quaternion algebra over S_0 , and Galois descent gives

$$B_0 \otimes_{S_0} E = C_s.$$

The restriction of $*$ to B_0 is canonical conjugation. This descent is valid in characteristic two as well, because E/S_0 is separable; see [17].

Fix B_s at this point, containing all places below ramification of B_0 and sufficiently large for the following integral properties outside B_s . The fields L and S_0 are unramified, the fixed algebras and embeddings have unramified integral models, and the involution at a nonsplit place of L/k is

given, up to scalar, by a unimodular Hermitian form. When S_0 splits, its two orthogonal Hermitian blocks are unimodular as well. These properties follow by spreading the fixed algebras, involution and separable embedding; the local classification of unitary involutions gives the stated Hermitian description [17].

We first construct an abstract quadratic extension that embeds in B_0 . Suppose the characteristic is odd. Choose $i_0 \in L^\times$ with $\text{Tr}_{L/k}(i_0) = 0$. For a scalar $c \in k^\times$, consider the quadratic algebra over S_0 with generator y and equation

$$y^2 = i_0^2 c.$$

An embedding of this algebra in B_0 has $y^* = -y$. The element $f = y/i_0 \in C_s$ therefore satisfies

$$f^* = f, \quad f^2 = c, \quad \text{Trd}_{C_s}(f) = 0.$$

In characteristic two, write $L = k(\iota)$ with $\iota^2 + \iota = b_L \in k$. Instead use the quadratic algebra over S_0 with equation

$$y^2 + y = c + b_L, \quad c \in k.$$

Its generator has reduced trace one, so $y^* = y + 1$ in B_0 . Then $f = y + \iota$ satisfies

$$f^* = f, \quad f^2 + f = c, \quad \text{Trd}_{C_s}(f) = 1.$$

These formulas specify how a quadratic embedding in B_0 produces a Hermitian quadratic embedding in C_s in either characteristic.

A quadratic separable field over S_0 embeds in B_0 precisely when it remains a field at every ramified place of B_0 . Indeed a quadratic local field extension kills the invariant $1/2$ of a quaternion division algebra, whereas the split algebra does not. The global Brauer invariant theorem then gives the splitting, and hence the embedding, criterion [35, 22]. The resulting local conditions on c occur above B_s and can always be met. In odd characteristic, the kernel of

$$k_v^\times / k_v^{\times 2} \longrightarrow S_{0,w}^\times / S_{0,w}^{\times 2}$$

has at most two elements when $S_{0,w}/k_v$ is quadratic, and one when $S_{0,w} = k_v$. The square-class group of an odd-characteristic local field has four elements. We can therefore choose c so that $i_0^2 c$ is nonsquare at each required place. In characteristic two the same argument applies to the restriction map on Artin–Schreier classes

$$k_v / \{a^2 + a : a \in k_v\} \longrightarrow S_{0,w} / \{a^2 + a : a \in S_{0,w}\}.$$

Its kernel has at most two elements, whereas its source is infinite. Thus $c + b_L$ can be required to have nonzero image. If S_0 has two places over v , both completions equal k_v and the exclusions at those places are identical.

At every $v \in T$, require that c define a split quadratic algebra over k_v : take c to be a nonzero square in odd characteristic and an Artin–Schreier value in characteristic two. At one further place v_* , split completely in E and outside $B_s \cup T$, require the quadratic algebra defined by c to be a field. Chebotarev supplies v_* , and weak approximation supplies c satisfying all these local conditions. The conditions are open in the local fields; for Artin–Schreier classes this follows also from the surjectivity of $a \mapsto a^2 + a$ on the maximal ideal. The field F' defined by c is disjoint from E : otherwise it would be one of the three quadratic subfields of E , all of which split at v_* . The associated algebra for y over S_0 is therefore a field, and the embedding criterion gives an embedding in B_0 . The preceding formulas give $F' \subset C_s$; moreover EF' is a field of degree two over E , hence a maximal subfield of C_s .

We next choose this embedding so that the two binary groups split at the places in T , using the integral properties secured when B_s was fixed.

Fix $v \in T$ and put $K = k_v$. When L splits at v , the algebra on each L -component is $M_4(K)$. Put $A_0 = S_0 \otimes_k K$. The four-dimensional module is free of rank two over A_0 , and its s -centralizer is $\text{End}_{A_0}(A_0^2)$. For the two roots $\lambda_1, \lambda_2 \in K$ of the chosen polynomial of f , take $f = \text{diag}(\lambda_1, \lambda_2)$ in this centralizer. Its two K -eigenspaces are copies of A_0 , of dimension two over K , whether A_0 is a field or split. Both special groups are SL_2 , and the involution specifies the paired L -component. After splitting A_0 , each eigenvalue occurs once in each of the two s -blocks, as required.

Suppose instead that $L \otimes_k K$ is the unramified quadratic field over K . If S_0 splits, decompose each of its two Hermitian blocks into orthogonal lines of unit norm values. Such decompositions follow by diagonalizing the reduction and lifting an orthogonal basis, in characteristic two as well. Assign to each f -eigenvalue one line from each s -block. Each resulting binary Hermitian space has unit determinant. All units of K are norms from the unramified quadratic extension, so its negative determinant is a norm and the space is hyperbolic. Its special unitary group is therefore split.

If S_0 is nonsplit, its completion equals the unique unramified quadratic extension $L \otimes_k K$. The two s -eigenspaces over that extension are dual totally isotropic two-spaces. Choose paired bases e_1, e_2 and e'_1, e'_2 with $\langle e_i, e'_j \rangle = \delta_{ij}$, and assign one f -eigenvalue to each hyperbolic plane $\langle e_i, e'_i \rangle$. This again makes both binary groups split.

All the local elements f just constructed are Hermitian and have each eigenvalue once in each C_s -block. They therefore give local embeddings in B_0 by the same descent formulas. For clarity, in characteristic two $y = f + \iota$ has reduced trace one and satisfies

$$y^* = y + 1 = \kappa(y),$$

so it is fixed by $* \circ \kappa$. Thus it really lies in B_0 . In odd characteristic the corresponding identity is $(i_0 f)^* = -i_0 f = \kappa(i_0 f)$.

Finally, local embeddings of the fixed quadratic algebra for y are conjugate by $B_0^\times$, by Skolem–Noether. Weak approximation in the open affine variety $B_0^\times$ over S_0 lets a global conjugator approximate all the prescribed local conjugators. The splitting property persists under sufficiently close approximation: the eigenspaces vary continuously and the determinant norm classes of their Hermitian forms are locally constant. The resulting global embedding has all the asserted properties. $\square$

For each field F' in Lemma 9.5, put $G_{F'} = \text{SU}(C_D(F'), *)$, a group over F' . Its restriction of scalars embeds in S , since over a separable closure it acts by special transformations on the two binary F' -eigenspaces. We will use the inclusion

$$(J_0 F' / F')^1 \subset (\text{Res}_{F'/k} G_{F'} \cap \text{U}(C_s, *))(k), \quad \text{Nrd}_{C_s}(b) = N_{J_0 F' / J_0}(b). \quad (9.8)$$

Here $(J_0 F' / F')^1$ denotes the elements of norm one. Indeed EF' is a maximal subfield of $C_D(F')$ as well as of C_s . Its norm to LF' computes the determinant in the binary algebra; the nontrivial automorphism of EF' / LF' restricts to the nontrivial automorphism of $J_0 F' / F'$. On this last field that automorphism is $*$. Thus norm one gives both the unitary and special conditions. The determinant in C_s is the norm from EF' to E , whose restriction to $J_0 F'$ is the norm displayed in Equation (9.8).

9.6 Correcting the determinant in degree four

We now isolate the local conditions on the factor q that suffice to remove its determinant. The neighborhoods in the next proposition are fixed before q is chosen. This will allow the preceding factorization to use them as ordinary local approximation conditions.

Proposition 9.6. *Assume the Margulis–Platonov assertion for binary special groups over every finite separable extension of k . Fix the finite quotient $S(k)/R$, of exponent dividing e , and retain the degree-four data $(D, *, S_0)$ above. Let B be a nonempty finite set of places of k containing B_s and all places below the rank-zero places of the group $\mathrm{SU}(C_s, *)$ over S_0 . There are identity neighborhoods W_v in $(\mathrm{U}(C_s, *) \cap S)(k_v)$, for $v \in B$, such that every*

$$q \in (\mathrm{U}(C_s, *) \cap S)(k), \quad q_v \in W_v \quad (v \in B),$$

has trivial image in $S(k)/R$.

Proof. First fix a field F_1 given by Lemma 9.5, with no extra local prescriptions, and fix its embedding in C_s . Write $G_i = \mathrm{SU}(C_D(F_i), *)$ when a field F_i has been specified. The field F_1 will remain fixed when the neighborhoods W_v are chosen; a second field F_2 will be chosen after q .

For an element q in the proposition put

$$r_0 = \mathrm{Nrd}_{C_s}(q) \in E^\times.$$

Let α and β be the automorphisms of E/k fixing S_0 and L , respectively. Unitarity gives $\alpha(r_0) = r_0^{-1}$, and ambient determinant one gives

$$N_{E/L}(r_0) = 1, \quad \beta(r_0) = r_0^{-1}.$$

Hence $\alpha\beta$ fixes r_0 , so $r_0 \in J_0^1$. Write a bar for the nontrivial automorphism of J_0/k . Choose once and for all $c_* \in J_0$ with $c_* + \bar{c}_* = 1$, and set

$$d = c_* + r_0 \bar{c}_*. \tag{9.9}$$

This formula is valid in both characteristics. It satisfies $r_0 \bar{d} = d$ and gives $d = 1$ when $r_0 = 1$. We will require q close enough to 1 at B that d is invertible, in which case $r_0 = d/\bar{d}$.

We describe the neighborhoods W_v before making any further choices. The local norm map for the fixed separable quadratic algebra $J_0 F_1/J_0$ is a submersion at 1: its differential is the surjective trace map, also in characteristic two. For d sufficiently close to 1, it therefore has solutions

$$N_{J_0 F_1/J_0}(x_{1,v}) = d, \quad x_{1,v} \longrightarrow 1 \quad \text{as } d \longrightarrow 1, \tag{9.10}$$

in the product of the local algebras above v . Because the field and embedding are fixed, the elements $b_{1,v} = x_{1,v}/x_{1,v}^*$ tend to 1 in the ambient algebra $D \otimes_k k_v$.

By the binary case of MP and Proposition 2.5, the quotient maps on $G_1(F_1)$ and on $\mathrm{SU}(C_s, *) (S_0)$ extend continuously to their respective products of rank-zero local groups. Choose W_v so small that the solutions in Equation (9.10) make $b_{1,v}$ belong to the needed identity neighborhoods at every rank-zero place of G_1 over v . At the same time require q and $b_{1,v}$ close enough to 1 that

$$(b_{1,v} b_{2,v})^{-1} q$$

belongs to the needed identity neighborhoods for $\mathrm{SU}(C_s, *)$ whenever $b_{2,v}$ is sufficiently close to 1 and the displayed element has determinant one in C_s . These requirements concern finitely many fixed groups and continuous multiplication maps, so they hold after shrinking W_v . They also ensure that d is invertible. For a later field F_2 , its local target at every $v \in B$ will be $x_{2,v} = 1$. Once F_2 is fixed, approximation to that target can be made arbitrarily close both in its binary group and in D ; no choice of F_2 enters the present neighborhoods.

Now let q satisfy these conditions, and hence fix r_0 and d . Say that a place $v \notin B$ is covered by F_i if

- (i) d is a norm from J_0F_i/J_0 at every place of J_0 above v ; and
- (ii) G_i is isotropic at every place of F_i above v .

All but finitely many places are covered by the fixed field F_1 . Indeed, away from the ramification and the divisors of d , norms from unramified quadratic algebras are surjective on units. Also G_1 is an unramified binary group at all but finitely many places, and is split there: a quaternion algebra over a local field with unramified integral model has trivial Brauer class, since the residue field is finite.

Let T be the finite set of places outside B not covered by F_1 . Apply Lemma 9.5 to choose F_2 split at every place of T , with both binary groups split there. The algebra J_0F_2/J_0 is then locally split at every place over T , and its norm is surjective. Thus every place outside B is covered by at least one of F_1, F_2 .

We seek invertible elements $x_i \in J_0F_i$ satisfying

$$N_{J_0F_1/J_0}(x_1)N_{J_0F_2/J_0}(x_2) = d. \quad (9.11)$$

At the places above B , take the local solutions $x_1 = x_{1,v}$ already specified and $x_2 = 1$. At any other k -place choose a field covering that entire place, let its variable carry the norm d at every J_0 -place above it, and set the other variable equal to 1. This gives local solutions everywhere. It also arranges the following useful condition: if G_i has a rank-zero place above $v \notin B$, then F_i cannot cover v , so the other field covers it and $x_i = 1$ on all components above v .

We justify both the Hasse principle and the required weak approximation for Equation (9.11), including in characteristic two. Put $K = J_0$, $E_i = J_0F_i$ and $y_2 = x_2^{-1}$. On the locus where the variables are invertible the equation becomes

$$N_{E_1/K}(x_1) - dN_{E_2/K}(y_2) = 0. \quad (9.12)$$

Both norm forms are nonsingular binary quadratic forms because E_i/K is separable; their direct sum in this display is a nonsingular four-variable quadratic form. This remains true in characteristic two, where its polar form is the direct sum of two nondegenerate alternating forms. Its projective zero locus $X_d \subset \mathbb{P}_K^3$ is consequently a smooth quadric surface, becoming $\mathbb{P}^1 \times \mathbb{P}^1$ over a separable closure.

Here is a local-global argument for this surface that works uniformly in the characteristic. Its two rulings determine a quadratic étale algebra K'/K . The automorphisms of $\mathbb{P}^1 \times \mathbb{P}^1$ act separately on the two factors or exchange them. Descent therefore expresses X_d as the restriction of scalars, from K' to K , of a Severi–Brauer conic over K' ; when $K' = K \times K$ this means a product of two such conics. A local point of X_d supplies points on the conic at every completion of K' . The Brauer local-global theorem over the global field K' , componentwise in the split case, makes the conic split. Hence $X_d(K)$ is nonempty; see [22] for the Brauer theorem.

A smooth quadric surface with a rational point is rational, by projection from that point. Its affine cone minus the vertex is rational as well: over a rational affine open of the surface the tautological line bundle is trivial, so the punctured cone is birational to $\mathbb{A}^2 \times \mathbb{G}_m$. Weak approximation follows on its smooth locus by using such a rational chart and density of a nonempty Zariski open in each local smooth variety. Remove also the two norm-zero loci. The local solutions above belong to the resulting open set; its rational points still approximate any finite collection of those local solutions. Inverting y_2 returns the same Hasse principle and weak approximation for Equation (9.11).

There are only finitely many rank-zero places of G_1 and G_2 . Apply this weak approximation at their underlying k -places and at B . At a rank-zero place outside B , the variable belonging to that group has local target 1, as arranged above. Approximate it closely enough that x_i/x_i^* lies

in an identity neighborhood killed by the continuous extension of the finite quotient on G_i . At B , approximate the specified $x_{1,v}$ and 1 closely enough to retain all the conditions used in choosing W_v , and also the analogous kernel conditions for G_2 .

For the resulting global solution set

$$b_i = x_i/x_i^* \quad (i = 1, 2).$$

Equation (9.8) places b_i in $(\text{Res}_{F_i/k} G_i \cap \text{U}(C_s, *))(k)$, and the local conditions just imposed show that both have trivial image in $S(k)/R$. Their determinants satisfy

$$\text{Nrd}_{C_s}(b_1) \text{Nrd}_{C_s}(b_2) = \frac{N_{E_1/K}(x_1)N_{E_2/K}(x_2)}{N_{E_1/K}(x_1)N_{E_2/K}(x_2)} = \frac{d}{d} = r_0.$$

Thus $q' = (b_1 b_2)^{-1} q$ belongs to $\text{SU}(C_s, *)(S_0)$. At all its rank-zero places it lies in the prescribed kernel neighborhoods, since those places lie above B . The binary case of MP therefore makes q' trivial in the quotient as well. The identity $q = b_1 b_2 q'$ proves the proposition. $\square$

9.7 Local factorizations near the identity

We return to even degree $n = 2m$. For $m = 2$, Proposition 9.6 supplies identity neighborhoods in $\text{U}(C_s) \cap S$ that can be fixed before q is chosen; for $m \geq 3$, the induction hypothesis supplies kernel neighborhoods in the fixed group $\text{SU}(C_s)$. It remains to choose a representative of each coset in V_0/R and solve the norm equations so that both factors w and q are killed. We first construct local open sets near 1 on which the norm equations are soluble and their solutions make both factors small in the ambient algebra.

Lemma 9.7. *Let $K = k_v$. Arbitrarily close to 1 in $S(K)$ there are nonempty open sets of elements $u \in \Omega(K)$ for which both equations in (9.6) have a solution a . The elements a, w, q may all be required to lie in any prescribed ambient identity neighborhoods. The same assertion holds when the second norm equation is omitted.*

Proof. Choose $X \in \text{Lie}(S)(K)$ such that the geometric matrix $X_{+-}X_{-+}$ is invertible and has separable characteristic polynomial. This is a nonempty Zariski open condition, as is seen in split coordinates, and the K -points of this vector space are Zariski dense. Smoothness supplies an analytic curve $u(\varepsilon)$ through 1 with derivative X . Formula (9.3) gives

$$(h - 1)/\varepsilon^2 \longrightarrow X_{+-}X_{-+}$$

on one block, with the same characteristic polynomial on the other. For small nonzero ε , therefore, $u(\varepsilon) \in \Omega$.

Identify the étale algebras generated by $(h - 1)/\varepsilon^2$ with the étale algebra generated by its limit. These identifications vary continuously: the characteristic polynomials have simple roots, and simple-root lifting, followed by the basis of powers, identifies the embeddings in the ambient algebra. The same holds after adjoining J_0 . Under these identifications, $h \rightarrow 1$ and $d_0 \rightarrow 1$.

Consider the simultaneous norm map from $(J_0 F)^\times$ to the torus of pairs (h, d_0) satisfying (9.5). This map is smooth and surjective as a morphism of tori. Indeed over a splitting field, write the coordinates of a as $(a_i, b_i)_{1 \leq i \leq m}$; then the map records

$$h_i = a_i b_i, \quad d_+ = \prod_i a_i, \quad d_- = \prod_i b_i, \quad \prod_i h_i = d_+ d_-.$$

Its kernel is the torus $b_i = a_i^{-1}$, $\prod_i a_i = 1$, of dimension $m - 1$. There is no inseparability in this assertion, even when the characteristic divides m . The local submersion theorem supplies solutions $a \rightarrow 1$. Formula (9.7) then gives $w, q \rightarrow 1$. If only the first norm is specified, the same argument uses the ordinary quadratic norm map. Around any sufficiently small fixed nonzero ε , these properties persist on an open set, by the same continuous étale identifications and norm submersion. $\square$

At auxiliary places split in LS_0 and in D , these neighborhoods can also prescribe a Frobenius cycle type for Ψ . In split coordinates take

$$u = \begin{pmatrix} I & \varepsilon I \\ \varepsilon X & I + \varepsilon^2 X \end{pmatrix}. \quad (9.13)$$

Its determinant is one and $h_+ = I + \varepsilon^2 X$. Choose X integral with invertible separable residue characteristic polynomial of the specified factorization type. At sufficiently large residue fields all cycle types are available. Taking ε small gives the norm solutions and smallness of Lemma 9.7 while retaining that type for the étale field generated by h .

9.8 The approximation conditions for the even step

Assume the induction hypothesis in degrees smaller than $2m$. We will choose a representative u of an arbitrary coset in V_0/R for which Lemma 9.4 can be applied globally. The choice must control the anisotropic places of the variable quaternion algebra Q , as well as the fixed group $SU(C_s)$.

Choose a finite set B containing A , the places below the anisotropic places of $SU(C_s)$, all exceptions to integral models of the fixed data, and an isotropic place v_0 of S . In degree four include B_s and apply Proposition 9.6 at this point. Denote this set by B_{corr} ; its field F_1 and neighborhoods W_v , $v \in B_{\text{corr}}$, are now fixed. Subsequent enlargements of B retain those neighborhoods on B_{corr} and impose no additional determinant-correction condition. At every added place we may still choose near-identity openings using Lemma 9.7.

Enlarge B by sufficiently large auxiliary places split in LS_0 and in D at which we impose, using (9.13), every permutation cycle type of degree m . These conditions force the splitting field of Ψ over LS_0 to have group S_m . To recall the group-theoretic argument, the transitive S_m -action on the cosets of a proper subgroup has a derangement, as follows by averaging the numbers of fixed points. An element conjugate into that subgroup fixes a coset, so the derangement's conjugacy class misses the subgroup. A subgroup containing every cycle type therefore cannot be proper.

At each place of B , use Lemma 9.7. Require u to lie sufficiently near 1 that its A -coordinate belongs to P_0 ; then these openings are compatible with every coset in V_0/R by Proposition 2.4. Require w to be small enough for Lemma 9.2. When $m \geq 3$, require q to be small enough to be killed in the fixed smaller group $SU(C_s)$. When $m = 2$, require $q_v \in W_v$ for $v \in B_{\text{corr}}$; at the other places of B we may require any sufficiently small identity neighborhood.

Outside B we allow either $u \notin \Omega$, or a point $u \in \Omega$ for which the required norm equations are soluble and Q is split at every local factor of F . We also allow a nonempty open set near 1 from Lemma 9.7, where the norm equations are soluble and w is small enough for Lemma 9.2; this second open set will only be used at the denominator places B_1 introduced in Proposition 3.4. Once such a place is known, we choose an indicated local solution a there. No extra condition on q is needed outside B , because all anisotropic places of the fixed group lie above B .

We verify the two remaining hypotheses of the approximation proposition: a codimension-two exclusion at integral reductions and nonempty algebraic tests at a single pole. This will ensure that the only nonsplit quaternion factors occur above $B \cup B_1$, where their elements are already small.

9.8.1 Integral reduction

Over a splitting field consider the four divisors

$$\det A' = 0, \quad \det B' = 0, \quad \det C' = 0, \quad \det D' = 0$$

in the split matrix group. Exclude their pairwise intersections. If $m \geq 3$, impose also the following conditions. Off all four divisors, Ψ must have at least one simple absolute root. On one divisor alone, require 0 to be a simple root for a diagonal divisor, and 1 to be a simple root for an off-diagonal divisor. The asserted root already occurs by the complementary-minor identities; for example

$$1 - A'(u^{-1})_{++} = B'(u^{-1})_{-+}.$$

For $m = 2$ there is no simple-root requirement.

These exclusions are invariant under descent. Exchanging s -labels interchanges the two diagonal divisors and the two off-diagonal divisors. The unitary diagram action does the same, because it acts by inverse transpose and complementary minors. More explicitly, the automorphisms of LS_0/k fixing L and S_0 each interchange both pairs, while their product fixes each divisor. The permutation character on either pair is therefore the quadratic character of J_0/k . The excluded closures define a k -subvariety Y .

We claim that Y has geometric codimension at least two. Work first in GL_{2m} . The four determinant hypersurfaces are distinct and irreducible, so their pairwise intersections have codimension at least two. On each one, the simple-root requirement holds generically: use independent two-dimensional transformations on m pairs, with one line in each block, and arrange exactly one of the resulting values to be 0 or 1. Off the divisors the matrix $B'D'^{-1}C'A'^{-1}$ is a submersive parameter. For $m \geq 3$, matrices with no simple characteristic root have codimension at least two. Indeed their distinct eigenvalues have at most $\lfloor m/2 \rfloor$ parameters, and each Jordan orbit has dimension at most $m^2 - m$. The resulting dimension is at most $m^2 - 2$. Finally all these tests are invariant under scalar multiplication of u , which carries the codimension statement to SL_{2m} . Enlarge B to make these statements valid on the integral models.

Let an integral point reduce outside Y , and assume it belongs to Ω . If J_0 is nonsplit at this place, its Frobenius interchanges both pairs of divisors. A rational residue point lying on one divisor would therefore lie on its mate, which is excluded. Thus it lies on none, and h and $1 - h$ are units in every local factor of F . The quadratic extension defined by J_0 is unramified, so $-h(1 - h)$ and h are norms of units. This proves that Q is split and gives a unit solution of the first equation in (9.6).

For $m \geq 3$ the second equation can be imposed as well. Given a unit solution of the first, its determinant discrepancy is a norm-one unit in J_0 . A simple residue root of Ψ gives an unramified factor of F over the local ground field. Write the discrepancy as $y/\bar{y}$ by Hilbert 90, taking y a unit; its valuation can be removed by a ground-field uniformizer. The norm on units from that unramified factor, after extending to J_0 , is surjective. Lift y through that norm and divide the lift by its conjugate. This changes the second norm by the required discrepancy without changing the first.

If J_0 splits, the quaternion algebra is split and the first equation is immediate. For the second equation, identify J_0 with two copies of the local field. Choose the first component of a with prescribed norm equal to the first component of d_0 , and determine the other component from h ; compatibility (9.5) gives its correct norm. Off the divisors, the prescribed norm is a unit and the simple residue root again supplies an unramified factor. On one divisor, the simple residue root 0 or 1 lifts to a degree-one factor of F , whose norm has no restriction on valuation. This proves all required local assertions for integral points reducing outside Y .

9.8.2 A single pole

Take the fixed Galois field M in Proposition 3.4 large enough to split all fixed data. A single pole then occurs only at a place split completely in M . In degree four, J_0 is consequently split, so there is no additional pole condition. For $m \geq 3$, it is enough to force Ψ to split, which makes both norm equations soluble.

Order the cocharacter exponents as $d_1 < \dots < d_{2m}$. At a single pole the split matrix expression is

$$u = g_L \operatorname{diag}(t^{ed_1}, \dots, t^{ed_{2m}}) g_R,$$

with integral invertible side factors and $\operatorname{ord}(t) > 0$. For the b -th elementary coefficient of Ψ , $1 \leq b \leq m$, require

$$\operatorname{ord}(c_b(\Psi)) = e \operatorname{ord}(t) \sum_{j=1}^b (d_j - d_{2m+1-j}). \quad (9.14)$$

These equalities follow from finitely many nonvanishing minor tests. Indeed write the coefficient as the exterior trace for $PuPu^{-1}P$. The uniquely lowest-weight term uses the first b indices in the positive exterior power and the last b in the inverse. Its coefficient is a product of opposite minors in the two rank- m projections

$$g_R P g_R^{-1}, \quad g_L^{-1} P g_L.$$

Requiring these minors to be nonzero on reduction gives (9.14). Such minors are generically nonzero on the variety of projections: one may use complementary subspaces in general position, or independent two-dimensional projections on the paired coordinates. The successive differences of the displayed valuations are strictly ordered. Every segment of the Newton polygon therefore has length one, and Ψ splits over the local field.

We must also check the order of the generic choices in Proposition 3.4. On the hyperplane belonging to a factor with conjugator k_j , put all other parameters equal to 1. Then, in the corresponding fixed bases, $g_R = k_j^{-1}$ and $g_L = x k_j$, where x is the initial rational point. For the two basic lists first choose the conjugators so that the right-hand minor tests hold. Next choose x , subject also to the prescribed local openings, so that all left-hand tests hold. These are finitely many nonempty Zariski open conditions. For every appended conjugator, x has already been fixed; both conditions are still nonempty open conditions on that conjugator, since conjugation sweeps out the full variety of rank- m projections. Impose the tests also on all Galois-conjugate hyperplanes. They form one finite algebraic family independent of the place, and inserting identity parameters in appended factors preserves all earlier tests. This is exactly the pole hypothesis of the approximation proposition.

9.8.3 Choosing the representative and solving the norms

All hypotheses of Proposition 3.4 are now verified. Apply it to a prescribed coset in V_0/R and the local openings at B . At the finite set B_1 of denominator places of its initial point, use the near-identity open sets already described, with local solutions a making w sufficiently small. At any further model or projection exceptions introduced in the proof of that proposition, use the integral-reduction open sets supplied by its original basic list. These later places therefore introduce no additional nonsplit quaternion factors requiring smallness. We obtain u in the prescribed coset, with

$$u \in \Omega, \quad \operatorname{Gal}(\Psi/LS_0) = S_m,$$

with all required norm equations locally soluble, and with Q split outside $B \cup B_1$.

The Galois condition implies that $F = k(h)$ is a field. If $\tilde{F}$ is its splitting field over k , then $\text{Gal}(\tilde{F}/k) \subset S_m$ already contains $\text{Gal}(\tilde{F}LS_0/LS_0) = S_m$. Thus this group is S_m and $\tilde{F} \cap LS_0 = k$. In particular the pair F, J_0 has the joint group required by Proposition 4.1.

For $m \geq 3$, apply that theorem to (9.6), using (9.5). It gives a global a approximating the selected local solutions at $B \cup B_1$. The binary factor w is killed: its quaternion algebra is split away from these places, and at every anisotropic factor above them its eigenvalues satisfy Lemma 9.2. The factor $q \in \text{SU}(C_s)$ is killed by the induction hypothesis and its smallness at the anisotropic places of this fixed degree- m group. Hence $u = wq$ is killed.

For $m = 2$, the first equation of (9.6) is a cyclic norm equation over F . The Hasse norm theorem gives a global solution [22]. Its solution variety is a torsor under the norm-one torus of a separable quadratic extension; after choosing a solution this torus is rational, so it has weak approximation. We may therefore approximate all selected local solutions at $B \cup B_1$. The binary factor w is killed by the same argument, and $q \in \text{U}(C_s) \cap S$ lies in the neighborhoods fixed in Proposition 9.6. That proposition kills q as well. This completes the even step.

Proof of Theorem 9.1. Proceed by induction on the degree, simultaneously over all global function fields. The inner and isotropic cases supply the base and all instances of those types. The odd-degree argument proves the assertion directly in division degree and by smaller plane groups otherwise. In even degree the preceding construction kills a representative of each coset in V_0/R , and hence $V_0/R = 1$.

To justify the form of the induction hypothesis used for smaller groups, recall that vanishing of the power defect for every e already gives their Margulis–Platonov assertion. A noncentral normal subgroup has finite index by Lemma 2.2; choosing e to kill the finite quotient puts the generated power subgroup inside it. The equality $V_0 = R$ makes this power subgroup open for the topology from the anisotropic places. A subgroup containing it is open and, by density, is the inverse image of its open normal closure. Thus each completed degree supplies precisely the hypothesis used in the next degree, including over finite separable extensions of k . $\square$

10 Proof of the main theorem

Proof of Theorem 1.1. The global Kneser–Tits theorem handles the isotropic case, and the known inner-type- A theorem handles anisotropic inner forms. As explained in Section 2, Harder’s theorem leaves only the anisotropic unitary groups $S = \text{SU}(D, *)$ of degree $n \geq 3$.

Let $N \triangleleft S(k)$ be noncentral. By Theorem 2.2, the group $S(k)/N$ is finite. Choose an integer $e \geq 1$ divisible by its exponent. With the notation (2.4), we have $R \subset N$, while Theorem 9.1 gives

$$R = V_0 = \delta_A^{-1}(P_0).$$

Thus N is a union of cosets of the open subgroup $\delta_A^{-1}(P_0)$. More explicitly, density of $\delta_A(S(k))$ induces an isomorphism of finite groups

$$S(k)/R \simeq H_A/P_0.$$

Let W be the inverse image in H_A of the normal subgroup corresponding to N/R . Then W is open and normal, and $N = \delta_A^{-1}(W)$. Density also shows that $W = \overline{\delta_A(N)}$, so this local subgroup is uniquely determined by N . If A is empty, H_A is trivial and the same argument gives $N = S(k)$. $\square$

A The finite simple-group obstruction

We prove Theorem 6.1, following the finite-group argument of [23, Section 6]. In addition to the double-coset obstruction proved there, we verify for each constructed set that its complement in an abelian subgroup meeting it is a subgroup. In characteristic two, this property supplies the finite abelian quotient used in Proposition 7.6. The resulting functions have additive period groups of finite index, which rules out the exceptional points left by the density argument.

The proof uses the classification of finite simple groups [1, Classification Theorem, p. 737]. Its common mechanism is to locate a part of a commuting pair on which centralizers are small. For alternating and linear groups, this part consists of two points or a line and a hyperplane. For the remaining classical groups it is a large irreducible subspace. Most exceptional and sporadic groups instead have an abelian subgroup whose nonidentity elements have exactly that subgroup as centralizer. We first establish the two elementary tests behind these constructions.

Throughout this section, $\mathcal{F}$ is a finite nonabelian simple group, $C(B) = C_{\mathcal{F}}(B)$, and

$$I(\mathcal{F}) = \{g \in \mathcal{F} : g^2 = 1\}, \quad k(\mathcal{F}) = \text{the number of conjugacy classes of } \mathcal{F}.$$

For commuting $W, Z \in \mathcal{F}$, the condition to be excluded is

$$ZW \in C(B)(ZW^{-1})C(BZ^2) \quad \text{for every } B \in \mathcal{F}. \quad (\text{A.1})$$

The choice of the nonempty proper conjugacy-invariant set $\mathcal{S}$ will be specified in each family. For every choice we must prove both that Equation (A.1) fails for some B whenever $W \in \mathcal{S}$ and $Z \notin \mathcal{S}$ commute, and that the complement condition in Theorem 6.1(ii) holds.

A.1 Two elementary tests

First take

$$\mathcal{S} = \{W \in \mathcal{F} : W^2 \neq 1\}. \quad (\text{A.2})$$

This set is nonempty, since a group of exponent two is abelian, and it does not contain the identity. For every abelian subgroup A , its complement is the subgroup $A[2] = \{a \in A : a^2 = 1\}$. Thus this choice automatically satisfies Theorem 6.1(ii). For commuting $W \in \mathcal{S}$ and $Z \notin \mathcal{S}$, put $v = ZW$. Then $Z^2 = 1$, $v^2 = W^2 \neq 1$, and Equation (A.1) becomes

$$v \in C(B)v^{-1}C(B) \quad \text{for every } B \in \mathcal{F}. \quad (\text{A.3})$$

Suppose that $\mathcal{F}$ acts on a set and that some point p satisfies $v^2p \neq p$. If $C(B)$ fixes p and vp pointwise, writing $v = av^{-1}b$ with $a, b \in C(B)$ gives $vp = av^{-1}p$. Applying a^{-1} gives $vp = v^{-1}p$, a contradiction. We call this the point-pair test. We will also use its incidence version: if $C(B)$ fixes a line p and preserves a subspace L , while $vp \subset L$ and $v^{-1}p \not\subset L$, the same equation is impossible.

The second test uses a self-centralizing abelian subgroup.

Lemma A.1. *Suppose that $C \subset \mathcal{F}$ is an abelian subgroup of odd order $c > 1$ such that $C_{\mathcal{F}}(t) = C$ for every $1 \neq t \in C$. If*

$$|\mathcal{F}| > k(\mathcal{F})c^4, \quad (\text{A.4})$$

then the union $\mathcal{S}$ of the conjugates of $C \setminus \{1\}$ satisfies both assertions of Theorem 6.1.

Proof. The set is nonempty and excludes the identity. If $W \in \mathcal{S}$ and Z commutes with W , then Z lies in the same conjugate of C as W . Consequently $Z \notin \mathcal{S}$ forces $Z = 1$. More generally, if an

abelian subgroup A meets $\mathcal{S}$, conjugating one member of $A \cap \mathcal{S}$ into C puts all of A in C ; hence $A \setminus \mathcal{S} = \{1\}$. This proves the complement assertion.

Conjugate W into C . Testing Equation (A.3) with every conjugate of a fixed nonidentity element of C shows that every conjugate v of W belongs to $Cv^{-1}C$. If $v = av^{-1}b$, $a, b \in C$, then

$$(vb^{-1})^2 = ab^{-1} \in C.$$

If this square is nonidentity, vb^{-1} centralizes a nonidentity element of C , so $v \in C$. Otherwise $v \in I(\mathcal{F})C$. Since $I(\mathcal{F})$ is conjugacy-invariant, $I(\mathcal{F})C = CI(\mathcal{F})$. Thus the whole conjugacy class of W lies in $CI(\mathcal{F})$, and

$$\frac{|\mathcal{F}|}{c} \leq c|I(\mathcal{F})|.$$

The Frobenius–Schur formula and Cauchy–Schwarz give

$$|I(\mathcal{F})| = \sum_{\chi \in \text{Irr}(\mathcal{F})} \nu_2(\chi)\chi(1) \leq \sum_{\chi} \chi(1) \leq \sqrt{k(\mathcal{F})|\mathcal{F}|}.$$

These inequalities contradict Equation (A.4). □

A.2 Alternating, linear, and sporadic groups

For alternating and projective special linear groups we use Equation (A.2). In A_n , choose p with $v^2p \neq p$. There is an even permutation B fixing exactly p, vp and having distinct odd cycle lengths greater than one on the remaining points, except when $n = 6$ or 8 . Indeed, for odd n use the single cycle of length $n - 2$, and for even $n \geq 10$ use lengths $3, n - 5$. Every permutation centralizing these moving cycles is even on their support. Therefore a centralizer element in A_n cannot interchange the two fixed points. The point-pair test applies. The two exceptions are $A_6 \simeq \text{PSL}_2(9)$ and $A_8 \simeq \text{PSL}_4(2)$.

Now let $\mathcal{F} = \text{PSL}_l(q)$, $l \geq 3$. Choose a line p such that $v^2p \neq p$. The three lines $p, vp, v^{-1}p$ are distinct. A hyperplane L can be chosen to contain vp and contain neither p nor $v^{-1}p$: in the quotient by vp , choose a linear functional nonzero on both remaining nonzero vectors. The union of two proper hyperplanes in the dual is not the entire dual, also over $\mathbb{F}_2$.

On the direct sum $p \oplus L$, take B to be the identity on p and an irreducible norm-one field multiplication on L . Such an element exists in the cyclic group of order $(q^{l-1} - 1)/(q - 1)$: a generator cannot lie in a proper subfield, since that group's order exceeds $q^{(l-1)/2} - 1$ when $l - 1 \geq 2$. A projective centralizer of B is an actual centralizer. Indeed, scalar multiplication must preserve its unique base-field eigenvalue, which is 1, and hence the multiplier is 1. The centralizer fixes p and preserves L , contradicting the incidence version of Equation (A.3).

For simple $\text{PSL}_2(q)$, the pointwise stabilizer of two points of the projective line is a split torus. It contains an element whose centralizer fixes both points, unless $q = 5$: its order is $q - 1$ for even q and $(q - 1)/2$ for odd q , and a noninvolution in it is regular with that torus as centralizer. The case $q = 5$ is A_5 . Choosing the torus for p, vp proves the assertion in all these cases.

For the sporadic groups other than M_{12} , and also for the Tits group, Table 1 gives a prime c for which a cyclic subgroup of order c is the centralizer of each of its nonidentity elements. The centralizer orders, class numbers, and group orders are those in the *Atlas* and the online *ATLAS of Finite Group Representations* [5, 46]. In each row, an element x in the class labeled cA in that group's conjugacy-class table has $C(x) = \langle x \rangle$ of order c . Since c is prime, every nonidentity power of x has the same centralizer. The displayed k is the exact class number. In every row the group order exceeds

Group	c	k	Group	c	k
M_{11}	5	10	O'N	31	30
M_{22}	11	12	Co ₁	23	101
M_{23}	23	17	Co ₂	23	60
M_{24}	23	26	Co ₃	23	42
J_1	7	15	Fi ₂₂	13	65
J_2	7	21	Fi ₂₃	23	98
J_3	19	21	Fi' ₂₄	29	108
J_4	43	62	HN	19	54
HS	11	24	Ly	67	53
McL	11	24	Th	31	48
He	17	33	$\mathbb{B}$	47	184
Ru	29	36	$\mathbb{M}$	71	194
Suz	13	43	${}^2F_4(2)'$	13	22

Table 1: Self-centralizing prime-order subgroups for the sporadic and Tits groups.

kc^4 , so Lemma A.1 applies. For example, the smallest margin is $10 \cdot 5^4 = 6250 < 7920 = |M_{11}|$. One may also check the other inequalities using $k \leq 200$, except that $k \leq 25$ suffices for M_{22}, M_{23}, J_1 .

For M_{12} use its sharply 5-transitive action of degree 12 and the *Atlas* class $8B$, of cycle shape $1^2 2 8$ [5, 46]; the online database labels this permutation representation $12a$ (the other degree-12 action interchanges the classes $8A$ and $8B$). An element centralizing $B \in 8B$ restricts to a cyclic shift on its eight-point orbit. This restriction is injective: its kernel fixes eight points, and an element fixing five points is the identity. The centralizer is therefore exactly $\langle B \rangle$, and fixes the two fixed points individually. Two-transitivity moves this pair to p, vp , so the point-pair test applies with Equation (A.2).

A.3 An auxiliary unitary count

The remaining classical groups need a torus argument on a large irreducible block. The relevant obstruction takes place in the full unitary isometry group, which need not itself be simple.

Lemma A.2. *Let $p \geq 3$ be an odd prime, let $H = \mathrm{U}_p(q)$ be the full isometry group of a nondegenerate Hermitian form over $\mathbb{F}_{q^2}$, and let $T \subset H$ be an irreducible torus of order $q^p + 1$. If $w \in T$ has field degree p over $\mathbb{F}_{q^2}$, then*

$$w \in T^h w^{-1} T^h \quad \text{for every } h \in H$$

is impossible.

Proof. Put

$$\overline{H} = H/Z(H), \quad \overline{T} = T/Z(H), \quad c = \frac{q^p + 1}{q + 1}, \quad d = (p, q + 1).$$

The actual centralizer of w is T . If g centralizes its projective image, then $gwg^{-1} = \lambda w$ for a scalar λ of norm one. Taking determinants gives $\lambda^p = 1$, so

$$|C_{\overline{H}}(\overline{w})| \leq dc. \tag{A.5}$$

Suppose the displayed condition in the lemma holds. Every conjugate v of w then lies in $Tv^{-1}T$. The calculation used in Lemma A.1 gives an element $u = vb^{-1}$ whose square lies in T . If u^2 is

nonscalar, it has field degree p , because p is prime; its centralizer is T , and hence $u \in T$. If u^2 is scalar, $\bar{u}$ has square one. Consequently the entire projective conjugacy class of $\bar{w}$ lies in $\overline{TI}(\overline{H})$.

The determinant quotient of $\overline{H}$ has order d , and $\overline{T}$ maps onto it. To see this, write T as the norm-one subgroup of $\mathbb{F}_{q^{2p}}^\times$ over $\mathbb{F}_{q^p}$. The determinant of a field multiplication is its norm to $\mathbb{F}_{q^2}$. On this torus, that norm takes a generator to a generator of the scalar norm-one group of order $q+1$; its exponent is congruent to c modulo q^p+1 . Modding out by scalar matrices mods the determinant out by p th powers, giving the quotient of order d .

All members of the conjugacy class have the same determinant in this quotient. For each fixed $i \in I(\overline{H})$, at most c/d elements $t \in \overline{T}$ have ti with that determinant. Combining this with Equation (A.5) gives

$$\frac{|\overline{H}|}{dc} \leq \frac{c}{d} |I(\overline{H})|, \quad \text{and therefore} \quad |\overline{H}| \leq c^2 |I(\overline{H})|.$$

We show that the reverse strict inequality always holds.

Every projective involution lifts to an involution in H . Indeed, if $g^2 = \alpha I$, then $\alpha^p = \det(g)^2$ in the scalar cyclic group of order $q+1$. As p is odd, α is a square in that group, and scalar rescaling gives an involution.

For odd q , an involution is determined up to conjugacy by the dimensions $a, p-a$ of its two nondegenerate eigenspaces. Projectively the unordered partitions are indexed by $1 \leq a < p/2$, and their centralizers have order at least

$$\frac{|U_a(q)| |U_{p-a}(q)|}{q+1}. \quad (\text{A.6})$$

For even q , the Jordan type is $2^a 1^b$, where $b = p - 2a$. Its full unitary centralizer has order

$$q^{a^2+2ab} |U_a(q)| |U_b(q)|. \quad (\text{A.7})$$

We recall the structure behind this formula, including in characteristic two; see also [45, Section 2.6, Case (A)(ii),(iv), pp. 34–36]. In the ambient general linear group the connected matrix centralizer has reductive quotient $\text{GL}_a \times \text{GL}_b$ and unipotent radical of dimension $a^2 + 2ab$. The Hermitian Frobenius gives the unitary forms on both multiplicity spaces. Equivalently, for $g = 1 + N$ of order two, $N = N^*$; the induced forms on $\ker N / \text{im } N$ and, using N , on the quotient by $\ker N$ are the forms on these two spaces. Nondegenerate Hermitian forms of a given dimension over a finite field are isometric. Connectedness and Lang's theorem give one full-unitary class for each occurring type and the stated order. Dividing by $q+1$ gives a lower bound for its projective centralizer.

The unitary order formula implies

$$|U_j(q)| = q^{j^2} \prod_{r=1}^j (1 - (-q)^{-r}) \geq q^{j^2}. \quad (\text{A.8})$$

For completeness, pair consecutive factors: their product is

$$(1 + q^{-(2r-1)})(1 - q^{-2r}) = 1 + q^{-2r}(q - 1 - q^{-(2r-1)}) > 1.$$

An unpaired final factor also exceeds one. Thus every nonidentity projective involution class has centralizer order at least

$$M = \frac{q^{(p^2+1)/2}}{q+1}.$$

There are at most $(p-1)/2$ such classes, and including the identity gives

$$\frac{|I(\overline{H})|}{|\overline{H}|} \leq \frac{p+1}{2M}.$$

When $p \geq 5$, we have $c < q^{p-1}$ and

$$\frac{M}{q^{2p-2}} = \frac{q^{(p^2-4p+5)/2}}{q+1} \geq \frac{2^p}{3} > \frac{p+1}{2}.$$

The first inequality uses $(p^2-4p+5)/2 \geq p$ and monotonicity in $q \geq 2$. This proves $c^2|I(\overline{H})| < |\overline{H}|$.

It remains to check $p = 3$. Now

$$c = q^2 - q + 1, \quad h = |\overline{H}| = q^3(q^3 + 1)(q^2 - 1).$$

There is one nonidentity involution class. In odd characteristic its centralizer lower bound is $M = q(q-1)(q+1)^2$, and $h/M = q^2c$. In even characteristic it is $M = q^3(q+1)$, and $h/M = (q^2-1)c$. In both cases $|I(\overline{H})| \leq 1 + q^2c$, whereas

$$h - c^2(1 + q^2c) = c((3q-4)q^4 + (q-2)q^2 + q - 1) > 0 \quad (q \geq 2).$$

This includes $U_3(2)$ and finishes the proof. $\square$

A.4 Symplectic and orthogonal groups

Let $\mathcal{F}$ be a projective symplectic or orthogonal simple group on its natural space of dimension $2l$ or $2l+1$. We first treat $l \geq 3$, using the standard low-rank identifications and treating PSp_4 separately below. Choose an odd prime

$$l/2 < p \leq l, \tag{A.9}$$

requiring $p < l$ in plus orthogonal type. For $l \geq 4$, set $m = \lfloor l/2 \rfloor \geq 2$. Bertrand's postulate gives $m < p < 2m$, hence $l/2 < p < l$, and this prime is odd. For $l = 3$ take $p = 3$, apart from plus type, which is PSL_4 and has already been treated. In odd-dimensional orthogonal type we may assume odd characteristic, since in even characteristic the group has the corresponding symplectic realization.

We use a nondegenerate block E of dimension $2p$ carrying a torus of order $q^p + 1$. In orthogonal type this block has minus type; the complement can be chosen with the sign needed for the ambient form. For minus type $p = l$ the block is the whole space, whereas the requirement $p < l$ in plus type leaves a complement of the necessary sign. The torus is the norm-one subgroup of $\mathbb{F}_{q^{2p}}^\times$ over $\mathbb{F}_{q^p}$, acting on E by field multiplication.

Define $\mathcal{S}$ to consist of those projective elements whose square, on the natural space, has an irreducible self-dual block of dimension $2p$. This condition is independent of the choice of a scalar lift and is conjugacy-invariant. The block is unique, because $4p > 2l + 1$, and is nondegenerate: its dual would otherwise require a second block of the same degree. The set excludes identity and is nonempty. Indeed, if t generates the torus of order $q^p + 1$, use t^2 on the block and identity on the complement. Squaring meets the orthogonal component and spinor conditions, whose quotients have exponent two. Moreover t^4 still has degree $2p$. To see this, the quotient of the norm-one torus by its intersection with $\mathbb{F}_{q^2}^\times$ has odd order

$$\frac{q^p + 1}{q + 1} > 1.$$

Thus $t^4 \notin \mathbb{F}_{q^2}$. The only other maximal proper subfield is $\mathbb{F}_{q^p}$, where a norm-one element is ± 1 and hence already lies in $\mathbb{F}_{q^2}$.

Suppose now that $W \in \mathcal{S}$, that $Z \notin \mathcal{S}$ commutes with W , and that Equation (A.1) holds. Lift W, Z to natural isometries, using Ω in orthogonal type. Their possible projective commutation multiplier has order at most two. They therefore commute actually with W^2 and preserve its unique marked block E . On E they act as field multiplications $w, z \in \mathbb{F}_{q^{2p}}^\times$ of norm one. Since $Z \notin \mathcal{S}$, z^2 lies in a proper subfield. The norm-one condition puts z^2 in $\mathbb{F}_{q^2}$, and then z itself is in $\mathbb{F}_{q^2}$: its degree over that field divides both p and 2.

There is a full unitary group $H = \mathrm{U}_p(q)$ acting on E and containing its field torus T . One way to see this directly is to write the invariant form in field coordinates. Its bilinear form has the shape

$$\mathrm{Tr}_{\mathbb{F}_{q^{2p}}/\mathbb{F}_q}(\kappa xy^*), \quad x^* = x^{q^p}.$$

Take the trace first to $\mathbb{F}_{q^2}$. If the resulting form is anti-Hermitian and the characteristic is odd, multiply it by $\delta \in \mathbb{F}_{q^2}^\times$ with $\delta^q = -\delta$; this makes it Hermitian without changing its isometry group. In characteristic two an anti-Hermitian form is already Hermitian. This gives the required unitary structure. In the quadratic case the form is

$$Q(x) = \mathrm{Tr}_{\mathbb{F}_{q^{2p}}/\mathbb{F}_q}(\kappa xx^*), \quad \kappa \in \mathbb{F}_{q^p}^\times.$$

In characteristic two this expression follows from the polarization and W^2 -invariance: two invariant quadratic forms with the same polarization differ by the square of an invariant linear form, and an irreducible block of degree greater than one has no such nonzero form. More explicitly, the Hermitian form $h(x, y) = \mathrm{Tr}_{\mathbb{F}_{q^{2p}}/\mathbb{F}_{q^2}}(\kappa xy^*)$ in the quadratic case satisfies $h(x, x) = Q(x)$: on $\mathbb{F}_{q^p}$ the trace to $\mathbb{F}_{q^2}$ equals the trace to $\mathbb{F}_q$, since p is odd. Thus H preserves the original form on E in both cases. The element z is a scalar of this unitary group.

For each $h \in H$, choose B acting on E by a square of a generator of T^h and as identity on the complement. This is an allowed ambient element. Even if h is not in Ω , conjugation by an isometry preserves Ω , so the orthogonal component condition is unchanged. Both B and BZ^2 have a unique marked square block E : multiplication of a full-degree element by a base-field scalar preserves its degree over $\mathbb{F}_{q^2}$, and the norm-one condition then gives full degree $2p$ over $\mathbb{F}_q$.

Each projective centralizer of B or BZ^2 commutes actually with its square, preserves E , and restricts there into T^h . This does not require the action on the complement to be semisimple; the complement is too small to have an irreducible factor of degree $2p$. Restricting Equation (A.1) to E and absorbing its scalar multipliers and the scalar z into T^h therefore gives

$$w \in T^h w^{-1} T^h \quad \text{for every } h \in H.$$

The element w has degree p over $\mathbb{F}_{q^2}$. This contradicts Lemma A.2.

To verify the complement assertion, let A be abelian and choose $W \in A \cap \mathcal{S}$. Its marked block E is now fixed. The preceding centralizer argument gives a well-defined homomorphism

$$\rho : A \longrightarrow T/(T \cap \mathbb{F}_{q^2}^\times). \quad (\text{A.10})$$

Indeed every lift restricts on E to a field multiplication, and changing the lift multiplies it by an ambient central scalar lying in $\mathbb{F}_{q^2}^\times$. The complement of E is too small to carry a marked block. For a restriction $z \in T$, its square has proper degree over $\mathbb{F}_q$ precisely when $z^2 \in \mathbb{F}_{q^2}$: the other maximal proper subfield is $\mathbb{F}_{q^p}$, whose norm-one elements are ± 1 . Because p is odd, $z^2 \in \mathbb{F}_{q^2}$ is equivalent to $z \in \mathbb{F}_{q^2}$. Full degree automatically gives self-duality by the norm-one equation. Thus $A \setminus \mathcal{S} = \ker \rho$, as required.

A.5 Projective unitary groups and rank two symplectic groups

Let $\mathcal{F} = \text{PSU}_l(q)$, $l \geq 3$, excluding the nonsimple pair $(l, q) = (3, 2)$. Choose an odd prime $l/2 < p \leq l$. Define $\mathcal{S}$ by requiring the square of a natural lift to have an irreducible block of degree p over $\mathbb{F}_{q^2}$. As above this block is unique and nondegenerate. If $l > p$, the determinant of a chosen torus element on the block can be corrected on its nonzero dimensional orthogonal complement. If $l = p$, use instead the determinant-one torus of order

$$c = \frac{q^p + 1}{q + 1}.$$

Its scalar intersection has order $d = (p, q + 1)$. Except for $(p, q) = (3, 2)$, $c > d$: for $p \geq 5$, $c \geq q^{p-2}(q - 1) + 1 \geq 2^{p-2} + 1 > p$, and for $p = 3, q \geq 3$, $c \geq 7 > 3$. As c is odd, a generator and its square are both nonscalar and hence have degree p . This proves that $\mathcal{S}$ is nonempty; it again excludes identity.

We check the projective-centralizer issue before applying the block argument. Suppose $gAg^{-1} = \lambda A$, where $A \in \text{SU}_l(q)$ has the marked square block. Multiplication by a base-field scalar preserves irreducible degrees, so g preserves the unique block. The determinant on the block gives $\lambda^p = 1$, and the total determinant gives $\lambda^l = 1$. If $l > p$, then $p < l < 2p$, so these conditions force $\lambda = 1$. If $l = p$ and $\lambda \neq 1$, multiplication by λ cycles all p distinct eigenvalues. Writing w for any one of them, their product is w^p , since p is odd, and determinant one gives $w^p = 1$. But λ has order p in the scalar group, so $p \mid q + 1$; all p th roots of unity then lie in $\mathbb{F}_{q^2}$, contradicting the full degree of w . Thus these projective centralizers are actual centralizers on the block.

For commuting $W \in \mathcal{S}$ and $Z \notin \mathcal{S}$, the restrictions are consequently field scalars w, z , with $z \in \mathbb{F}_{q^2}$ by the same odd-degree argument as before. For every conjugate T^h in the full block unitary group choose B with irreducible square on that torus, correcting its determinant on the complement if necessary. The square of BZ^2 retains full degree. Applying the preceding projective-centralizer argument to B and BZ^2 , Equation (A.1) restricts to the relation prohibited by Lemma A.2.

For the complement assertion, fix $W \in A \cap \mathcal{S}$ in an abelian subgroup A . The projective-centralizer argument just proved gives a restriction homomorphism as in Equation (A.10), now with T the torus on the p -dimensional Hermitian block. A restriction z has square of degree less than p precisely when z^2 , equivalently z , is in $\mathbb{F}_{q^2}$. Since the complementary subspace has dimension less than p , this characterizes nonmembership in $\mathcal{S}$. Once again $A \setminus \mathcal{S} = \ker \rho$.

For $\text{PSp}_4(q)$ take $\mathcal{S}$ to be the set whose square is irreducible on the natural four-dimensional space. It is nonempty by the torus of order $q^2 + 1$ and excludes identity. Lifting a commuting pair $W \in \mathcal{S}$, $Z \notin \mathcal{S}$ gives norm-one field scalars $w, z \in \mathbb{F}_{q^4}^\times$. Since $z^2 \in \mathbb{F}_{q^2}$, the norm-one condition gives $z^2 = \pm 1$. Thus $Z^2 = 1$ projectively, and $v = ZW$ still has an irreducible square.

Choose a nonzero vector x such that x, vx span an isotropic plane L . Such an x exists in the field description of the alternating form: the condition $\text{Tr}_{\mathbb{F}_{q^4}/\mathbb{F}_q}(\kappa x(vx)^*) = 0$ is one homogeneous $\mathbb{F}_q$ -linear condition on $xx^* \in \mathbb{F}_{q^2}$. It has a nonzero solution, and the field norm $x \mapsto xx^*$ is surjective. Irreducibility ensures that x, vx are independent and that $v^{-1}x \notin L$; otherwise v would satisfy a quadratic polynomial.

Choose a regular unipotent B whose unique line-plane flag is $\langle x \rangle \subset L$. Its existence in every characteristic can be seen in a symplectic flag basis from the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

for the alternating form with nonzero upper anti-diagonal entries 1, 1. This matrix preserves the form and has one Jordan block, also in characteristic two. Its centralizer preserves $\ker(B - 1) = \langle x \rangle$ and $\ker((B - 1)^2) = L$. A projective centralizer is actual, since a scalar multiple of a unipotent matrix can have the same eigenvalues only when the scalar is one. The incidence test now contradicts Equation (A.3). If an abelian A meets this $\mathcal{S}$, fix the irreducible block using any member of the intersection. Every other member restricts to a norm-one field scalar z , and the calculation above gives

$$z^2 \text{ has proper degree} \iff z^2 \in \mathbb{F}_{q^2} \iff z^2 = \pm 1.$$

The last condition says precisely that its projective image has square one. Thus $A \setminus \mathcal{S} = A[2]$. The nonsimple group $\mathrm{Sp}_4(2)$ is not needed; its simple derived group is A_6 .

A.6 Small tori in exceptional groups

We have proved the obstruction for all classical simple groups. For most exceptional families we apply Lemma A.1 to a rational maximal torus. We use the usual simply connected algebraic realization followed by its central quotient. Rational tori are obtained by Weyl-group twisting, and their orders are determinants of the Frobenius action minus one on their character lattices; see [42] and [19, Propositions 25.1–25.3, pp. 219–220]. Semisimple centralizers in a simply connected semisimple group are connected [41, Remark 2.10]. The tori below and their self-centralizing property in the simple quotient occur in [39, Corollary 8.2 and Lemma 8.7]. We verify that property here by a root-lattice estimate before applying Lemma A.1. Their orders outside type G_2 are also recorded in [12, Section 3]; the two G_2 orders follow directly from its order-three and order-six Weyl actions.

Write Φ_j for the j th cyclotomic polynomial. Use the tori in Table 2, before passing to the central quotient. For $G_2(q)$ choose the sign avoiding a factor 3, and for $q = 3$ choose the smaller order 7.

Group	Torus order	Weyl action
$G_2(q), q \geq 3$	$q^2 \pm q + 1$	Order 3 or 6
${}^3D_4(q), F_4(q)$	$\Phi_{12}(q)$	Characteristic polynomial Φ_{12}
${}^\epsilon E_6(q)$	$\Phi_9(\epsilon q)$	Φ_9 , with diagram sign
$E_8(q)$	$\Phi_{30}(q)$	Coxeter action

Table 2: Tori used for the exceptional simple groups; $\epsilon = 1$ denotes split E_6 and $\epsilon = -1$ twisted E_6 .

These actions are irreducible over $\mathbb{Q}$. For G_2, F_4, E_8 use a Coxeter element or an appropriate power. In triality D_4 , let b be the central node, a an outer node, and γ the diagram cycle of the outer nodes. The action $s_a s_b \gamma$ has characteristic polynomial $X^4 - X^2 + 1$. For E_6 , the regular order-nine Weyl element in Springer’s tables [40, Section 5.4, Table 1] has a primitive ninth-root eigenvalue. Its rational characteristic polynomial, of degree six, is therefore Φ_9 . Its negative belongs to the nontrivial diagram coset and gives the twisted case.

After passing to the central quotient, dividing by $(3, q - \epsilon)$ in type E_6 , every prime divisor of these torus orders is a primitive cyclotomic prime. Their respective lower bounds are

$$7, \quad 13, \quad 19, \quad 31.$$

Indeed a prime dividing $\Phi_m(q)$ either has multiplicative order m for q modulo that prime, or is an exceptional prime dividing m . The displayed polynomials have no such exceptional factor except for 3 in $\Phi_9(\epsilon q)$; when present this factor has valuation one and is exactly removed by the center. All resulting torus orders are odd and greater than one.

We next prove that *every* nonidentity element of each resulting torus has centralizer exactly that torus. It suffices to do so for an element t of prime order r upstairs, with r one of these primitive primes. If a root α vanished on t , Frobenius invariance would make every root in its twisted Weyl orbit vanish on t . By irreducibility that orbit spans the rational character space. Choose a linearly independent subset. The lattice it generates has full rank in the weight lattice, and its index must be divisible by r , since evaluation at t is a nontrivial character of order r trivial on this sublattice.

Normalize long roots to have squared length two. Hadamard's inequality bounds these lattice indices by

$$2\sqrt{3}, \quad 8, \quad 8\sqrt{3}, \quad 16 \quad (\text{A.11})$$

in the respective rows of Table 2. The respective weight-lattice covolumes are

$$1/\sqrt{3}, \quad 1/2, \quad 1/\sqrt{3}, \quad 1.$$

The second value holds for both D_4 and F_4 . Each bound is smaller than the corresponding r . Thus t is regular. Connectedness of its algebraic centralizer makes that centralizer exactly the torus.

Any nonidentity element of a torus in the simple quotient has a nontrivial prime-order power of this kind. The argument survives projective centralization: one can lift that power with order r coprime to the center, and a conjugate differing from it by a central element must have that central factor equal to one, by preservation of its order. Therefore its projective centralizer is precisely the torus in the quotient. The hypotheses on C in Lemma A.1 are now verified.

For the numerical inequality, we use

$$k(\mathcal{F}) \leq 100q^l, \quad (\text{A.12})$$

where l is the absolute rank. This follows, with a smaller constant, from [10, Theorem 1.1, p. 3024] for the simply connected fixed-point groups; taking a quotient cannot increase the number of conjugacy classes. The standard group order formulas [19, Tables 24.1–24.2, pp. 208, 211] give

$$\frac{\mathcal{F}}{|\mathcal{F}|} > \begin{array}{c|ccccc} \mathcal{F} & G_2(q) & {}^3D_4(q) & F_4(q) & {}^\epsilon E_6(q) & E_8(q) \\ \hline & 0.7q^{14} & 0.7q^{28} & q^{52}/2 & q^{78}/6 & q^{248}/2. \end{array}$$

In the two small G_2 cases the direct comparisons are

$$\begin{aligned} |G_2(3)| &= 4245696 > 100 \cdot 3^2 \cdot 7^4, \\ |G_2(4)| &= 251596800 > 100 \cdot 4^2 \cdot 13^4. \end{aligned}$$

For $q \geq 5$ the chosen G_2 torus has order $c \leq 1.24q^2$, and $0.7q^{14} > 100q^2(1.24q^2)^4$ already at $q = 5$, with increasing ratio thereafter. In triality type $c < q^4$, so it is enough that $0.7q^8 > 100$, true at $q = 2$. For F_4, E_6, E_8 use respectively $c < q^4, c < 2q^6, c < 2q^8$; the displayed lower bounds exceed $100q^l c^4$ already at $q = 2$, and thereafter with increasing ratio. This proves the desired inequality in every row, so Lemma A.1 completes these families.

A.7 Type E_7 and the Suzuki–Ree groups

For E_7 we transfer the torus obstruction from a suitable E_6 subgroup. In the simply connected algebraic group choose a maximal-rank subgroup M geometrically a Levi subgroup with derived group $D_M = [M, M]$ of type E_6 and one-dimensional center. Both rational forms E_6 and 2E_6 can be obtained: the ambient Weyl group contains -1 , which normalizes this subsystem and induces its nontrivial diagram option, so rational Weyl twisting gives the second form. Choose the sign ϵ such that $(3, q - \epsilon) = 1$.

Conjugation on the derived group gives a map π from $M(q)$ to adjoint ${}^eE_6(q)$. Its image is the simple group of that type. Indeed the simply connected derived subgroup already maps onto it: both the finite center and the Frobenius cohomology of its order-three algebraic center vanish under the chosen coprimality condition. Denote this finite simple image by $\mathcal{F}_0$, and let $C_0 \subset \mathcal{F}_0$ be the torus used above. Geometrically M is $(E_{6,\text{sc}} \times \mathbb{G}_m)/\mu_3$; the central one-dimensional torus has Frobenius action ϵq on its character lattice. The kernel $K_M = \ker \pi$ is therefore central, with order dividing a product of $q - \epsilon$ and powers of 3. It has no prime factor in common with $|C_0|$. Since $\pi(D_M(q)) = \mathcal{F}_0$, every element of $M(q)$ differs from an element of $D_M(q)$ by an element of K_M . Thus

$$M(q) = D_M(q)K_M, \quad M(q)/D_M(q) \simeq K_M/(K_M \cap D_M(q)).$$

In particular, for every primitive prime r dividing $|C_0|$, this quotient has order prime to r , and every element of r -power order in $M(q)$ belongs to $D_M(q)$.

Define $\mathcal{S}$ in projective $E_7(q)$ to be the union of the conjugates of the images of those elements of $M(q)$ whose projection belongs to $C_0 \setminus \{1\}$. The preimage in M of the algebraic torus containing C_0 is a maximal torus of M , and also of E_7 . Thus all these elements are semisimple. The set is nonempty by surjectivity, and it excludes identity: the ambient central elements project trivially to adjoint E_6 .

We need a prime-order power of each such element to be regular not only in E_6 but in E_7 . For $W \in M(q)$ with nonidentity projection in C_0 , choose a prime r dividing the order of $\pi(W)$ and a power t of W of order r . Because K_M has order prime to r , $\pi(t) \neq 1$; the quotient calculation above also puts t in $D_M(q)$. Every E_7 root restricts to a nonzero weight on this E_6 . To check nonvanishing, the fundamental weight perpendicular to the E_6 subsystem has squared norm $3/2$; a root proportional to it would have rational proportionality factor of square $4/3$, impossible. The restriction has length at most $\sqrt{2}$. Its orbit under the irreducible twisted Weyl action spans the six-dimensional weight space. The same index argument as in Equation (A.11) bounds the resulting weight-lattice index by $8\sqrt{3} < 19$, smaller than the primitive prime. Thus no root vanishes on t , and its algebraic centralizer in E_7 is precisely the maximal torus containing it.

Now conjugate $W \in \mathcal{S}$ into the specified torus in $M(q)$. Every element commuting with it projectively belongs to this torus: the center of simply connected E_7 has order at most two, so the primitive-prime power is actually centralized, and its connected centralizer is the torus. In particular a commuting $Z \notin \mathcal{S}$ lies in $M(q)$ and projects to 1 in $\mathcal{F}_0$; a nonidentity projection would lie in C_0 and put Z in $\mathcal{S}$.

For each conjugate of a nonidentity element of C_0 in $\mathcal{F}_0$, choose a lift B in $M(q)$. Both B and BZ^2 have that nonidentity projection, so the preceding regularity argument applies to each. Their projective centralizers are toral and project into the corresponding conjugate of C_0 . Projecting Equation (A.1) to $\mathcal{F}_0$ therefore puts the nonidentity projection w of W in

$$C_0^h w^{-1} C_0^h \quad \text{for every } h \in \mathcal{F}_0.$$

The counting proof of Lemma A.1, already verified for this E_6 group, excludes this. This proves the obstruction in E_7 .

We must also identify the complement inside an abelian subgroup A meeting $\mathcal{S}$. Fixing a member of the intersection conjugates A into the image $\bar{T}$ of the torus just used. The map $\pi : T \rightarrow C_0$ descends to $\bar{T}$, since the ambient center acts trivially on adjoint E_6 . Every element with nonidentity projection belongs to $\mathcal{S}$ by definition. Conversely, every element of $\mathcal{S}$ has order divisible by a prime dividing $|C_0|$: its nonidentity projection has such order, and passage to projective E_7 removes at most a factor two. No element of the image of K_M has such a prime divisor. Therefore even

conjugacy into a different torus cannot put an element of $\ker \pi$ into $\mathcal{S}$, and

$$A \setminus \mathcal{S} = A \cap \ker(\pi : \bar{T} \rightarrow C_0).$$

This completes both assertions in type E_7 .

For the simple Suzuki and rank-one Ree groups, use Equation (A.2) and their doubly transitive rank-one BN-pair action. A pair stabilizer has a torus of order $q - 1$, containing a prime-order element of order greater than three. In Suzuki type $q = 2^{2a+1} \geq 8$, and $q - 1$ is odd and not divisible by three. In Ree type $q = 3^{2a+1} \geq 27$, the number $q - 1$ has 2-adic valuation one and an odd factor greater than one, not divisible by three.

Such a prime-order element t is regular. On the standard Frobenius-stable pair torus, the exceptional Frobenius F acts on roots by $F^*\alpha = \ell^j \alpha'$, where ℓ is the characteristic and α' is a root of the opposite length; the factor ℓ^j includes the field-Frobenius part. If $\alpha(t) = 1$, then $F(t) = t$ gives $\alpha'(t)^{\ell^j} = 1$. Since the order of t is prime to ℓ , also $\alpha'(t) = 1$. The two roots are nonparallel, because a reduced root system has no parallel roots of different lengths. The index of the lattice they span in the weight lattice is at most $2\sqrt{2}$ in B_2 and at most 2 in G_2 , by the same length and covolume calculation as above. Neither index can be divisible by the chosen prime. Its centralizer is therefore the pair torus, which fixes both points individually. Choose the pair to be p, vp and apply the point-pair test.

Finally consider ${}^2F_4(q)$, $q = 2^{2a+1} \geq 8$. Let γ be the normalized diagram reversal in F_4 . The twisted Weyl action $s_1 s_2 \gamma$ has square $s_1 s_2 s_4 s_3$, a Coxeter element. The resulting finite torus C is contained in the Frobenius-square torus of order $\Phi_{12}(q)$. Its order is one of the two factors

$$q^2 \pm \sqrt{2q^3 + q} \pm \sqrt{2q} + 1,$$

with the signs chosen together; these are Malle's two cyclic tori, as recorded in [12, Theorem 8], and their orders multiply to $\Phi_{12}(q)$. Equivalently, the determinant formula bounds its order between $(\sqrt{q} - 1)^4$ and $4q^2$. It is odd and greater than one. Every prime divisor is at least thirteen, and the F_4 root-lattice argument proves that every nonidentity element has centralizer precisely C . The standard group order gives $|{}^2F_4(q)| > q^{26}/2$. Also $k({}^2F_4(q)) \leq 100q^4$; the sharper bound $q^2 + 4q + 17$ is given in [10, Table 1, p. 3049]. Thus

$$|{}^2F_4(q)| > 100q^4(4q^2)^4 \geq k({}^2F_4(q))|C|^4 \quad (q \geq 8),$$

and Lemma A.1 applies.

Completion of the proof of Theorem 6.1. The classification consists of the alternating groups, the classical and exceptional groups of Lie type, and the sporadic groups. All families have been covered above. The low-rank orthogonal groups have their usual linear, unitary, or symplectic realizations; in particular plus type $l = 3$ is PSL_4 , and rank two symplectic type was treated separately. The exceptional small derived simple groups are $G_2(2)' \simeq \mathrm{PSU}_3(3)$, ${}^2G_2(3)' \simeq \mathrm{PSL}_2(8)$, and the Tits group, already included in Table 1. Every chosen $\mathcal{S}$ is nonempty, proper, and conjugacy-invariant, and for it Equation (A.1) has been excluded for every commuting pair with $W \in \mathcal{S}$, $Z \notin \mathcal{S}$. The complement assertion was established with each construction: it is $A[2]$ for the square sets, $\{1\}$ for the self-centralizing torus sets, and the kernel of a restriction or projection homomorphism in the remaining cases. Finally, if $s \in \mathcal{S}$ but $s^{-1} \notin \mathcal{S}$, the complement subgroup in $\langle s \rangle$ would contain s^{-1} and hence s , a contradiction. This proves invariance under inversion and completes Theorem 6.1. $\square$

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